Special Relativity and Classical Field Theory

Notes on Selected Topics for the Course

"Klassische Feldtheorie"

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1 Introduction

1.1 Overview

These are notes on selected topics covered in the 3rd year (6th semester) course "Klassische Feldtheorie". Prerequisites for this course are:

- Basic Calculus and Linear Algebra
- Basics of Special Relativity
- Maxwell Theory (Electrodynamics)
- Lagrangian Mechanics and Action Principle

In general the new subjects covered in this course are (usually a strict subset of) those indicated in the table of contents:

1. At the beginning of the course I give a lightning review of the physical foundations of special relativity (definition of inertial systems, Galilean relativity principle, propagation of light, Maxwell, Michelson-Morley, Lorentz, Einstein etc.). However, since this is 1st year undergraduate material, I do not cover it in these notes, and I assume familiarity with these topics.

2. The first aim of these notes is to arrive at a Lorentz covariant formulation of special relativity and the laws of classical phyics (primarily mechanics and electrodynamics or Maxwell theory) in terms of what are known as Lorentz tensors.

   After all, special relativity is (regardless of what you may have been taught) not fundamentally a theory about people changing trains erratically, running into barns with poles, or doing strange things to their twins; rather, it is a theory of a fundamental symmetry principle of physics, namely that the laws of physics are invariant under Lorentz transformations. They should therefore also be formulated in a way which makes this symmetry manifest. This is achieved by the use of objects which transform in a simple (multi-)linear way under Lorentz transformations, and such objects are called Lorentz tensors.

3. The second aim of these notes is to provide an introduction to classical Lagrangian field theory, in order to introduce some fundamental concepts involved in the modern formulation of theoretical physics, like the Noether theorem for field theories, the energy-momentum tensor, and the idea of minimal coupling.

4. Moreover, I usually end with some remarks and reflections on gravity and relativity, as an outlook on general relativity. This is described in detail in the first part of my (voluminous) Lecture Notes on General Relativity and will therefore also not be covered in these notes.

5. Sections 7 and 8 contain supplementary and more advanced material that will not be covered in the course.

1.2 Notation and Conventions

Please do not be scared off by this section. Notation is mainly a book-keeping device, a language that one needs to get used to and that one learns by using it.

- Good notation is one that is at the same time informative, unambiguous (in the situation at hand), and easy to use.
- Bad notation is one in which objects that appear are undefined, ill-defined, or one that is uninformative or difficult to understand or remember and therefore difficult to use.

How detailed or specific the notation should be will very much depend on the context (and the person using it) and should therefore permit a certain amount of flexibility: it should be sufficiently precise to be able to perform the task at hand in an efficient and accident-free manner, but it does not have to be more precise than that.

Having said this, here are some notational conventions that I will (try to more or less consistently) adhere to in the following:

- As is common in physics, instead of using some abstract coordinate-free notation (beloved by mathematicians) we will usually work in components that refer to a specific (orthonormal) basis or (Cartesian) coordinate system.

  I usually use lower-case Roman letters from the beginning of the alphabet $(a, b, c, \ldots)$ for spacetime indices, and Roman letters from the middle of the alphabet $i, j, k, \ldots$ for spatial indices. In particular, Cartesian coordinates for a point x of the Euclidean space $\mathbb{R}^3$ are denoted by

  $$\vec{x} = (x^i) = (x^1, x^2, x^3) \quad \text{with} \quad i, j, \ldots \in \{1, 2, 3\} \ , \tag{1.1}$$

  and inertial spacetime coordinates of an event in Minkowski spacetime will be denoted by

  $$(x^a) = (x^0 = ct, x^i) \equiv (x^0, \vec{x}) \quad \text{with} \quad a, b, \ldots \in \{0, 1, 2, 3\} \ . \tag{1.2}$$

- You see that, as is customary, we have already tacitly (and now explicitly) identified a point x in $\mathbb{R}^3$, given by the coordinates $(x^i) = (x^1, x^2, x^3)$, with the position vector $\vec{x}$ (pointing from the origin to the point x).

  Once one has decided to denote the components of the position vector $\vec{x}$ by x^i, it is reasonable to extend this notation to other vectors $\vec{v} \in \mathbb{R}^3$, i.e. to denote its components by $\vec{v} = (v^i) = (v^1, v^2, v^3)$, with "upper" indices.

- We will often deal with (linear) transformations of coordinates or vectors. In this case, one needs a notation to distinguish the new from the old coordinates. Here there are several options, and which one is the most useful may depend on the circumstances (recall the discussion above), but may also be a question of personal taste.

  – In vectorial notation, one can try to distinguish the new coordinates from the old coordinates $\vec{x}$, by writing something like $\vec{x}'$ or or $\bar{\vec{x}}$, but this can quickly become

somewhat inconvenient (and is also not ideal on the blackboard, unless the backboard is really clean). Thus, in vectorial notation, it is often more convenient to use a new letter for the new coordinates, such as $\vec{y}$ or $\vec{z}$ etc. This is at least easy to read.

- In components, with initial coordinates x^i, one can also follow the above convention and simply denote the new coordinates by y^i. However, in that case it is also occasionally convenient to just use "barred" or "primed" x-coordinates instead, such as $\bar{x}^i$ (which is easy to read).

  For certain purposes, it is also useful to employ a different kind or range of indices for different coordinate systems, say $x^i, x^j, \ldots$ for the original coordinates, and something like $\bar{x}^m, \bar{x}^n, \ldots$ or $y^m, y^n, \ldots$ for the new coordinates. This has the advantage that writing something like v^i makes it clear that these are the components of a vector $\vec{v}$ with respect to the original basis, while something like v^m or $\bar{v}^m$ would then obviously refer to the coordinates of the same vector $\vec{v}$ with respect to the new basis.

- I will be rather pedantic about the positioning of indices (up/down, left/right). There are many good reasons for this (and many good reasons for not being sloppy about this; most undergraduate textbooks make a total mess of these things, even textbooks which are very good in other respects). You will (have to) get used to this, and perhaps you will also learn to appreciate the immense usefulness of paying attention to these issues.

  In particular, and at its most elementary level:

  - Care should be taken that the positioning and labelling of indices on both sides of an equation (or among different terms in an equation) is consistent. I.e. an equation like $v^a = w^a$ makes sense, but something like $v^a = w_b$ does not.
  - Summation over indices (as in matrix multiplication or in the action of a matrix on a vector, say) will usually, i.e. unless explicitly indicated otherwise, be a summation over one lower and one equal upper index, and summation over such an index pair is understood (occasionally this is called the *Einstein summation convention*).

  Thus, for the action of a matrix R on a vector $\vec{v}$ say, $\vec{w} = R\vec{v}$, the notation in components (with indices) could be

$$\vec{w} = R\vec{v} \quad \Leftrightarrow \quad w^i = \sum_k R^i{}_k v^k \equiv R^i{}_k v^k \quad (\text{OK! } \checkmark) \ , \tag{1.3}$$

  but we would not allow something (without further explanation) like

$$w^i = R_{ik} v^k \quad \text{or} \quad w_i = R_{ik} v_k \quad (\text{illegal! } \boldsymbol{\times}) \tag{1.4}$$

  At a more fundamental level, as you will learn in section 2, the positioning of indices is used to indicate and provide valuable information in a very compact way, namely how an object transforms under certain linear transformations. This is the basis of the enormously efficient and useful formalism formalism of tensor algebra and tensor calculus that we will use to formulate the Lorentz-invariant laws of physics.

- My (general relativity rather than particle physics) convention for the *Minkowski metric* is the "mostly plus" convention, i.e.

$$(\eta_{ab}) = \text{diag}(-1, +1, +1, +1) \ , \tag{1.5}$$

2 Minkowski Space(-Time) and Lorentz Tensor Algebra

2.1 Einstein Principle of Relativity as an Invariance Principle

Considerations regarding the principle of relativity (equivalence of inertial systems) on the one hand and the observed properties of the propagation of light (invariance of the velocity of light) on the other show that these properties are not compatible with the Galilean transformations between inertial systems. Since it is unreasonable to believe that there is a principle of relativity for mechanics but not for electromagnetic processes (after all, many mechanical forces are of electromagnetic origin), the Galilei transformations (and the Galilean invariant Newtonian mechanics) need to be modified in such a way as to ensure the validity of a relativity principle for Maxwell theory (electrodynamics) and mechanics.

Thus the new starting point is the premise that there *is* a principle of relativity for all physical processes, but the tacit (and, as shown by Einstein, unwarranted) assumption of a universal time should be replaced by the invariance of the speed of light (i.e. that in vacuum it has the same measured velocity in any inertial system, and independently of the velocity of the source).

Our first aim is thus to find the new correct transformations respecting the above requirements. There are many ways to do this, either by making an inspired ansatz (guess) and trial and error, or more axiomatically and systematically, or ...

For our purposes, the most useful and efficient (and in my opinion also physically most plausible) starting point is the invariance of the wave operator

$$\Box = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \Delta = -\frac{\partial^2}{(\partial x^0)^2} + \sum_{i=1}^{3}\frac{\partial^2}{(\partial x^i)^2} \tag{2.1}$$

(variously also known as the d'Alembert operator or simply "Box") describing the propagation of waves with speed c. I.e. our aim is to determine those transformations of the coordinates

$$x^a \to \bar{x}^a(x) \tag{2.2}$$

which leave $\Box$ invariant, i.e. which are such that $\bar{\Box} = \Box$,

$$-\frac{\partial^2}{(\partial \bar{x}^0)^2} + \sum_{i=1}^{3}\frac{\partial^2}{(\partial \bar{x}^i)^2} \stackrel{!}{=} -\frac{\partial^2}{(\partial x^0)^2} + \sum_{i=1}^{3}\frac{\partial^2}{(\partial x^i)^2} \quad . \tag{2.3}$$

As we will see, this approach will immediately lead us to the description of special relativity and Lorentz transformations in terms of a 4-dimensional spacetime, namely Minkowski space, and its geometry.

Remarks:

1. A priori, the invariance of $\Box$ is not sufficient for the transformation $x \to \bar{x}$ to be a transformation between inertial systems. I.e. it could be that there are transformations that leave $\Box$ invariant but that do not map stright line trajectories of massive particles to straight lines. However, this does not happen - the transformations turn out to automatically be *affine* transformations (the definition of affine transformations will be recalled below).

2. A priori, the invariance of $\Box$ is also not necessary for the transformation $x \to \bar{x}$ to be a transformation between inertial systems. I.e. it could be that there are more general transformations that do not leave the wave operator $\Box$ itself invariant, but that do leave the wave equation $\Box f = 0$ invariant, and that do map inertial systems to inertial systems, but very conveniently and cooperatively this does also not happen.

   Indeed the requirement of the invariance of $\Box$ turns out to lead to precisely a 10-parameter family of transformations generalising the Galilean group consisting of 3 rotations + 3 Galilean boosts (velocity transformations) + (3+1) space and time translations. In this sense, invariance of $\Box$ is really an optimal and optimised requirement.

3. Just as an aside, an example of a transformation leaving $\Box f = 0$ invariant but not $\Box$ itself is the dilatation

$$x^a \to \bar{x}^a = \lambda x^a \quad \Rightarrow \quad \Box \to \bar{\Box} = \lambda^{-2}\Box \ . \tag{2.4}$$

   However, dilatations do not map an inertial system to a physically equivalent and indistinguishable reference system, and neither do the other ("conformal") transformations under which the equation $\Box f = 0$ is invariant.

2.2 Warm-Up: Euclidean Geometry, Euclidean Group and the Laplace Operator

As a warm-up exercise for our task of determining the transformations under which $\Box$ is invariant and understanding the consequences and implications of this, we take a look at Euclidean geometry and its relation to the Laplace operator. The material in this section is very elementary and should be familiar to you, but perhaps it provides a slightly new perspective on things that you already know.

Our starting point is Euclidean space $\mathbb{R}^3$, equipped with standard Cartesian coordinates

$$\vec{x} = (x^1, x^2, x^3) = (x^i) \tag{2.5}$$

and equipped with the Euclidean line element

$$ds^2 = d\vec{x}^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2 \ . \tag{2.6}$$

It will be convenient to also introduce the Euclidean metric with components δ_{ij}, in terms of which the Euclidean line element can be written as

$$ds^2 = \sum_{i,j=1}^{3} \delta_{ij} dx^i dx^j \equiv \delta_{ij} dx^i dx^j \ . \tag{2.7}$$

In the last step I have employed the (so-called Einstein) summation convention, in which a summation over a lower and an equal upper index is implied.

Remarks:

1. At its most basic, δ_{ij} equips the vector space $\mathbb{R}^3$ with a scalar product,

$$\vec{v}, \vec{w} \in \mathbb{R}^3 \quad \to \quad <\vec{v}, \vec{w}> \equiv \vec{v}.\vec{w} = \delta_{ij} v^i w^j \ , \tag{2.8}$$

and hence in particular also with a notion of norm $|v|$ of a vector,

$$|v|^2 = \vec{v}.\vec{v} \geq 0 \ , \tag{2.9}$$

and with a notion of an angle α between vectors, by the usual formula

$$\vec{v}.\vec{w} = |v|\,|w| \cos\alpha \ . \tag{2.10}$$

2. The metric or line element also defines (or encodes the information about) the *geometry* of the space, such as distances between two points with coordinate differences Δx^i,

$$\Delta s^2 = \delta_{ij} \Delta x^i \Delta x^j \ , \tag{2.11}$$

the length of a curve γ,

$$L(\gamma) = \int_\gamma ds \ , \tag{2.12}$$

and likewise areas and volumes. Note that s here, and in the line element ds^2, refers to the *arc-length*, that is to the parametrisation of the curve $x^i = x^i(s)$, such that the tangent vector has unit length,

$$x^i = x^i(s): \quad \frac{d\vec{x}}{ds}.\frac{d\vec{x}}{ds} \equiv \delta_{ij} \frac{dx^i}{ds}\frac{dx^j}{ds} = 1 \ . \tag{2.13}$$

By *definition*, this equation is equivalent to the definition (2.7) of the line element, i.e.

$$\delta_{ij} \frac{dx^i}{ds}\frac{dx^j}{ds} = 1 \quad \Leftrightarrow \quad ds^2 = \delta_{ij} dx^i dx^j \ . \tag{2.14}$$

We now consider transformations of the Cartesian coordinates to (a priori arbitrary) other coordinates,

$$x^i \to \bar{x}^i(x) \equiv y^i \quad \text{or} \quad \vec{x} \to \vec{y} \ . \tag{2.15}$$

Under such a transformations differentials and partial derivatives transform with the corresponding Jacobi matrix and its inverse,

$$dy^i = \frac{\partial y^i}{\partial x^k} dx^k \quad , \quad \frac{\partial}{\partial y^i} = \frac{\partial x^k}{\partial y^i}\frac{\partial}{\partial x^k} \ . \tag{2.16}$$

I hope that you are familiar with the following three facts:

1. Affine Transformations

   The most general coordinate transformations that transform straight lines into straight lines are the so-called *affine* transformations, i.e. transformations of the form

$$\vec{y} = A\vec{x} + \vec{b} \tag{2.17}$$

   where A is an arbitrary constant matrix and $\vec{b}$ is an arbitrary constant vector. In components we write this as

$$y^i = A^i{}_k x^k + b^i \ . \tag{2.18}$$

2. Invariance of the Euclidean Line Element

   The most general coordinate transformations that leave the Euclidean line element invariant,

$$d\vec{y}^2 = d\vec{x}^2 \quad \Leftrightarrow \quad \delta_{ij} dy^i dy^j = \delta_{ij} dx^i dx^j \tag{2.19}$$

   are affine transformations

$$\vec{y} = R\vec{x} + \vec{b} \tag{2.20}$$

   where R is an orthogonal transformation, i.e. it satisfies $R^T R = \mathbb{1}$, where R^T denotes the transpose matrix and $\mathbb{1}$ the unit matrix. It is more instructive to write this condition more explicitly as the statement that R leaves the unit matrix (the Euclidean metric) invariant, by writing it as

$$R^T \mathbb{1} R = \mathbb{1} \ . \tag{2.21}$$

   In components, one has

$$y^i = R^i{}_k x^k + b^i \quad \text{with} \quad \delta_{ij} R^i{}_k R^j{}_m = \delta_{km} \ . \tag{2.22}$$

   The linear part of these transformations can be characterised by the statement that they are precisely those linear transformations that leave the length (or distance from the origin) invariant, i.e. for $y^i = A^i{}_k x^k$ one has

$$\delta_{ij} y^i y^j = \delta_{km} x^k x^m \quad \forall\, x \quad \Leftrightarrow \quad \delta_{ij} A^i{}_k A^j{}_m = \delta_{km} \ . \tag{2.23}$$

3. Invariance of the Laplace Operator

   The transformations found above are also precisely the transformations that leave the Laplace operator invariant, i.e.

$$\sum_{i=1}^{3} \frac{\partial^2}{(\partial y^i)^2} = \sum_{i=1}^{3} \frac{\partial^2}{(\partial x^i)^2} \quad \Leftrightarrow \quad \vec{y} = R\vec{x} + \vec{b} \ . \tag{2.24}$$

The proof of the assertions 2 and 3 will be given at the end of this section.

In any case, the upshot of this dicussion is that Euclidean geometry can equivalently be characterised by either the Euclidean line element (and its invariances) or the invariance of the Laplace operator,

$$\text{Euclidean Geometry:} \quad \text{Invariance of} \quad ds^2 \quad \Leftrightarrow \quad \text{Invariance of} \quad \Delta \ . \tag{2.25}$$

And therefore either requirement leads uniquely to the transformations (2.22) which form the symmetry group of Euclidean geometry (the Euclidean group - cf. below).

Remarks:

1. Rotations and Reflections

   The condition $R^T R = \mathbb{1}$ for an orthogonal transformation implies

$$\det(R^T R) = (\det(R))^2 = +1 \ . \tag{2.26}$$

   Transformations with $\det(R) = +1$ are rotations, those with $\det(R) = -1$ are a composition of a reflection and a rotation.

2. Infinitesimal Rotations

   Infinitesimal rotations are rotations with R of the form $R = \mathbb{1} + \alpha$, with α infinitesimal and with

$$(\mathbb{1} + \alpha)^T \mathbb{1} (\mathbb{1} + \alpha) = \mathbb{1} \quad \Rightarrow \quad \alpha + \alpha^T = 0 \ . \tag{2.27}$$

   Thus α is anti-symmetric. In components, an infinitesimal rotation therefore has the form

$$\delta x^i = \alpha^i{}_k x^k \tag{2.28}$$

   with

$$\alpha_{ik} \equiv \delta_{ij} \alpha^j{}_k = -\alpha_{ki} \ . \tag{2.29}$$

   It describes an infinitesimal rotation in the (ik)-plane.

   (a) As a prototypical example, consider a rotation $R(\theta)$ by the angle θ in $\mathbb{R}^2$,

$$R(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \tag{2.30}$$

   For small (infinitesimal) θ this reduces to

$$R(\theta) \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \theta \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix} \ , \tag{2.31}$$

   displaying explicitly the anti-symmetric generator of rotations.

   (b) In 3 dimensions, but only in 3 dimensions (!), one can equivalently think of a rotation in a plane as a rotation around an axis, namely the axis orthogonal to that plane, by parametrising α_{ik} as $\alpha_{ik} = \epsilon_{ikl} v^l$. Then infinitesimal rotations can be written in the (more clumsy but perhaps also more familiar vector product) form

$$\delta \vec{x} = \vec{x} \times \vec{v} \ . \tag{2.32}$$

3. Euclidean Group

   The transformations $\vec{y} = R\vec{x} + \vec{b}$ form a group. In particular, from

$$\vec{z} = S\vec{y} + \vec{c} = (SR)\vec{x} + (S\vec{b} + \vec{c}) \tag{2.33}$$

   one has the semi-direct product composition (multiplication) law

$$(S, \vec{c}).(R, \vec{b}) = (SR, S\vec{b} + \vec{c}) \ . \tag{2.34}$$

   This group is called the Euclidean Group and it is the symmetry group of Euclidean geometry.

Proofs:

- Properties of Jacobi Matrices

  It is often convenient to distinguish different coordinate systems by their indices. Thus we consider a coordinate transformation $x^i \to y^m$, and in the following indices $i, j, \ldots$ refer to the x-coordinates, and indices $m, n, \ldots$ to the y-coordinates.

  Associated with this coordinate transformation we have the Jacobi matrices

$$J_i^m = \frac{\partial y^m}{\partial x^i} \quad , \quad J_m^i = \frac{\partial x^i}{\partial y^m} \quad . \tag{2.35}$$

  These matrices are inverses to each other, i.e. they satisfy

$$J_i^m J_n^i = \delta_n^m \quad \text{and} \quad J_i^m J_m^j = \delta_i^j \quad . \tag{2.36}$$

  The Jacobi matrices are in general x-dependent (unless the coordinate transformation is at most linear), but the one crucial property that sets them apart from generic x-dependent matrices is that they satisfy

$$\frac{\partial}{\partial x^j} J_i^m = \frac{\partial^2 y^m}{\partial x^j \partial x^i} = \frac{\partial^2 y^m}{\partial x^i \partial x^j} = \frac{\partial}{\partial x^i} J_j^m \tag{2.37}$$

  (and likewise for the inverse Jacobi matrices). Abbreviating the partical derivatives by ∂_i etc., we write this identity as

$$\partial_j J_i^m = \partial_i J_j^m \quad . \tag{2.38}$$

- Proof of Assertion 2

  Invariance of the Euclidean line element,

$$\delta_{mn} dy^m dy^n = \delta_{mn} J_i^m J_j^n dx^i dx^j \stackrel{!}{=} \delta_{ij} dx^i dx^j \quad , \tag{2.39}$$

  is equivalent to

$$\delta_{mn} J_i^m J_j^n = \delta_{ij} \quad . \tag{2.40}$$

  The aim is to show that this implies that the matrix J_i^m is constant.

  Note that in general a matrix satisfying the above condition does not have to be constant: take any orthogonal matrix which describes a rotation by an angle θ, say, which satisfies the above equation; if you then make θ an arbitrary function of $\vec{x}$, $\theta = \theta(\vec{x})$, it will still satisfy the above condition because it is a purely algebraic constraint. What the argument below will show is that no such matrix can arise as the Jacobi matrix of a coordinate transformation.

  To that end, let us act on this equation with ∂_k. Using the property (2.38) twice, one deduces

$$\begin{aligned}
0 &= \delta_{mn}[(\partial_k J_i^m) J_j^n + J_i^m \partial_k J_j^n] \\
&= \delta_{mn}[(\partial_i J_k^m) J_j^n + J_i^m \partial_j J_k^n] \\
&= \partial_i(\delta_{mn} J_k^m J_j^n) - \delta_{mn} J_k^m \partial_i J_j^n + \partial_j(\delta_{mn} J_i^m J_k^n) - \delta_{mn}(\partial_j J_i^m) J_k^n \\
&= -\delta_{mn} J_k^m \partial_i J_j^n - \delta_{mn}(\partial_j J_i^m) J_k^n \\
&= -2\delta_{mn} J_k^m \partial_i J_j^n
\end{aligned} \tag{2.41}$$

(where in the last step the symmetry of δ_{mn} was used to exchange the indices m, n). Since δ and J are invertible matrices we conclude

$$\delta_{mn} J_i^m J_j^n = \delta_{ij} \quad \Rightarrow \quad \partial_i J_j^n = 0 \ . \tag{2.42}$$

Therefore the coordinate transformation must be affine, and then the linear part must be an orthogonal transformation,

$$\delta_{mn} dy^m dy^n = \delta_{ij} dx^i dx^j \quad \Rightarrow \quad y^m = R_i^m x^i + b^m \quad \text{with} \quad \delta_{mn} R_i^m R_j^n = \delta_{ij} \ . \tag{2.43}$$

- Proof of Assertion 3

  We write the Laplace operator in x-coordinates as

  $$\Delta = \delta^{ij} \partial_i \partial_j \ , \tag{2.44}$$

  where δ^{ij} is the inverse matrix to δ_{ij}, i.e. $\delta^{ij}\delta_{jk} = \delta^i{}_k$ etc. Using the chain rule

  $$\partial_i = J_i^m \partial_m \tag{2.45}$$

  one finds that

  $$\delta^{ij} \partial_i \partial_j = \delta^{ij} \partial_i (J_j^n \partial_n) = \delta^{ij} J_i^m J_j^n \partial_m \partial_n + \delta^{ij} (\partial_i J_j^n) \partial_n \ . \tag{2.46}$$

  Requiring the invariance of the Laplace operator, i.e. that this be equal to $\delta^{mn}\partial_m\partial_n$,

  $$\delta^{ij} \partial_i \partial_j \stackrel{!}{=} \delta^{mn} \partial_m \partial_n \ , \tag{2.47}$$

  leads to the 2 conditions

  $$\delta^{ij} J_i^m J_j^n \stackrel{!}{=} \delta^{mn} \quad \text{and} \quad \delta^{ij} (\partial_i J_j^n) = 0 \ . \tag{2.48}$$

  But as in the proof above, the first condition alone already implies that the Jacobi matrix has to be constant (and an orthogonal matrix), and then the second condition is identically satisfied. Thus we conclude

  $$\delta^{ij} \partial_i \partial_j = \delta^{mn} \partial_m \partial_n \quad \Rightarrow \quad y^m = R_i^m x^i + b^m \quad \text{with} \quad \delta_{mn} R_i^m R_j^n = \delta_{ij} \ . \tag{2.49}$$

2.3 From Invariance of $\Box$ to Minkowski Geometry and Lorentz Transformations

We now return to the issue of determining the new transformations between inertial systems by starting with the invariance of the wave operator $\Box$. By analogy with what we did above in the case of Euclidean geometry, this will immediately not only provide us with the required transformations, but also with their geometric interpretation.

Thus we look for those transformations $x^a \to \bar{x}^a(x)$ which satisfy

$$\bar{\Box} = \Box \quad \Leftrightarrow \quad -\frac{\partial^2}{(\partial \bar{x}^0)^2} + \sum_{i=1}^3 \frac{\partial^2}{(\partial \bar{x}^i)^2} = -\frac{\partial^2}{(\partial x^0)^2} + \sum_{i=1}^3 \frac{\partial^2}{(\partial x^i)^2} \ . \tag{2.50}$$

By analogy with the Euclidean story recalled above, we have the following facts:

1. Transformations that leave $\Box$ invariant are also precisely those transformations that leave the *Minkowski line-element*

$$ds^2 = -c^2dt^2 + d\vec{x}^2 = -(dx^0)^2 + \sum_{i=1}^{3}(dx^i)^2 \tag{2.51}$$

invariant,

$$\bar{\Box} = \Box \quad \Leftrightarrow \quad -(d\bar{x}^0)^2 + \sum_{i=1}^{3}(d\bar{x}^i)^2 = -(dx^0)^2 + \sum_{i=1}^{3}(dx^i)^2 \ . \tag{2.52}$$

As in the Euclidean case, it will be convenient to write this line element in terms of a metric, the *Minkowski metric* η_{ab}, as

$$ds^2 = \eta_{ab}dx^a dx^b \ . \tag{2.53}$$

Thus η_{ab} is a diagonal matrix with entries

$$\eta = (\eta_{ab}) = \mathrm{diag}(-1, +1, +1, +1) \ , \tag{2.54}$$

or, more explicitly but clumsily, with components

$$\eta_{00} = -1 \quad , \quad \eta_{i0} = \eta_{0i} = 0 \quad , \quad \eta_{ik} = \delta_{ik} \ , \tag{2.55}$$

or in matrix form

$$(\eta_{ab}) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix} \tag{2.56}$$

(thus we are using the "mostly plus" convention).

2. Transformations satisfying either of the above (equivalent) requirements are automatically affine transformations (thus they qualify as transformations between inertial systems),

$$\bar{x}^a = L^a_{\ b}x^b + b^a \quad , \tag{2.57}$$

where the matrices L are constrained by the condition that they leave η invariant,

$$L^T\eta L = \eta \quad \Leftrightarrow \quad \eta_{ab}L^a_{\ c}L^b_{\ d} = \eta_{cd} \ . \tag{2.58}$$

These transformations are called *Poincaré transformations*. The linear transformations $\bar{x}^a = L^a_{\ b}x^b$ are called *Lorentz transformations*.

Lorentz transformations are thus also precisely those linear transformations that leave the Minkowski length (or distance from the origin)

$$\eta_{ab}x^a x^b = -c^2t^2 + \vec{x}^2 \tag{2.59}$$

invariant, i.e. for $\bar{x}^a = L^a_{\ b}x^b$ one has

$$\eta_{ab}\bar{x}^a\bar{x}^b = \eta_{cd}x^c x^d \quad \forall\ x \quad \Leftrightarrow \quad \eta_{ab}L^a_{\ c}L^b_{\ d} = \eta_{cd} \ . \tag{2.60}$$

The proofs of these assertions are formally precisely analogous to those given in the Euclidean case in the previous section, with the replacement of δ by η.

Lorentz and Poincaré transformations form groups. Here are some of their basic properties.

1. Lorentz Group

   Lorentz transformations

   $$\bar{x}^a = L^a_{\ b} x^b \quad \text{with} \quad \eta_{ab} L^a_{\ c} L^b_{\ d} = \eta_{cd} \quad \Leftrightarrow \quad L^T \eta L = \eta \tag{2.61}$$

   form a group called the *Lorentz group*.

   Since the conditions impose 10 constraints on the a priori 16 independent parameters of a (4×4)-matrix $L^a_{\ b}$, this is a 6-parameter group. It generalises the 6-parameter Galilean transformations

   $$\vec{y} = R\vec{x} - \vec{v}t \tag{2.62}$$

   consisting of 3 rotations (or orthogonal transformations) and 3 Galilean boosts.

   The defining equations for Lorentz transformations imply

   $$\begin{aligned} L^T \eta L = \eta \qquad &\Rightarrow \quad \det(L) = \pm 1 \\ (L^T \eta L)_{00} = \eta_{00} \qquad &\Rightarrow \quad |L^0_{\ 0}| \geq 1 \end{aligned} \tag{2.63}$$

   Thus, in addition to rotations and boosts, a general Lorentz transformation can also contain time- or space-reflections (and, in particular, a transformation with $L^0_{\ 0} \leq -1$ corresponds to a time reflection). The transformations with $\det L = +1$ and $L^0_{\ 0} \geq 1$ form a connected subgroup of the Lorentz group, consisting only of rotations and boosts but no reflections. For the time being, we will not consider reflections and we will simply refer to this subgroup (technically the group of proper orthochronous Lorentz transformations) as the Lorentz group.

   Infinitesimal Lorentz rotations, i.e. Lorentz transformations with L of the form $L = \mathbb{1} + \omega$, ω infinitesimal, are characterised by

   $$(\mathbb{1} + \omega)^T \eta (\mathbb{1} + \omega) = \eta \quad \Rightarrow \quad (\eta\omega) + (\eta\omega)^T = 0 \ . \tag{2.64}$$

   Thus the matrix $\eta\omega$ is anti-symmetric. In components, an infinitesimal Lorentz transformation therefore has the form

   $$\delta x^a = \omega^a_{\ b} x^b \quad \text{with} \quad \omega_{ab} \equiv \eta_{ac} \omega^c_{\ b} = -\omega_{ba} \ . \tag{2.65}$$

2. Poincaré Group

   The transformations

   $$\bar{x}^a = L^a_{\ b} x^b + b^a \tag{2.66}$$

   are called *Poincaré transformations*, and they generate the *Poincaré group*. It is the 10-dimensional symmetry group of Minkowskian geometry, and as such it is simultaneously

the 4-dimensional spacetime counterpart of the Euclidean group and the correct special relativistic generalisation of the 10-parameter Galilean group consisting of rotations, Galilean boosts and space and time translations.

Analogously to the Euclidean group, the Poincaré group is a semi-direct product of the Lorentz group and the group of translations.

Any two inertial systems in the sense of the equivalence principle of special relativity are related by a Poincaré transformation.

2.4 Example: Lorentz Transformations in (1+1) Dimensions (Review)

To illustrate the above, we consider Lorentz transformations in (1+1) dimensions, i.e. in a spacetime with coordinates (x^0, x^1). With one spatial dimension, there are no rotations, and therefore the Lorentz group consists of boosts (in the x^1-direction) and time and space reflections. The latter are represented by the matrices

$$T = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix} \quad , \quad P = \begin{pmatrix} +1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2.67}$$

(and they will play no role in the following).

In terms of the time and space coordinates $(t, x = x^1)$, the equation for a Lorentz boost to an inertial system traveliing with velocity v in the (positive) x^1-direction takes the (hopefully familiar) form

$$\begin{aligned} \bar{t} &= \frac{1}{\sqrt{1 - v^2/c^2}}(t - (v/c^2)x) \\ \bar{x} &= \frac{1}{\sqrt{1 - v^2/c^2}}(x - vt) \ . \end{aligned} \tag{2.68}$$

Written in this way, it is obvious that this transformation reduces to a standard Galilean boost in the "non-relativistic" (better: Galilean relativistic) limit $v/c \to 0$,

$$v/c \to 0 \quad \Rightarrow \quad \bar{t} = t \quad , \quad \bar{x} = x - vt \ . \tag{2.69}$$

The asymmetry between the two equations in (2.68) is due to the fact that t and x have different dimensions, so that the conversion factor c is needed to relate one to the other. It is thus much more convenient to use $x^0 = ct$ instead of t. Then only dimensionless parameters can appear in the transformations of $(x^0, x = x^1)$. Specifically, the transformations now take the form

$$\begin{aligned} \bar{x}^0 &= \gamma(v)(x^0 - \beta(v)x^1) \\ \bar{x}^1 &= \gamma(v)(x^1 - \beta(v)x^0) \end{aligned} \tag{2.70}$$

where the dimensionless parameters $\beta(v)$ and $\gamma(v)$ are

$$\beta(v) = v/c \qquad \gamma(v) = (1 - \beta(v)^2)^{-1/2} \ . \tag{2.71}$$

Note in particular that these equations immediately imply things like time dilation and Lorentz contraction:

- Time Dilation:

  Consider a single clock at rest ($\Delta x = 0$) in the inertial system with coordinates (t, x), sending out signals at time intervals Δt. In the inertial system with coordinates $(\bar{t}, \bar{x})$, the measured time interval is

  $$\Delta\bar{t} = \gamma(v)(\Delta t - (v/c^2)\Delta x) = \gamma(v)\Delta t > \Delta t \ , \tag{2.72}$$

  measured by two distinct (synchronised) clocks at a spatial distance

  $$\Delta\bar{x} = \gamma(v)(\Delta x - v\Delta t) = -\gamma(v)v\Delta t \ . \tag{2.73}$$

  This is usually phrased as something like "moving clocks run slower than clocks at rest" (or whatever words you want to attach to the above equations). Note, however, that these words can be misleading because they suggest an immediate contradiction:

  > *But from the viewpoint of the 2nd inertial system it is the 1st one that is moving, therefore one should find $\Delta t > \Delta\bar{t}$, in contradiction with the result $\Delta\bar{t} > \Delta t$ derived above; hence I have shown that Einstein was wrong, that I am much smarter than Einstein, and that all of 20th century physics is a big conspiracy.*

  This (unfortunately all too common but faulty) reasoning ignores the fact that in the derivation given above there is a clear asymmetry between the experimental procedures in the two inertial systems: in the 1st inertial system, there is a single clock at a fixed position x, in the 2nd intertial system one needs two distinct clocks at two different positions!

  Time measurements requiring just a single clock are clearly more intrinsic and less arbitrary than those referring to a comparison of different clocks at different places (in particular, they do not require any prescription for the synchronisation of clocks at different places), and this will lead us to the definition of *proper time* in section 2.5 below.

- Lorentz Contraction

  If one considers an object of length L in the original inertial system. i.e.

  $$\Delta x^1 = L \quad \text{at} \qquad \Delta x^0 = 0 \tag{2.74}$$

  (length measurements are defined by simultaneously measuring the position of the two ends!), then in the new inertial system one has

  $$\bar{L} = \Delta\bar{x}^1 \quad \text{at} \quad \Delta\bar{x}^0 = 0 \ , \tag{2.75}$$

  and

  $$\Delta\bar{x}^0 = 0 \quad \Rightarrow \quad \Delta x^0 = \beta\Delta x^1 = \beta L \ , \tag{2.76}$$

  leading to

  $$\bar{L} = \Delta\bar{x}^1 = \gamma(\Delta x^1 - \beta\Delta x^0) = \gamma(1-\beta^2)L = \gamma^{-1}L < L \tag{2.77}$$

  (and again you can try to attach more or less misleading words to these unambiguous equations).

However, I want to emphasise that there is nothing fundamental about these Lorentz contractions or similar effects (even though they are often misrepresented in this way): they just arise when combining the effects of Lorentz transformation with a prescription or convention for measuring lengths, based on the synchronisation of clocks in a given inertial system.

Therefore, let us quickly return to more interesting things. Since β and γ are not independent,

$$\gamma(v)^2 - \gamma(v)^2\beta(v)^2 = 1 \ , \tag{2.78}$$

it is convenient to parametrise the transformation in terms of the *rapidity* α, defined by setting

$$\gamma(v) = \cosh\alpha(v) \quad , \quad \gamma(v)\beta(v) = \sinh\alpha(v) \quad \Rightarrow \quad \beta(v) = \tanh\alpha(v) \ . \tag{2.79}$$

In terms of the rapidity α, the boost can be written as a hyperbolic rotation

$$\begin{pmatrix} \bar{x}^0 \\ \bar{x}^1 \end{pmatrix} = \underbrace{\begin{pmatrix} \cosh\alpha & -\sinh\alpha \\ -\sinh\alpha & \cosh\alpha \end{pmatrix}}_{L(\alpha)} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix} \tag{2.80}$$

or

$$\bar{x}^a = L(\alpha)^a_{\ b} x^b \ . \tag{2.81}$$

For small (infinitesimal) rapidities α, $L(\alpha)$ reduces to

$$L(\alpha) \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \alpha \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \ . \tag{2.82}$$

Note that the second term is not (yet) anti-symmetric, but in accordance with the general result (2.65) above, its product with η is,

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix} \ . \tag{2.83}$$

In order to illustrate how useful it can be to rephrase Lorentz transformations in this way, as hyperbolic rotations, here are two simple applications:

1. Painless Derivation of the Relativistic Velocity Addition Formula

   Under consecutive boosts (along the same axis), the rapidities (and not the velocities) are additive,

   $$L(\alpha_1)L(\alpha_2) = L(\alpha_1 + \alpha_2) \ . \tag{2.84}$$

   The standard addition formula for hyperbolic functions then implies

   $$\alpha_3 = \alpha_1 + \alpha_2 \quad \Rightarrow \quad \beta_3 = \tanh(\alpha_1 + \alpha_2) = \frac{\tanh\alpha_1 + \tanh\alpha_2}{1 + \tanh\alpha_1 \tanh\alpha_2} \tag{2.85}$$

   and thus the relativistic velocity addition formula

   $$v_3 = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}} \ . \tag{2.86}$$

Note that this is as unstrange or unmysterious as the fact that under successive spatial rotations $R(\theta)$, say, angles are additive,

$$R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2) \ , \tag{2.87}$$

but slopes $s = \tan\theta$ are not,

$$\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2} \quad \Leftrightarrow \quad s_3 = \frac{s_1 + s_2}{1 - s_1 s_2} \ . \tag{2.88}$$

And just as for small angles slopes are approximately additive, for small rapidities velocities are approximately additive.

2. Painless Derivation of the Relativistic Doppler Effect

Under a Lorentz transformation, the lightcone coordinates

$$x^{\pm} = x^0 \pm x^1 \tag{2.89}$$

transform as

$$\bar{x}^{\pm} = \mathrm{e}^{\mp\alpha} x^{\pm} \qquad \mathrm{e}^{-\alpha} = \sqrt{\frac{1 - v/c}{1 + v/c}} \tag{2.90}$$

In an inertial system with coordinates $(x^0 = ct, x^1)$, a lightray with frequency ω is described by the wave

$$\mathrm{e}^{-i(\omega/c)x^-} = \mathrm{e}^{-i(\omega/c)(ct - x^1)} \ . \tag{2.91}$$

In terms of the inertial coordinates $\bar{x}^a$ of a boosted observer, this can be written as

$$\mathrm{e}^{-i(\omega/c)x^-} = \mathrm{e}^{-i(\bar{\omega}/c)\bar{x}^-} \tag{2.92}$$

with

$$\bar{\omega} = \mathrm{e}^{-\alpha}\omega \ . \tag{2.93}$$

(For a more general derivation, see the end of section 3.2).

2.5 Minkowski Space, Light Cones, Wordlines, Proper Time (Review)

Die Anschauungen über Raum und Zeit, die ich Ihnen entwickeln mchte, sind auf experimentell-physikalischem Boden erwachsen. Darin liegt ihre Stärke. Ihre Tendenz ist eine radikale. Von Stund' an sollen Raum für sich und Zeit für sich völlig zu Schatten herabsinken und nur noch eine Art Union der beiden soll Selbständigkeit bewahren.

([...] *Henceforth space by itself and time by itself are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.*)

(H. Minkowski, 1907)

It follows from the considerations in section 2.3 that the arena for special relativity is a four-dimensional spacetime, known as Minkowski spacetime or (henceforth) *Minkowski space* for short (ever since Minkowski's visionary 1907 talk, the union of space and time is implied by uttering the word "Minkowski"). It is the space of events, labelled by inertial coordinates x^a, and equipped with a geometry (in particular a prescription for measuring distances) encoded in the Minkowski line element

$$ds^2 = \eta_{ab} dx^a dx^b \quad . \tag{2.94}$$

This line element provides us with a notion of distance. It also equips Minkowski space with a causal structure (in particular a distinction between the future and the past of an event). Since this is basic material, I will be telegraphic:

1. Distance & Causal Structure

   (a) The Minkowski metric defines the Lorentz (and Poincaré) invariant distance

   $$(\Delta x)^2 = \eta_{ab}(x^a_P - x^a_Q)(x^b_P - x^b_Q) \tag{2.95}$$

   betwen two events P and Q with coordinates x^a_P and x^a_Q respectively.

   (b) Depending on the sign of $(\Delta x)^2$, the two events P, Q are called, spacelike, lightlike (null) or timelike separated,

   $$(\Delta x)^2 = \begin{cases} > 0 & \text{spacelike} \quad \text{separated} \\ = 0 & \text{lightlike} \quad \text{separated} \\ < 0 & \text{timelike} \quad \text{separated} \end{cases} \tag{2.96}$$

   (c) The set of events that are lightlike separated from P define the lightcone at P. It consists of two components (joined at P), the future and the past lightcone, distinguished by the sign of $x^0_Q - x^0_P$ (positive for Q on the future lightcone, $x^0_Q > x^0_P$, negative for Q on the past lightcone).

2. Curves and Tangent Vectors

   (a) A parametrised curve is given by a map $\lambda \mapsto x^a(\lambda)$. The tangent vector to the curve at the point $x(\lambda_0)$ has components

   $$x'^a(\lambda_0) = \frac{d}{d\lambda} x^a(\lambda)|_{\lambda=\lambda_0} \quad . \tag{2.97}$$

   It is called spacelike, lightlike (null) or timelike, depending on the sign of $\eta_{ab} x'^a x'^b$,

   $$\eta_{ab} x'^a x'^b \quad \begin{cases} > 0 & \text{spacelike} \\ = 0 & \text{lightlike} \\ < 0 & \text{timelike} \end{cases} \tag{2.98}$$

   This sign (and hence this classification) depends only on the image of the curve, not its parametrisation.

(b) A curve whose tangent vector is everywhere timelike is called a timelike curve (and likewise for lightlike and spacelike curves). A curve whose tangent vector is everywhere timelike or null (i.e. non-spacelike) is called a causal curve. Worldlines of massive particles are timelike curves, those of massless particles (light) are null curves.

3. Proper Time

(a) For timelike separated events and timelike curves, proper time τ, defined by

$$ds^2 = -c^2 d\tau^2 \quad \Leftrightarrow \quad d\tau^2 = -c^{-2}\eta_{ab}dx^a dx^b \tag{2.99}$$

provides one with a Lorentz invariant notion of the temporal distance τ_{PQ} along a timelike worldline connecting 2 events P and Q,

$$\tau_{PQ} = \int_P^Q d\tau \ . \tag{2.100}$$

Its physical interpretation is that it is the time shown by a single clock in the restframe (inertial or not) of the observer travelling along that worldline. As such, it clearly and almost tautologically cannot depend on a choice of inertial system.

Likewise spacelike curves are naturally parametrised by proper distance ds.

(b) While this τ_{PQ} is Lorentz invariant, it depends on the choice of world line connecting P and Q. This can be made more explicit. In any inertial system with coordinates $(t, \vec{x})$, the worldline can be written as $\vec{x} = \vec{x}(t)$, and then the above integral can be written as (and evaluated from)

$$\tau_{PQ} = \int_{t_P}^{t_Q} dt\sqrt{1 - \vec{v}^2/c^2} \ , \tag{2.101}$$

where $\vec{v} = d\vec{x}/dt$ is the coordinate velocity. This shows very clearly that τ_{PQ} depends on the velocity $\vec{v}(t)$ of the path $\vec{x}(t)$ connnecting the two events P and Q. In particular, the proper time measured by an inertial observer (e.g. with $\vec{v} = 0$) will always be larger than that measured by a non-inertial observer.

There is absolutely nothing paradoxical about this: just as it would not ever surprise you that the spatial distance between two points depends on the path taken, it should not shock you that the same is true for the temporal distance (there is no "twin paradox", just a "twin fact").

As in the brief discussion of time dilation in section 2.4 above, confusion (or, more often, deliberate obfuscation) only arises if one willfully ignores the asymmetry between the two twins / observers: one stays at all times in a fixed inertial system, the other does not. End of story ...

(c) A natural Lorentz-invariant parametrisation of timelike curves is thus provided by the Lorentz-invariant proper time τ along the curves,

$$x^a = x^a(\tau) \ , \tag{2.102}$$

with

$$cd\tau = \sqrt{-\eta_{ab}dx^a dx^b} \quad \Rightarrow \quad \eta_{ab}\frac{dx^a(\tau)}{d\tau}\frac{dx^b(\tau)}{d\tau} = -c^2 \ . \tag{2.103}$$

(d) The derivative with respect to proper time will be denoted by an overdot,

$$\dot{x}^a(\tau) = \frac{d}{d\tau} x^a(\tau) \ . \tag{2.104}$$

Because τ is Lorentz-invariant, $\bar{\tau} = \tau$, tangent vectors $\dot{x}^a$ of τ-parametrised curves transform linearly under Lorentz transformations,

$$\dot{\bar{x}}^a(\tau) = \frac{d}{d\tau} \bar{x}^a(\tau) = \frac{\partial \bar{x}^a}{\partial x^b} \frac{d}{d\tau} x^b(\tau) = L^a_{\ b} \dot{x}^b(\tau) \ . \tag{2.105}$$

These objects will be the starting point of our discussion of relativistic mechanics, and are the prototypes of what are called Lorentz vectors or, more generally, Lorentz tensors.

2.6 Lorentz Vectors and Minkowski Geometry

Our aim is to reformulate Lorentz invariant laws of physics in such a way that their invariance under Lorentz transformations is manifest. To that end, we use as building blocks objects that transform in a simple (linear, multilinear) manner under Lorentz transformations. The prototype of such objects (Lorentz tensors) are so-called Lorentz vectors.

Lorentz vectors (or *4-vectors*) are simply objects with components v^a which, under Lorentz transformations

$$\bar{x}^a = L^a_{\ b} x^b \ , \tag{2.106}$$

transform with the matrix $L^a_{\ b}$ (to be thought of as the Jacobian of the transformation relating $\bar{x}^a$ and x^a),

$$\bar{v}^a = \frac{\partial \bar{x}^a}{\partial x^b} v^b = L^a_{\ b} v^b \ . \tag{2.107}$$

It is natural to equip a vector space $\mathbb{V}$ of such 4-vectors with the (indefinite) Minkowski scalar product $\eta = (\eta_{ab})$, with

$$v.w \equiv \eta(v,w) = \eta_{ab} v^a w^b = -v^0 w^0 + v^1 w^1 + v^2 w^2 + v^3 w^3 \ . \tag{2.108}$$

Then the following properties are evident:

1. If $v^a = 0$ in one inertial systems (this means $v^a = 0 \quad \forall\ a$), then $\bar{v}^a = 0$ in any inertial system. In particular the assertion $v^a = 0$ is Lorentz invariant.

2. If v^a is a Lorentz vector, then its (Minkowski) norm

$$v^2 \equiv \eta_{ab} v^a v^b \tag{2.109}$$

is Lorentz invariant,

$$\eta_{ab} \bar{v}^a \bar{v}^b = \eta_{ab} v^a v^b \ . \tag{2.110}$$

Depending on the sign of its Minkowski norm, a Lorentz vector is called spacelike ($v^2 > 0$), lightlike (or null, $v^2 = 0$) or timelike ($v^2 < 0$).

3. If v^a and w^a are Lorentz vectors, then their (Minkowski) scalar product

$$v.w = \eta_{ab} v^a w^b \tag{2.111}$$

is Lorentz invariant,

$$\eta_{ab} \bar{v}^a \bar{w}^b = \eta_{ab} v^a w^b \ . \tag{2.112}$$

Here are, for the sake of illustration, two simple consequences of these definitions:

- If v is timelike and $v.w = 0$ then w is spacelike.
- Any timelike vector v can be written as the sum of 2 lightlike vectors,

$$\forall\, v \quad \text{with} \quad v^2 < 0 \quad \exists \quad w_1, w_2 \quad \text{with} \quad (w_1)^2 = (w_2)^2 = 0 \quad \text{such that} \quad v = w_1 + w_2 \ . \tag{2.113}$$

One way to prove such statements is to note that, because these statements are Lorentz invariant, it suffices to prove them in one conveniently chosen inertial system in order to establish the validity of these statements in all inertial systems. For a timelike vector v, such a convenient choice of inertial system is one where v has components

$$v = (v^0 \neq 0, v^1 = 0, v^2 = 0, v^3 = 0) \ . \tag{2.114}$$

Then the first statement follows immediately, because in this inertial system w will have only spatial components,

$$v.w = 0 \quad \Rightarrow \quad w^0 = 0 \ . \tag{2.115}$$

The second statement can be established by decomposing v e.g. as

$$v = \tfrac{1}{2}(v^0, v^0, 0, 0) + \tfrac{1}{2}(v^0, -v^0, 0, 0) \tag{2.116}$$

both of which are evidently null. Thus you can send a message to yourself in the future by bouncing light off a mirror ...

Beware however, that other seemingly plausible geometric statements about Minkowskian geometry need not be true. E.g.

- The sum of two spacelike vectors is not necessarily spacelike (take $v = (1, 2, 0, 0)$ and $w = (1, -2, 0, 0)$)
- The sum of two timelike vectors is not necessarily timelike (however, this becomes a true statement if one adds the condition that the two vectors are pointing towards the future).
- The sum of two null vectors is not necessarily null (as seen above).

Much more fun can be had along these lines, but this ends the (for our purposes more than sufficient) brief excursion into the realm of Minkowskian analytic geometry.

2.7 Lorentz Scalars and Lorentz Covectors

Lorentz vectors are just one particular example of objects that transform in a nice multilinear way under Lorentz transformations.

Actually the simplest objects are so-called *Lorentz Scalars*. Lorentz scalars are objects that are invariant under Lorentz transformations. Examples are e.g. the proper time τ and scalar products and norms of Lorentz vectors. (In particular, therefore, the scalar product is a scalar, a terminological convenience ...).

Another class of objects that transform in an as simple way as Lorentz vectors, and that occur quite naturally are so-called *Lorentz Covectors*. Lorentz covectors are objects u_a that transform under Lorentz transformations with the dual (contragredient = inverse transpose) transformation, i.e.

$$\bar{u}_a = \Lambda_a{}^b u_b \quad \text{with} \quad \Lambda_a{}^b L^a{}_c = \delta^b{}_c \quad \Leftrightarrow \quad \Lambda = (L^T)^{-1} \ . \tag{2.117}$$

Since by definition L satisfies $L^T \eta L = \eta$, Λ can equivalently be written as

$$\Lambda = (L^T)^{-1} \quad \Leftrightarrow \quad \Lambda = \eta L \eta^{-1} \tag{2.118}$$

(the component version of this equation will be derived below). In particular, therefore, given a Lorentz transformation L, Λ can be obtained from L without having to explicitly invert the matrix L.

The characteristic (and defining) feature of Lorentz covectors is that their "contraction" (pairing) with a Lorentz vector gives a Lorentz scalar,

$$\bar{u}_a \bar{v}^a = \Lambda_a{}^b u_b L^a{}_c v^c = \delta^b{}_c u_b v^c = u_a v^a \ . \tag{2.119}$$

Remarks:

1. Thus covectors can naturally be regarded as elements of the dual $\mathbb{V}^*$ of the space $\mathbb{V}$ of 4-vectors, with u_a defining the Lorentz-invariant linear mapping

$$u: \ v \in \mathbb{V} \mapsto u(v) = u_a v^a \in \mathbb{R} \ . \tag{2.120}$$

   In general, the finite dimensional vector spaces $\mathbb{V}$ and $\mathbb{V}^*$ are isomorphic, but there is no natural isomorphism between them. However, if $\mathbb{V}$ has been equipped with a scalar product (as in our case), there *is* a natural identification $\mathbb{V}^* \cong \mathbb{V}$, given by

$$v \in \mathbb{V} \mapsto v^* \in \mathbb{V}^*: \quad v^*(w) = \eta(v, w) \ . \tag{2.121}$$

   In components, this is the statement that if v^a is a Lorentz vector, then

$$v^*_a \equiv \eta_{ab} v^b \tag{2.122}$$

   is a covector. Since already the index position indicates that this is a covector (an element of the dual space), one usually omits the *, and writes this simply as

$$v_a = \eta_{ab} v^b \ . \tag{2.123}$$

(We will verify below directly that v_a indeed transforms as, hence is, a covector.) Thus v^a refers to v thought of as an element of $\mathbb{V}$ (one also says that these are the contravariant components of v), while v_a refers to v thought of as an element v^* of $\mathbb{V}^*$ with the help of the scalar product or metric (and one also refers to these as the covariant components of v). Thus the covariant components v_a of v are related to the contravariant components v^a by

$$(v_0, v_1, v_2, v_3) = (-v^0, v^1, v^2, v^3) \ . \tag{2.124}$$

And physicists also refer to this operation $v^a \mapsto v_a$ as "using the metric to lower the index".

2. Note that in Euclidean geometry, with $\eta_{ab} \to \delta_{ik}$ (or $\eta \to \mathbb{1}$) and $L \to R$, orthogonal transformations, one has

$$(R^T)^{-1} = R \ , \tag{2.125}$$

and thus the dual transformation is the same as the original transformation. Moreover, numerically the contravariant components are equal to the covariant components of a vector $\vec{v}$,

$$v_i = \delta_{ik} v^k \quad \Rightarrow \quad (v_1, v_2, v_3) = (v^1, v^2, v^3) \ . \tag{2.126}$$

Therefore one usually does not make a distinction between vectors and covectors in that context. However, conceptually it would make sense to do so, because one still uses the Euclidean metric to (tacitly) identify $\mathbb{R}^3$ with its dual $\mathbb{R}^{3*}$.

3. Clearly, if η_{ab} allows us to transform vectors into covectors, then its inverse can be used to map covectors to vectors. To be consistent with the conventions for the positioning of indices we have adopted, we denote the inverse metric by η^{ab},

$$\eta^{ab}\eta_{bc} = \delta^a_{\ c} \ , \tag{2.127}$$

and then we have the statement that if u_a is a covector then

$$u^a \equiv \eta^{ab} u_b \tag{2.128}$$

is a vector (and the inverse metric is used to "raise the index").

4. Just like the Minkowski norm of a vector is a scalar, so is the Minkowski norm of a covector,

$$\eta^{ab}\bar{u}_a\bar{u}_b = \eta^{ab}u_a u_b \ . \tag{2.129}$$

Note that, with the convention for raising and lowering indices, this can equivalently be written as

$$\eta^{ab}u_a u_b = u_a u^a = u^a u_a = \eta_{ab}u^a u^b \ . \tag{2.130}$$

5. One way to establish directly that $v_a = \eta_{ab}v^b$ is a covector is to calculate

$$\bar{v}_a = \bar{\eta}_{ab}\bar{v}^b = \eta_{ab}L^b_{\ c}v^c = (\eta_{ab}L^b_{\ c}\eta^{cd})v_d \ , \tag{2.131}$$

where we made use of the invariance of the Minkowski metric, i.e. $\bar{\eta}_{ab} = \eta_{ab}$. Thus v_a transforms with $\eta_{ab}L^b_{\ c}\eta^{cd}$, but these are just the components of the matrix $\eta L \eta^{-1}$ which, as we have seen in (2.118), is precisely the matrix Λ,

$$\Lambda_a^{\ d} = \eta_{ab}L^b_{\ c}\eta^{cd} \ . \tag{2.132}$$

6. If one extends the convention of raising and lowering indices with the Minkowski metric and its inverse to the Lorentz transformation matrices themselves (one can but does not have to do that), then one can write

$$\Lambda_a^{\ d} = \eta_{ab} L^b_{\ c} \eta^{cd} = L_a^{\ d} \ , \tag{2.133}$$

with

$$L_a^{\ b} L^a_{\ c} = \delta^b_{\ c} \ . \tag{2.134}$$

Obviously this requires being really careful with the relative up-down and left-right positioning of indices, and is therefore only recommended if you are comfortable with this. It does however have the advantage, that the transformation behaviour of a covector follows trivially from that of a vector (so that one does not need to postulate them seperately),

$$\bar{v}^a = L^a_{\ b} v^b \quad \Rightarrow \quad \bar{v}_a = L_{ab} v^b = L_a^{\ b} v_b \ . \tag{2.135}$$

2.8 Higher Rank Lorentz Tensors and Tensor Algebra

With Lorentz vectors and Lorentz covectors at our disposal, we can now also easily construct objects that transform in a slightly more general (multilinear) way. For example, if v^a and w^a are Lorentz vectors, then their direct product $v^a w^b$ does not transform as a vector but like the product of two vectors, with two matrices L, and likewise for other direct products of vectors and covectors.

We formalise this by defining general *Lorentz tensors*.

Lorentz (p,q)-tensors are objects $T^{a_1 \ldots a_p}_{c_1 \ldots c_q}$ that transform under Lorentz transformations like a product of p vectors and q covectors,

$$T^{a_1 \ldots a_p}_{c_1 \ldots c_q} \to \bar{T}^{a_1 \ldots a_p}_{c_1 \ldots c_q} = L^{a_1}_{\ b_1} \ldots L^{a_p}_{\ b_p} \Lambda_{c_1}^{\ d_1} \ldots \Lambda_{c_q}^{\ d_q} \, T^{b_1 \ldots b_p}_{d_1 \ldots d_q} \ . \tag{2.136}$$

With this terminology, Lorentz vectors are (1,0)-tensors, Lorentz covectors are (0,1)-tensors and Lorentz scalars are (0,0)-tensors.

It is clear from the definition that

- linear combinations of (p,q)-tensors are again (p,q)-tensors;
- the direct product of a (p_1,q_1)-tensor with a (p_2,q_2)-tensor is a (p_1+p_2, q_1+q_2)-tensor.

Thus tensors form an algebra.

This tensor algebra comes equipped with two more useful algebraic operation, namely contraction and (anti-)-symmetrisation:

1. Contraction

   The contraction between (summation over) one upper and and lower index maps a (p,q)-tensor to a $(p-1, q-1)$-tensor. Examples:

(a) If v^a is a vector, and u_a is a covector, then $u_a v^b$ is the prototype of a (1,1)-tensor, and its contraction $u_a v^a$ is (as we have already seen) a scalar or (0,0)-tensor. More generally, then, if $T^a_{\ b}$ is any (1,1)-tensor, its "trace" $T^a_{\ a}$ is a scalar.

(b) If v^a is a vector and T_{ab} a (0,2)-tensor, $T_{ab}v^c$ is a (1,2)-tensor, and the contraction $T_{ab}v^b$ is a covector or (0,1)-tensor:

$$\bar{T}_{ab}\bar{v}^b = \Lambda_a^{\ c}\Lambda_b^{\ d}T_{cd}L^b_{\ e}v^e = \Lambda_a^{\ c}\delta^d_{\ e}T_{cd}v^e = \Lambda_a^{\ c}T_{cd}v^d \quad . \tag{2.137}$$

The rule and upshot of this is that the tensor type can always be read off from the number and positioning of the free indices, a huge calculational simplification.

2. Symmetrisation and anti-Symmetrisation

A $(0,2)$-tensor T_{ab}, is said to be symmetric if $T_{ab} = T_{ba}$ and anti-symmetric if $T_{ab} = -T_{ba}$. This is well-defined because it is a Lorentz-invariant notion: a tensor is symmetric in all inertial systems iff it is symmetric in one inertial system, etc.

Given any $(0,2)$-tensor T_{ab}, one can decompose it into its symmetric and anti-symmetric parts as

$$T_{ab} = \tfrac{1}{2}(T_{ab} + T_{ba}) + \tfrac{1}{2}(T_{ab} - T_{ba}) \equiv T_{(ab)} + T_{[ab]} \quad . \tag{2.138}$$

The decomposition into symmetric and anti-symmetric parts is invariant under Lorentz transformations. In particular, when T_{ab} is a tensor, also $T_{(ab)}$ and $T_{[ab]}$ are tensors, and thus (anti-)symmetrisation is yet another linear operation that one can perform on tensors.

The factor $\frac{1}{2}$ is chosen such that the symmetrisation of a symmetric tensor is the same as the original tensor,

$$T_{ab} = T_{ba} \quad \Rightarrow \quad T_{(ab)} = T_{ab} \quad , \quad T_{[ab]} = 0 \tag{2.139}$$

(and likewise for the anti-symmetrisation of anti-symmetric tensors).

This can be generalised to the (anti-)symmetrisation of any pair of (contravariant or covariant) indices; e.g.

$$T_{(ab)c} = \tfrac{1}{2}(T_{abc} + T_{bac}) \tag{2.140}$$

is the symmetrisation of T_{abc} in its first and second index.

It can also be generalised to the total (anti-)symmetrisation of a higher-rank tensor; e.g.

$$T_{(abc)} \equiv \tfrac{1}{3!}(T_{abc} + T_{bac} + T_{cba} + T_{bca} + T_{acb} + T_{cab}) \tag{2.141}$$

is totally symmetric, i.e. symmetric under the exchange of any pair of indices, and

$$T_{[abc]} \equiv \tfrac{1}{3!}(T_{abc} - T_{bac} - T_{cba} + T_{bca} - T_{acb} + T_{cab}) \tag{2.142}$$

is totally anti-symmetric. The prefactor $\frac{1}{6}$ is again there to ensure that the total symmetrisation of a totally symmetric tensor is the original tensor (and likewise for the total anti-symmetrisation of totally anti-symmetric tensors). This generalises in an evident way to higher rank p tensors, with the combinatorial prefactor $1/p!$.

A special case of this, which will appear in the context of Maxwell theory, is the total anti-symmetrisation of a tensor T_{abc} that is already anti-symmetric in two of its indices,

say $T_{abc} = T_{a[bc]}$. In that case, three out of the six terms in the above expression are superfluous and total anti-symmetrisation reduces to cyclic permutation,

$$T_{abc} = T_{a[bc]} \quad \Rightarrow \quad T_{[abc]} = \tfrac{1}{3}(T_{abc} + T_{cab} + T_{bca}) \ . \tag{2.143}$$

Remarks:

1. A (1,1)-tensor $T^a_{\ b}$ can be thought of as an element of $\mathbb{V} \otimes \mathbb{V}^*$, and thus as a linear map

$$T = (T^a_{\ b}) : \quad \mathbb{V} \to \mathbb{V} \ , \tag{2.144}$$

given by

$$v^a \mapsto T^a_{\ b} v^b \tag{2.145}$$

(which, by our rules, is indeed again a vector). The trace defined above is then really just the usual trace of a linear map. However, given a (0,2)-tensor T_{ab}, say, something like $\sum_a T_{aa}$ is *not* a Lorentz scalar. This is reflected in the fact that T_{ab} can be thought of as an element of $\mathbb{V}^* \otimes \mathbb{V}^*$ or as a linear map

$$T = (T_{ab}) : \quad \mathbb{V} \to \mathbb{V}^* : \quad v^a \mapsto T_{ab} v^b \ , \tag{2.146}$$

between two different vector spaces. For such maps, there is no natural definition of a trace. However, given the metric (scalar product), we do of course have an identification $\mathbb{V} \cong \mathbb{V}^*$, and indeed with the help of the metric we can define a Lorentz invariant trace of T_{ab} by

$$T_{ab} \to T^a_{\ b} = \eta^{ac} T_{cb} \to T^a_{\ a} = \eta^{ac} T_{ca} = \eta^{ab} T_{ab} \ . \tag{2.147}$$

2. If, as in the above equation, one extends the convention of raising and lowering indices with the Minkowski metric to higher rank tensors, then some care is required with the relative positioning of upper (contravariant) and lower (covariant) indices. E.g. $T^a_{\ b} = \eta^{ac} T_{cb}$ (raising the first index) is not the same as $T_b^{\ a} = \eta^{ac} T_{bc}$ (raising the second index) unless T_{ab} is symmetric.

3. Frequently it will be of interest to know how to construct a Lorentz scalar from some Lorentz tensor, perhaps with the help of the Minkowski metric (which is always available). We have seen various and prototypical examples of this in the above, like taking a trace $T_{ab} \to \eta^{ab} T_{ab}$ or taking a norm, $v^a \to \eta_{ab} v^a v^b$, and all this generalises in various ways to higher rank tensors. For example, from a (1,3)-tensor $R^a_{\ bcd}$ one could construct the scalar

$$R \equiv R^a_{\ bad} \eta^{bd} \tag{2.148}$$

which is linear in the tensor, or the norm

$$K \equiv R^a_{\ bcd} \eta_{ae} \eta^{bf} \eta^{cg} \eta^{dh} R^e_{\ fgh} \equiv R_{abcd} R^{abcd} \ , \tag{2.149}$$

or something intermediate like

$$R^a_{\ bcd} \to R_{bd} = R^a_{\ bad} \to R^{ab} R_{ab} = \eta^{ac} \eta^{bd} R_{cd} R_{ab} \ , \tag{2.150}$$

etc etc. (This example is not as crazy or random as it looks - you will encounter it if you study general relativity: $R^a_{\ bcd}$ is the Riemann curvature tensor, R_{bd} the Ricci tensor, R the Ricci scalar, K the Kretschmann scalar ...).

4. The number of independent components of a general (p, q)-tensor in 4 dimensions is 4^{p+q}. The number of independent components is reduced if the tensor has some symmetry properties. Thus

   - a symmetric (0,2)- or (2,0)-tensor has $4 \times 5/2 = 10$ independent components,
   - an anti-symmetric (0,2)- or (2,0)-tensor has $4 \times 3/2 = 6$ independent components,
   - a totally anti-symmetric $(0, 3)$-tensor T_{abc} has $4 \times 3 \times 2/(2 \times 3) = 4$ independent components,
   - and a totally anti-symmetric $(0, 4)$-tensor T_{abcd} has only got one independent component, namely T_{0123} (all the others being determined by anti-symmetry).

5. One argument that we will frequently make use of is that if S_{ab} is symmetric and A^{ab} is anti-symmetric then $S_{ab}A^{ab} = 0$,

$$S_{ab} = S_{(ab)} \quad , \quad A^{ab} = A^{[ab]} \quad \Rightarrow \quad S_{ab}A^{ab} = 0 \ . \tag{2.151}$$

   There are several ways to prove this:

   (a) The most pedestrian way is to write out the contraction explicitly, and to use the (anti-)symmetry properties, in particular also $A^{11} = 0$, to conclude

$$S_{ab}A^{ab} = S_{11}A^{11} + S_{12}A^{12} + S_{21}A^{21} + \ldots = 0 + S_{12}A^{12} - S_{12}A^{12} + \ldots = 0 \ . \tag{2.152}$$

   (b) More abstractly, one can simply exchange the summation indices to conclude

$$S_{ab}A^{ab} = S_{ba}A^{ba} = -S_{ab}A^{ab} \quad \Rightarrow \quad S_{ab}A^{ab} = 0 \ . \tag{2.153}$$

   (c) In matrix language, this is the statement that the trace of a product of a symmetric matrix S and an anti-symmetric matrix A is zero,

$$\mathrm{tr}(SA) = \mathrm{tr}(SA)^T = \mathrm{tr}(A^T S^T) = -\,\mathrm{tr}(AS) = -\,\mathrm{tr}(SA) \quad \Rightarrow \quad \mathrm{tr}(SA) = 0 \ . \tag{2.154}$$

   More generally, when T^{ab} is an arbitrary tensor, only its symmetric part will contribute to the contraction with S_{ab}, and only the anti-symmetric part will contribute to the contraction with A_{ab},

$$S_{ab}T^{ab} = S_{ab}T^{(ab)} \quad , \quad A_{ab}T^{ab} = A_{ab}T^{[ab]} \ , \tag{2.155}$$

   so that e.g.

$$S_{ab}u^a v^b = \tfrac{1}{2}S_{ab}(u^a v^b + u^b v^a) \quad , \quad A_{ab}u^a v^b = \tfrac{1}{2}A_{ab}(u^a v^b - u^b v^a) \ . \tag{2.156}$$

2.9 Lorentz Tensor Fields and the Lorentz-invariance of Tensorial Equations

So far, we have defined tensors purely algebraically. In physical applications, however, we will usually deal with tensors that are e.g. defined along the worldline of a particle or that are functions of the Minkowski (inertial) coordinates.

We formalise this by defining a *tensor field* to be a map from Minkowski space to a space of tensors. I.e. a tensor field assigns to each point of Minkowski space a tensor

$$T: \quad x \mapsto T^{a_1\ldots a_p}_{c_1\ldots c_q}(x) \tag{2.157}$$

(with the obvious modification for a tensor field along a curve etc.). Under a Lorentz transformation $x^a \mapsto \bar{x}^a = L^a_{\ b} x^b$, the tensor field transforms as

$$\bar{T}^{a_1\ldots a_p}_{c_1\ldots c_q}(\bar{x}) = L^{a_1}_{\ b_1}\ldots L^{a_p}_{\ b_p}\Lambda^{\ d_1}_{c_1}\ldots\Lambda^{\ d_q}_{c_q}\ T^{b_1\ldots b_p}_{d_1\ldots d_q}(x) \ . \tag{2.158}$$

- In particular, a *scalar field* is an object $f(x)$ satisfying

  $$\bar{f}(\bar{x}) = f(x) \ . \tag{2.159}$$

  Even more explicitly, writing both sides of functions of x, this is

  $$\bar{f}(\bar{x}(x)) = f(x) \ . \tag{2.160}$$

  While this may look baroque, this is nothing other than a special case of the standard transformation behaviour of a function under coordinate transformations, namely that (up to the obvious change of argument) the function does not transform under changes of coordinates.

  At the risk of over-explaining this, here is a simple example to illustrate this point. Take $\mathbb{R}^2$ with Cartesian coordinates (x^1, x^2) and consider e.g. the function $f(x) = (x^1)^2 + (x^2)^2$. In terms of polar coordinates $\bar{x}^1 = r, \bar{x}^2 = \phi$, this is the function $\bar{f}(\bar{x}) = (\bar{x}^1)^2$, and thus

  $$\bar{f}(\bar{x}(x)) = (\bar{x}^1(x))^2 = (x^1)^2 + (x^2)^2 = f(x) \ . \tag{2.161}$$

- Liekwise, a *vector field* is now an object $V^a(x)$ satisfying

  $$\bar{V}^a(\bar{x}) = L^a_{\ b} V^b(x) \ , \tag{2.162}$$

  a *covector field* is an object $U_a(x)$ that transforms as

  $$\bar{U}_a(\bar{x}) = \Lambda_a^{\ b} U_b(x) \ , \tag{2.163}$$

  etc.

- Given a vector field $V^a(x)$, say, $\eta_{ab}V^a(x)V^b(x)$ is then an example of a scalar field, and, as we will see below, given a scalar field $f(x)$, its partial derivatives give a *covector field*

  $$U_a(x) = \partial_{x^a} f(x) \ . \tag{2.164}$$

What is of central importance for us is the fact that tensorial equations of the form

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(x) = 0 \tag{2.165}$$

are automatically Lorentz invariant in the sense that they are satisfied in one inertial system if and only if they are satisfied in all inertial systems.

The proof of this is straightforward. Since the matrices L and Λ are invertible, the tensorial transformation behaviour (2.158) implies that

$$\bar{T}^{a_1\ldots a_p}_{c_1\ldots c_q}(\bar{x}) = 0 \quad \forall\, \bar{x} \quad \Leftrightarrow \quad T^{b_1\ldots b_p}_{d_1\ldots d_q}(x) = 0 \quad \forall\, x \ . \tag{2.166}$$

This is precisely the statement that the equation $T^{\cdots}_{\cdots}(x) = 0$ is satisfied in one inertial system if and only if it is satisfied in all inertial systems.

We see that with Lorentz tensor fields as building blocks, it is utterly straightforward to write down Lorentz-invariant equations. This is the main justification for introducing and working with such objects.

2.10 Lorentz-invariant Integration

In order to write down equations of motion, Lagrangians, actions etc., we need not just the purely algebraic operations we have discussed so far, but we also need to be able to differentiate and integrate in a way compatible with Lorentz invariance. We start with integration. This is required e.g. when we want to write down actions for particles (mechanics) or fields (Maxwell theory etc.).

In the former case, in Galilean meachnics, actions are writen as integrals over the (absolute) coordinate time t. This is not a good starting point for us. Rather, as already mentioned above, it will be naturally to parametrise the worldlines of particles by their Lorentz invariant proper time τ, and to consider integrals $\int d\tau(\ldots)$ instead. Indeed, if $f(\tau)$ is a Lorentz invariant scalar along the worldline $x(\tau)$, then

$$S_f = \int d\tau f(\tau) \tag{2.167}$$

is manifestly Lorentz invariant.

When it comes to field theory, we shall integrate over all of Minkowski space. By the usual rules of calculus, under a coordinate transformation $x \to \bar{x}(x)$ the volume element d^4x transforms as

$$d^4\bar{x} = \left|\frac{\partial \bar{x}}{\partial x}\right| d^4x \ , \tag{2.168}$$

where

$$\left|\frac{\partial \bar{x}}{\partial x}\right| = \left|\det(\partial \bar{x}^a/\partial x^b)\right| \tag{2.169}$$

is the determinant of the Jacobi matrix. Now for a Lorentz transformation one has

$$\left|\frac{\partial \bar{x}}{\partial x}\right| = |\det(L^a_{\ b})| = +1 \ , \tag{2.170}$$

and thus d^4x is Lorentz invariant. Then the integral of a scalar field $F(x)$

$$S_F = \int d^4x\ F(x) \tag{2.171}$$

is also manifestly Lorentz invariant.

By the same token as above, for mechanics we have a natural Lorentz invariant differential operator, namely $d/d\tau$, which we will use to define Lorentz tensorial velocities,

$$\dot{x}^a(\tau) = \frac{d}{d\tau} x^a(\tau) \ . \tag{2.172}$$

accelerations etc., as in (2.104).

Turning now to the differentiation of tensor fields, we first need to determine how partial derivatives with respect to inertial coordinates transform under Lorentz transformation. We will see that, just as differentials transform like vectors,

$$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad d\bar{x}^a = L^a_{\ b} dx^b \ , \tag{2.173}$$

partial derivatives transform inversely, i.e. like covectors,

$$\frac{\partial}{\partial \bar{x}^a} = \Lambda_a^{\ b} \frac{\partial}{\partial x^b} \ . \tag{2.174}$$

Proof: Set

$$\frac{\partial}{\partial \bar{x}^a} = M_a^{\ b} \frac{\partial}{\partial x^b} \ . \tag{2.175}$$

We will show that $M = \Lambda$. To that end, use the chain rule to write

$$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad \frac{\partial}{\partial x^b} = \frac{\partial \bar{x}^c}{\partial x^b} \frac{\partial}{\partial \bar{x}^c} = L^c_{\ b} \frac{\partial}{\partial \bar{x}^c} \ , \tag{2.176}$$

and plug this into the previous equation,

$$\frac{\partial}{\partial \bar{x}^a} = M_a^{\ b} L^c_{\ b} \frac{\partial}{\partial \bar{x}^c} \ , \tag{2.177}$$

to conclude

$$M_a^{\ b} L^c_{\ b} = \delta^c_{\ a} \quad \Leftrightarrow \quad M_a^{\ b} = \Lambda_a^{\ b} \ . \tag{2.178}$$

It follows that the partial derivative of a scalar field f, i.e. $\bar{f}(\bar{x}) = f(x)$, is a covector field, and consistently with our conventions we will abbreviate it by $\partial_a f$ etc.,

$$\frac{\partial}{\partial x^a} f(x) = \partial_a f(x) \ . \tag{2.179}$$

More generally, the partial derivatives of the components of a (p,q)-tensor field,

$$T^{a_1 \ldots a_p}_{c_1 \ldots c_q}(x) \quad \rightarrow \quad \partial_a T^{a_1 \ldots a_p}_{c_1 \ldots c_q}(x) \tag{2.180}$$

are the components of a $(p, q+1)$-tensor field (and again we can always just read off the tensor structure from the positioning of the free indices). For example, if V^a is a vector field, $\partial_b V^a$ is a (1,1)-tensor, $\partial_b \partial_c V^a$ is a (1,2)-tensor, etc.

In particular, since ∂_a transforms like a covector, we can use the standard recipes to construct Lorentz scalars from it. For example, if $V^a(x)$ is a Lorentz vector field, then

$$V \equiv V^a \partial_a \tag{2.181}$$

is a Lorentz invariant 1st order differential operator, the directional derivative along the vector field.

Moreover, if $J^a(x)$ is a Lorentz vector field, then its *4-divergence* $\partial_a J^a(x)$ is a scalar field. To see what this 4-divergence means, parametrise the vector field ("4-current") as

$$(J^a(x)) = (c\rho(x), j^i(x)) \ . \tag{2.182}$$

Then one has

$$\partial_a J^a(x) = \frac{\partial}{\partial t}\rho + \partial_i j^i = \frac{\partial}{\partial t}\rho + \vec{\nabla}.\vec{j} \ . \tag{2.183}$$

Thus the "continuity equation"

$$\frac{\partial}{\partial t}\rho + \vec{\nabla}.\vec{j} = 0 \quad \Leftrightarrow \quad \partial_a J^a(x) = 0 \tag{2.184}$$

(which arises in many different contexts) is Lorentz invariant, provided that the current $J^a(x)$ indeed transforms as a 4-vector. This will typically not be the case. However, we will verify much later that, cooperatively, for Maxwell theory the electric charge density ρ and electric current $\vec{j}$ indeed combine precisely into such a Lorentz vector, and then we can immediately conclude that the continuity equation of Maxwell theory (which is implied by the Maxwell equations) is Lorentz invariant. This will be the first step in our programme to reformulate Maxwell theory in a manifestly Lorentz invariant (i.e. Lorentz tensorial) way.

Finally, we know of another way to construct a scalar from a covector u_a, namely to take its norm $\eta^{ab}u_a u_b$. Applying this to ∂_a, we thus get the Lorentz invariant differential operator $\eta^{ab}\partial_a\partial_b$. What is this operator? Well, of course, this is just the wave operator

$$\eta^{ab}\partial_a\partial_b = -\frac{\partial^2}{(\partial x^0)^2} + \sum_{i=1}^{3}\frac{\partial^2}{(\partial x^i)^2} = \Box \ , \tag{2.185}$$

that was the starting point for our investigations at the beginning of this section. Using the conventions for raising and lowering indices also for ∂_a, $\Box$ can also be (and frequently is) written as

$$\Box = \eta^{ab}\partial_a\partial_b = \partial^a\partial_a = \partial_a\partial^a \ . \tag{2.186}$$

So we have come full circle. We originally defined Lorentz transformations by the requirement of invariance of $\Box$, and we have now ended up with a formalism in which this invariance is manifest! This is always the sign of a good formalism:

With the right formalism, things that should be simple or obvious are indeed simple or obvious!

3 Lorentz-Covariant Formulation of Relativistic Mechanics

3.1 Covariant Formulation of Relativistic Kinematics and Dynamics

As our first application of the formalism developed in the previous section, we consider relativistic mechanics. It is clear that the Newtonian description of the motion of a particle in terms of $\vec{x}(t)$ is a suboptimal starting point for relativistic mechanics. Instead, as alluded to several times above, our starting point for describing the motion of massive particles will be the parametrisation

$$x^a = x^a(\tau) \tag{3.1}$$

of the position of a particle in Minkowski space by its proper time τ. Here are the subsequent (tensorial, specifically vectorial) building blocks.

1. 4-Velocity

   We define the 4-velocity to be

   $$u^a(\tau) = \frac{dx^a(\tau)}{d\tau} \quad . \tag{3.2}$$

   This is manifestly a Lorentz vector (along the worldline of the particle),

   $$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad \bar{u}^a = L^a_{\ b} u^b \quad . \tag{3.3}$$

   The proper time τ is related to the coordinate time t in an inertial system by

   $$d\tau = \sqrt{1 - \vec{v}^2/c^2}\, dt \equiv \gamma(v)^{-1} dt \quad \Leftrightarrow \quad \frac{d}{d\tau} = \gamma(v) \frac{d}{dt} \quad . \tag{3.4}$$

   Therefore, the components of u^a in such an inertial system can be written as

   $$(x^a) = (ct, \vec{x}(t)) \quad \Rightarrow \quad (u^a) = (\gamma(v)c, \gamma(v)\vec{v}) \quad . \tag{3.5}$$

   The important thing to note is that the ubiquitous γ-factors in traditional less covariant presentations of the subject arise only if and when one insists on expressing things in terms of the coordinate time in some inertial system. Once one does that, however, it is not at all obvious that a quantity like $(\gamma(v)c, \gamma(v)\vec{v})$, which is non-linear in $\vec{v}$, transforms in a nice way under Lorentz transformations (whereas from the covariant point of view this is completely obvious and by now a triviality).

   Now let us consider the norm $\eta_{ab} u^a u^b$ of the 4-velocity. What is it? By construction this is a Lorentz scalar that has the dimension of (velocity)^2, so one can anticipate that the result is $\sim c^2$ (with a negative constant of proportionality, because u^a is timelike). Indeed, one has precisely

   $$u_a u^a \equiv \eta_{ab} u^a u^b = -c^2 \quad . \tag{3.6}$$

   The uninsightful way to check this is to start from (3.5) and to calculate

   $$\eta_{ab} u^a u^b = -(u^0)^2 + \ldots = -\gamma(v)^2 c^2 + \gamma(v)^2 v^2 = -c^2 \quad . \tag{3.7}$$

While this calculation shows that (3.6) is correct, it sheds no light on why it is correct. The more intelligent and insightful way to derive (3.6) is to note that this is just the definition of proper time,

$$-c^2 d\tau^2 = \eta_{ab} dx^a dx^b \quad \Leftrightarrow \quad \eta_{ab} \frac{dx^a}{d\tau} \frac{dx^b}{d\tau} = -c^2 \ . \tag{3.8}$$

An important consequence of this is that only 3 of the 4 components of u^a are independent. This is as it should be. After all, simply because we choose to describe the motion of a particle in terms of $x^a(\tau)$ rather than $\vec{x}(t)$, we are not introducing new degrees of freedom.

2. 4-Acceleration

Continuing in this spirit, we define the 4-acceleration of a massive particle by

$$a^a(\tau) = \frac{du^a(\tau)}{d\tau} = \frac{d^2 x^a(\tau)}{d\tau^2} \ . \tag{3.9}$$

Again this is manifestly a Lorentz vector along the worldline,

$$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad \bar{a}^a = L^a_{\ b} a^b \ . \tag{3.10}$$

It follows from differentiating $u_a u^a = -c^2$ (3.6) that

$$u_a u^a = -c^2 \quad \Rightarrow \quad u_a a^a = 0 \ . \tag{3.11}$$

In particular, because u^a is timelike, a^a is spacelike.

The components of a^a, when expressed in terms of the coordinates of a particular inertial system, are related in a non-trivial and non-obviuous way to the components of the coordinate acceleration $\vec{b} = d\vec{v}/dt$. For example, for the spatial components one finds, from differentiating $\gamma(v)\vec{v}$, that

$$a^i = \gamma(v)^2 b^i + \gamma(v)^4 \vec{v}.\vec{b}\, v^i \ . \tag{3.12}$$

Note how unpleasant it would be to have to prove directly that these are the spatial components of a Lorentz vector, whereas this fact is built into our formalism.

Armed with this, we now have a plausible candidate for the manifestly Lorentz invariant equation of motion of a free particle, namely

$$a^a(\tau) = \frac{d^2 x^a(\tau)}{d\tau^2} = 0 \ . \tag{3.13}$$

Let us check that in any inertial system this just reduces to the usual statement that the coordinate acceleration is zero,

$$\frac{d^2 x^a(\tau)}{d\tau^2} = 0 \quad \Rightarrow \quad \vec{b} = \frac{d\vec{v}}{dt} = 0 \ . \tag{3.14}$$

To that end, let us write this equation more explicitly as

$$\frac{d}{d\tau}(\gamma(v)c, \gamma(v)\vec{v}) = \gamma(v) \frac{d}{dt}(\gamma(v)c, \gamma(v)\vec{v}) = 0 \ . \tag{3.15}$$

From the time-component we infer that $\gamma(v)$ is constant, and then from the spatial components we indeed infer that $\vec{v}$ is constant.

3.2 Energy-Momentum 4-Vector

A plausible candidate for the definition of a momentum 4-vector, generalising the Newtonian definition $m\vec{v}$, is

$$p^a = mu^a \quad . \tag{3.16}$$

Here m refers to the rest mass of the particle, i.e. the mass in its rest frame (and as such it is tautologically a Lorentz scalar). Thus p^a is again a 4-vector,

$$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad \bar{p}^a = L^a_{\ b} p^b \quad . \tag{3.17}$$

We will confirm this definition in section 3.4 below, where we show that the momentum derived from the Lagrangian is the covector $p_a = mu_a$. Explicitly, its components are

$$(p^a) = (m\gamma(v)c, m\gamma(v)\vec{v}) \equiv (E/c, \vec{p}) \quad . \tag{3.18}$$

Here $\vec{p} = \gamma(v)m\vec{v}$ is the relativistic generalisation of the Newtonian momentum $m\vec{v}$ (to which it reduces for small velocities) and requires no further discussion. The quantity

$$E = cp^0 = m\gamma(v)c^2 \quad . \tag{3.19}$$

is called the relativistic energy

There are various reasons for calling E the energy:

1. First of all, for small v it reduces to

$$E \approx mc^2 + \tfrac{1}{2}mv^2 + \ldots \tag{3.20}$$

   It thus generalises the usual kinetic energy but, famously, also includes the rest energy

$$E_0 = E(v = 0) = mc^2 \quad . \tag{3.21}$$

2. By the equations of motion for a free particle, $\vec{p}$ and E are conserved quantities. The former is just the relativistic generalisation of momentum conservation, and since E is also a conserved quantity one may as well call it the energy. Note by the way that Lorentz invariance and $\vec{p}$-conservation alone already imply that E must also be conserved, because under Lorentz transformations E and $\vec{p}$ transform into each other.

3. Moreover, as we will see in section 3.4, E is really just the Legendre transform of the Lagrangian of a free particle, i.e. the Hamiltonian, $E = H$.

4. A final justification for calling E the energy is that it is (via Noether's theorem) the conserved quantity associated to time-translation invariance (see section 3.6). In fact, the p^α are the conserved quantities associated to spacetime translation invariance.

From the point of view of conserved quantities, a priori it may be debatable whether or not the "constant" $E_0 = mc^2$ should (or has to) be included in the definition of E. In fact, if it were true that the total rest energy $\sum E_0 = \sum mc^2$ (summed over all particles) were always

individually conserved in any multi-particle scattering process (but of course you know that it is not), then one could define $\tilde{E} = E - E_0$, and $\tilde{E}$ would then also be conserved. However, even then this would be an illogical and silly thing to do: E_0 is a scalar, and therefore $(E - E_0)/c$ would neither be a scalar nor the 0-component of any Lorentz 4-vector.

After all it is E and not $\tilde{E}$ that mixes with $\vec{p}$ under Lorentz transformations. In particular, from the transformation under boosts in the x^1-direction (cf. section 2.4) with velocity w^1, say,

$$\bar{p}^1 = \gamma(w)(p^1 - \beta(w)p^0) = \gamma(w)(p^1 - (w^1/c^2)E) \tag{3.22}$$

we see that $E_0 = mc^2$ is the essential part of E to ensure that this reduces to the Galilean transformation of momenta in the non-relativistic limit,

$$w \ll c \quad \Rightarrow \quad \bar{p}^1 = p^1 - w^1 E_0/c^2 = p^1 - mw^1 \ . \tag{3.23}$$

After this excursion, let us return to what we may now confidently call the energy-momentum 4-vector p^a. Since $u_a u^a = -c^2$, we have

$$p_a p^a = \eta_{ab} p^a p^b = -m^2 c^2 \ , \tag{3.24}$$

or

$$(p^0)^2 - \vec{p}^2 = m^2 c^2 \ . \tag{3.25}$$

Thus the momenta of a massive particle lie on a hyperboloid in momentum space, called the *mass shell* by particle physicists.

Plugging in the above components of p^a, one obtains the well-known Pythagorean relation

$$E^2 = m^2 c^4 + \vec{p}^2 c^2 \tag{3.26}$$

among energy, mass and momentum (which we now again understand as a consequence of the definition of proper time τ).

For massless particles, travelling at the speed of light, p^a is of course not timelike but lightlike,

$$p^a p_a = 0 \ , \tag{3.27}$$

and thus their momenta lie on the lightcone in momentum space. Using the usual (de Broglie) relations, the momentum 4-vector is related to the wave 4-vector k^a by

$$p^a = \hbar k^a \ , \tag{3.28}$$

with

$$k_a x^a = -\omega t + \vec{k}.\vec{x} \quad \Rightarrow \quad p^0 = \hbar k^0 = \hbar(\omega/c) = (\hbar\omega)/c = E/c \tag{3.29}$$

(note that here the identification of p^0 with E/c is immediate) and

$$p_a p^a = 0 \quad \Leftrightarrow \quad E = c|\vec{p}| \quad \Leftrightarrow \quad \omega = c|\vec{k}| \ . \tag{3.30}$$

As an application of this, we can rederive and generalise the derivation of the relativistic Doppler effect, discussed in a (1+1)-dimensional setting at the end of section 2.4. Thus, in an inertial system with coordinates (t, x^i), consider a light ray described by the wave vector

$$(k^a) = (\omega/c, k^i) \ , \tag{3.31}$$

with $\omega = c|\vec{k}|$. The frequency observed by an inertial observer in this inertial system, with 4-velocity

$$(u^a) = (c, 0, 0, 0) \tag{3.32}$$

is ω, which can be written as the Lorentz invariant expression

$$\omega = -u_a k^a \ . \tag{3.33}$$

Then, in complete generality, the frequency $\bar{\omega}$ seen by any other observer with 4-velocity $\bar{u}^a$ is the component of k^a along that obervers 4-velocity, namely

$$\bar{\omega} = -\bar{u}_a k^a \ . \tag{3.34}$$

In particular, for a lightray travelling in the x^1-direction and an observer boosted in the x^1-direction,

$$(k^a) = (\omega/c, \omega/c, 0, 0) \quad , \quad \bar{u}^a = (\gamma(v)c, \gamma(v)v, 0, 0) \ , \tag{3.35}$$

one finds

$$\bar{\omega} = -\bar{u}_a k^a = (\gamma(v)c)(\omega/c) - (\gamma(v)v)(\omega/c) = \omega \frac{1 - v/c}{\sqrt{1 - v^2/c^2}} = \omega \sqrt{\frac{1 - v/c}{1 + v/c}} \ , \tag{3.36}$$

in complete agreement with (2.93),

3.3 Minkowski Force? (how not to introduce forces and interactions)

Here is a brief comment on how (not) to include forces or interactions among particles in special relativity. The standard way to do this in Newtonian mechanics is to introduce a force term via Newton's equation

$$m \frac{d^2 \vec{x}}{dt^2} = \vec{F} \ . \tag{3.37}$$

If one naively tries to extend this covariantly, one will be led to something like

$$m \frac{d^2 x^a}{d\tau^2} = K^a \tag{3.38}$$

for some Lorentz vector K^a (known as the Minkowski force vector). However, for a variety of reasons this is not a particularly useful or intelligent way of introducing forces or interactions among particles in the setting of fundamental Lorentz invariant forces and interactions.

First of all, we learn from the fact that the 4-acceleration is orthogonal to the 4-velocity that necessarily

$$m a^a = K^a \quad \Rightarrow \quad K^a u_a = 0 \ . \tag{3.39}$$

Thus the force has to be orthogonal to (and therefore in particular has to depend on) the velocity u^a. This automatically disqualifies all the usual velocity independent phenomenological forces one considers in non-relativistic mechanics (as well as, of course, friction forces proportional to the velocity).

It should come as no surprise, however, that there is one potential exception to this, namely the Lorentz force

$$\vec{F} = e(\vec{E} + \vec{v} \times \vec{B}) \tag{3.40}$$

of Maxwell theory (our prime candidate for a Lorentz invariant field theory), describing the force acting on a charged massive particle in an electromagnetic field. We will verify later on, that the Lorentz force can indeed be described in terms of a Minkowski force K^a (cf. section 4.11).

However, more importantly this example teaches us how to introduce Lorentz invariant interactions among and forces on particles: such forces require a mediator, a *field*, and therefore K^a should not be introduced phenomenologically, but should rather be deduced from the (Lorentz invariant) coupling of relativistic particles to a (Lorentz invariant) field theory. All this is best done not at the level of equations of motion, but at the level of actions or Lagrangians. Again we will see later on (section 4.12) how to accomplish this in the case of Maxwell theory.

3.4 Lorentz-invariant Action Principle for a Free Relativistic Particle

We now want to construct an action principle for a relativistic particle, from which its equation of motion follows as the Euler-Lagrange equation. The general strategy in setting up an action principle is to

- define or identify the space of dynamical variables (fields)
- specify the symmetries one wants the theory to have
- and to then construct the simplest (or general) local functional of the fields and their derivatives that has these symmetries.

Here "local" means that the functional is given by the integral of a Lagrangian. Moreover, if one wants the resulting Euler-Lagrange equations to be at most 2nd order differential equations, then the choice of functional is further restricted by the requirement that it should be at most linear in 2nd derivatives and/or quadratic in 1st derivatives of the fields.

In the case at hand, the dynamical fields are the trajectories / worldlines $x^a(\tau)$, and the symmetry that we want to require is Poincaré invariance. Our ansatz for a local action is thus (cf. section 2.10)

$$S[x] = \int d\tau L(x^a, \dot{x}^a) \tag{3.41}$$

where $\dot{x}^a = u^a = dx^a/d\tau$. This is Lorentz invariant if L is a scalar under Lorentz transformation, and it is moreover translation (and thus Poincaré) invariant if L does not depend explicitly on the x^a, $L = L(\dot{x}^a)$. At this point we have reduced the task to that of constructing a scalar from $\dot{x}^a = u^a$, but this seems to leave no interesting possibilities since we already know that the obvious candidate is just

$$\eta_{ab}\dot{x}^a\dot{x}^b = -c^2 \quad . \tag{3.42}$$

However, looking more closely at the integration measure $d\tau$, we realise that this already depends quadratically on the differentials dx^a: after all,

$$d\tau = d\tau(x) = \sqrt{-c^{-2}\eta_{ab}dx^a dx^b} \ . \tag{3.43}$$

Therefore we get a candidate action by simply choosing L to be some constant. Then, up to this constant, the action $S[x]$ would just be the total proper time between the endpoints of the world line, and solutions to the resulting Euler-Lagrange equations would be those world lines that extremise the proper time. This is in complete agreement with the observation made back in section 2.5 that the proper time is maximal for inertial observers.

Thus our refined ansatz for the action is

$$S[x] = \alpha \int d\tau(x) \tag{3.44}$$

for some constant α. The resulting Euler-Lagrange equations will of course be independent of α, but we may as well choose this α in a nice and convenient way. First of all, in order for this action to have the dimension of an action (energy $\times$ time), α should have the dimensions of an energy, and for a particle with rest mass m we can set $\alpha \sim mc^2$. Comparison with the non-relativistic limit will then fix the proportionality factor (to be (-1)), and anticipating this we write the action as

$$S[x] = -mc^2 \int d\tau(x) \ . \tag{3.45}$$

Our main task will be to show (confirm) that this action is indeed extremised by solutions to the equations of motion for a free particle,

$$\delta S[x] = 0 \quad \forall\, \delta x^a \quad \Rightarrow \quad \ddot{x}^a(\tau) = 0 \ . \tag{3.46}$$

The variation here refers to variations of the path

$$x^a(\tau) \to x^a(\tau) + \delta x^a(\tau) \ . \tag{3.47}$$

Since under this variation the velocities vary as

$$\dot{x}^a(\tau) \to \dot{x}^a(\tau) + \frac{d}{d\tau}\delta x^a(\tau) \ , \tag{3.48}$$

one sees that the variation of the velocities is simply

$$\delta\dot{x}^a(\tau) \equiv \delta\left(\frac{d}{d\tau}x^a(\tau)\right) = \frac{d}{d\tau}\delta x^a(\tau) \ , \tag{3.49}$$

i.e. "δ and $d/d\tau$ commute". This is the defining and characteristic property of what one means by variations.

Moreover, as usual in variational calculus, one should also fix the integration domain and restrict the variations to those vanishing on the boundary of this domain (i.e. in the case at hand: one fixes the endpoints of the path). Therefore, let us state once and for all (without indicating this explicitly in the equations) that we are considering paths between an initial event with coordinates x_i^a and a final event with coordinates x_f^a, and therefore with variations that vanish at these endpoints,

$$x^a(\tau_{i,f}) = x^a_{i,f} \quad \Rightarrow \quad \delta x^a(\tau_{i,f}) = 0 \ . \tag{3.50}$$

Before embarking on the calculation, it will be extremely convenient to make the dependence of the Lagrangian on the velocities more explicit. For that we temporarily introduce an arbitrary new parameter $\lambda = \lambda(\tau)$ (which is then also Lorentz invariant) with

$$d\tau = \frac{d\tau}{d\lambda} d\lambda \tag{3.51}$$

and with $d\tau/d\lambda > 0$ (so that the transformation $\tau \to \lambda$ is invertible). We can thus consider the paths as functions of λ, $x^a = x^a(\lambda)$, and we have the corresponding velocities

$$x'^\alpha(\lambda) = \frac{d}{d\lambda} x^a(\lambda) \ . \tag{3.52}$$

Then

$$c d\tau = \sqrt{-\eta_{ab} x'^a x'^b}\, d\lambda \ , \tag{3.53}$$

and in terms of these quantities, the dependence of the action and its Lagrangian on the velocities x'^a is now much more manifest and transparent. The action is

$$S[x] = -mc^2 \int d\lambda \sqrt{-c^{-2} \eta_{ab} x'^a x'^b} \equiv \int d\lambda \; L_\lambda(x'^a) \ , \tag{3.54}$$

and thus for any choice of λ one has the simple and explicit Lagrangian

$$L_\lambda(x'^a) = -mc^2 \frac{d\tau}{d\lambda} = -mc(-\eta_{ab} x'^a x'^b)^{1/2} \ . \tag{3.55}$$

In order to obtain the equations of motion, one can either use the Euler-Lagrange equations

$$\frac{d}{d\lambda} \frac{\partial L_\lambda}{\partial x'^a} = \frac{\partial L_\lambda}{\partial x^a} \tag{3.56}$$

(see below), or one can directly vary the action. Let us do the latter:

- The first step is

$$\begin{aligned} \delta S[x] &= -mc \int d\lambda \; \tfrac{1}{2}(-\eta_{cd} x'^c x'^d)^{-1/2} (-2\eta_{ab} x'^a \delta x'^b) \\ &= mc \int d\lambda \; (-\eta_{cd} x'^c x'^d)^{-1/2} (\eta_{ab} x'^a \frac{d}{d\lambda} \delta x^b) \\ &= mc \int d\lambda \; (\frac{1}{c} \frac{d\lambda}{d\tau}) \eta_{ab} \frac{dx^a}{d\lambda} \frac{d}{d\lambda} \delta x^b \end{aligned} \tag{3.57}$$

  where we have used

$$\delta(\eta_{ab} x'^a x'^b) = \eta_{ab} \delta x'^a x'^b + \eta_{ab} x'^a \delta x'^b = 2\eta_{ab} x'^a \delta x'^b \tag{3.58}$$

  and (3.53).

- Writing $d\lambda = (d\lambda/d\tau) d\tau$ and and then switching back from λ- to τ-derivatives everywhere, one finds

$$\delta S[x] = m \int d\tau \; (\frac{d\lambda}{d\tau})^2 \eta_{ab} \frac{dx^a}{d\lambda} \frac{d}{d\lambda} \delta x^b = m \int d\tau \; \eta_{ab} \frac{dx^a}{d\tau} \frac{d}{d\tau} \delta x^b \ . \tag{3.59}$$

- Now we can integrate by parts, and drop the boundary term (because $\delta x^b = 0$ there),

$$\delta S[x] = -m \int d\tau \; \eta_{ab} \frac{d^2 x^a}{d\tau^2} \delta x^b \quad . \tag{3.60}$$

- This finally implies

$$\delta S[x] = 0 \quad \forall \, \delta x \quad \Leftrightarrow \quad \eta_{ab} \frac{d^2 x^a}{d\tau^2} = 0 \quad \Leftrightarrow \quad \frac{d^2 x^a}{d\tau^2} = 0 \quad . \tag{3.61}$$

as was to be shown.

Remarks:

1. In order to derive this result (perhaps more directly) from the Euler-Lagrange equations, the first thing one needs to calculate are the momenta $\partial L_\lambda / \partial x'^a$. Explicit calculation shows that these agree precisely with the covariant 4-momenta $p_a = m u_a$ already introduced in section 3.2, i.e.

$$\frac{\partial L_\lambda}{\partial x'^a} = p_a = m \eta_{ab} \frac{dx^b}{d\tau} \quad , \tag{3.62}$$

independently of the choice of λ. Since L_λ does not depend explicitly on the x^a, the Euler-Lagrange equations then reduce to

$$\frac{d}{d\lambda} \frac{\partial L_\lambda}{\partial x'^a} = \frac{d}{d\lambda} p_a = 0 \quad \Leftrightarrow \quad \frac{d}{d\tau} p_a = 0 \quad \Leftrightarrow \quad \ddot{x}^a = 0 \quad . \tag{3.63}$$

2. In an inertial coordinate system with coordinates (t, x^i), a natural choice for λ is the coordinate time $\lambda = t$. With this choice, the Lagrangian takes the simple and explicit form

$$L_t = -mc^2 \sqrt{1 - \vec{v}^2/c^2} \quad . \tag{3.64}$$

There are at least two fun things one can do with or learn from this Lagrangian:

   (a) In the non-relativistic (better: Galilean relativistic) limit $v \ll c$, L_t reduces to

$$L_t = -mc^2 + \tfrac{1}{2} m \vec{v}^2 + \dots \quad . \tag{3.65}$$

   Thus (up to the constant mc^2) one recovers the well-known non-relativistic Lagrangian, namely the kinetic energy. It is pleasing to see this arise from the proper time of the relativistic particle.

   (b) Given the standard Lagrangian L_t and action $S[x] = \int dt \; L_t$, one can in the usual way define the canonical momenta $p_i^{(c)}$

$$p_i^{(c)} = \frac{\partial L_t}{\partial v^i} \tag{3.66}$$

   and then the Hamiltonian H via the Legendre transform,

$$H = p_i^{(c)} v^i - L_t \quad . \tag{3.67}$$

For the former one finds

$$p_i^{(c)} = m\gamma(v)v_i = p_i \tag{3.68}$$

(which should not come as a surprise in view of (3.62)), and for the Hamiltonian one then finds

$$H = m\gamma(v)c^2 = E \ , \tag{3.69}$$

precisely the quantity we called the relativistic energy before (and this provides one rationale for referring to E as the energy).

3. A caveat may be in order here. When we first introduced the parameter $\lambda = \lambda(\tau)$, then we were really just thinking of this as a reparametrisation of the worldline, and if λ is really just a function of τ and nothing else, then λ is of course also a Lorentz scalar. In particular, in that case not just

$$-mc^2 d\tau = L_\lambda d\lambda \tag{3.70}$$

is Lorentz invariant, but L_λ itself is Lorentz invariant. We can also choose (as we did just above) $\lambda = t$, and even though along a given path we can relate t to τ, by solving

$$d\tau = \sqrt{1 - \vec{v}^2/c^2}dt \quad \Rightarrow \quad \tau = \tau(t, \vec{x}(t)) \ , \tag{3.71}$$

and then inverting this to obtain t as a function of τ, this relation is path dependent. While we can do this along a given path, of course we know that t as such is not Lorentz invariant, and therefore neither is L_t. Rather, the Lagrangians associated to t and $\bar{t}$ (coordinate time in some other inertial system) are related by

$$-mc^2 d\tau = L_t dt = L_{\bar{t}} d\bar{t} \ . \tag{3.72}$$

4. The equation $p_i v^i - L_t = E$ obtained above can be rewritten as

$$p_i v^i - L_t = E \quad \Leftrightarrow \quad p_0 \frac{dx^0}{dt} + p_i \frac{dx^i}{dt} - L_t = p_a \frac{dx^a}{dt} - L_t = 0 \ . \tag{3.73}$$

This equation is true not just for $\lambda = t$ but for any λ, i.e. the covariant Hamiltonian or Legendre transform $\mathcal{H}_\lambda$ of the Lagrangian L_λ is equal to zero,

$$\mathcal{H}_\lambda = p_a x'^a - L_\lambda = 0 \ . \tag{3.74}$$

This reflects the fact that the 4 components of the momenta are not independent, since $p_a p^a = -m^2 c^2$,

$$p_a x'^a - L_\lambda = \frac{1}{m}\frac{d\tau}{d\lambda} p_a p^a + mc^2 \frac{d\tau}{d\lambda} = \frac{1}{m}\frac{d\tau}{d\lambda}(p_a p^a + m^2 c^2) = 0 \ . \tag{3.75}$$

Ultimately the vanishing of $\mathcal{H}_\lambda$ is due to the reparametrisation invariance of the action, expressed as

$$d\tau = (d\tau/d\lambda)d\lambda = (d\tau/d\sigma)d\sigma \tag{3.76}$$

(but it would lead too far to explain this last assertion here).

3.5 Noether Theorem and Conservation Laws (Review)

Let us quickly recall and rederive Noether's (first) theorem for classical mechanics. Here, in order to hopefully make this look more familiar, we use the notation commonly used in that context, i.e. q^a are (generalised) coordinates on some configuration space Q, and the dynamical variables are paths $q^a = q^a(t)$ on Q. In applications to relativistic mechanics (in section 3.6 below), all we then have to do is replace $q^a(t) \to x^a(\lambda)$.

Now, given any function of $q^a(t)$ and $\dot{q}^a(t)$, and perhaps other variables, e.g. a Lagrangian $\mathcal{L}(q, \dot{q}, t)$, its variation under variations of the path

$$q^a(t) \to q^a(t) + \delta q^a(t) \tag{3.77}$$

is

$$\delta\mathcal{L}(q^a(t), \dot{q}^a(t), t) = \frac{\partial\mathcal{L}}{\partial q^a(t)}\delta q^a(t) + \frac{\partial\mathcal{L}}{\partial\dot{q}^a(t)}\delta\dot{q}^a(t) \ . \tag{3.78}$$

Using the defining and characteristic property of variations, namely

$$\delta\dot{q}^a(t) \equiv \delta(\tfrac{d}{dt}q^a(t)) = \tfrac{d}{dt}\delta q^a(t) \ , \tag{3.79}$$

this can be written as

$$\delta\mathcal{L}(q^a(t), \dot{q}^a(t), t) = \left(\frac{\partial\mathcal{L}}{\partial q^a(t)} - \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}^a(t)}\right)\delta q^a(t) + \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}^a(t)}\delta q^a(t)\right) \ . \tag{3.80}$$

This is what I will refer to as the *Variational Master Equation* (VME).

What makes this equation so useful is that it relates 3 apparently quite different objects. On the left-hand side, one has a variation, the Euler-Lagrange equations appear in the 1st term on the right-hand side, and the 2nd term on the right-hand side is a total time-derivative, so that structurally the equation looks like

$$\text{Variation} = \text{Euler-Lagrange Equations} + \text{Total Time-Derivative} \ . \tag{3.81}$$

Thus, if we can eliminate or constrain one of the terms in these equations, then we obtain a potentially non-trivial and interesting relation between the other two. This can be achieved by selecting appropriate variations or classes of variations and/or by integrating the VME.

Concretely,

1. by integrating and choosing the variations to preserve the end-points of the path, one eliminates the 2nd term on the right-hand side and obtains a 1-line proof of Hamilton's principle that Lagrangian dynamics is such that the path is a stationary point of the action;

2. by choosing special variations $\delta_s q^a$ that leave the Lagrangian invariant ($\delta_s\mathcal{L} = 0$, infinitesimal symmetries) or invariant up to a total time-derivative, one constrains the left-hand side and obtains a 1-line proof of Noether's (first) theorem;

3. by restricting to solutions of the Euler-Lagrange equations and variations among them one eliminates the 1st term on the right-hand side and obtains a simple proof of the Hamilton-Jacobi relations which relate the time- and space-derivatives of the "classical" action to energy and momentum respectively.

Our interest here will be in the second option, but just for completeness here is the argument for the first item: we integrate the VME over a time interval $I = [t_1, t_2]$ and consider only variations that vanish at the end points, $\delta q^a(t_1) = \delta q^a(t_2) = 0$. Then from the left-hand side of (3.80) we obtain the variation of the action, and therefore

$$\delta S[q] \equiv \delta \int_I dt\ \mathcal{L} = \int_I dt \left(\frac{\partial \mathcal{L}}{\partial q^a(t)} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \right) \delta q^a(t) + \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \delta q^a(t) \right) |_{t_1}^{t_2} \tag{3.82}$$

Since the boundary term vanishes, we obtain the result that the action is extremised by solutions to the Euler-Lagrange equations,

$$\delta S[q] = 0 \quad \forall\ \delta q^a \quad \Leftrightarrow \quad \frac{\partial \mathcal{L}}{\partial q^a(t)} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} = 0 \ . \tag{3.83}$$

Now we turn to the second option mentioned above. Thus, let $q^a \to q^a + \delta_s q^a$ be a variation that leaves the Lagrangian invariant for all paths $q^a(t)$,

$$\delta_s \mathcal{L}(q, \dot{q}, t) = \frac{\partial \mathcal{L}}{\partial q^a(t)} \delta_s q^a(t) + \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \delta_s \dot{q}^a(t) = 0 \ . \tag{3.84}$$

We will refer to such a transformation as an infinitesimal symmetry of the Lagrangian. Then there is a corresponding conserved quantity, namely

$$P_\delta = \frac{\partial \mathcal{L}}{\partial \dot{q}^a} \delta_s q^a = p_a \delta_s q^a \ , \tag{3.85}$$

i.e. P_δ is constant along any solution to the Euler-Lagrange equations.

Proof: $\delta_s \mathcal{L} = 0$ implies

$$0 = \left(\frac{\partial \mathcal{L}}{\partial q^a(t)} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \right) \delta_s q^a(t) + \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \delta_s q^a(t) \right) \ . \tag{3.86}$$

Thus

$$\frac{\partial \mathcal{L}}{\partial q^a(t)} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} = 0 \quad \Rightarrow \quad \frac{d}{dt} P_\delta = 0 \ . \tag{3.87}$$

In particular, if q^1, say, is a *cyclic* variable, i.e. if $\mathcal{L}$ does not depend explicitly on q^1, then the Lagrangian is invariant under (infinitesimal) translations of q^1, and this leads to momentum conservation,

$$\delta_s q^1 = \epsilon^1 \quad \Rightarrow \quad P_\delta = \epsilon^1 p_1 \quad \Rightarrow \quad \frac{d}{dt} p_1 = 0 \ . \tag{3.88}$$

Here is a minor (and obvious but useful) variant and generalisation of the above:

Let $q^a \to q^a + \delta_s q^a$ be a variation that leaves the Lagrangian *quasi-invariant* for all paths $q^a(t)$, i.e. invariant up to a total time-derivative,

$$\delta_s \mathcal{L}(q, \dot{q}, t) = \frac{\partial \mathcal{L}}{\partial q^a(t)} \delta_s q^a(t) + \frac{\partial \mathcal{L}}{\partial \dot{q}^a(t)} \delta_s \dot{q}^a(t) = \frac{d}{dt} \mathcal{F}_\delta(q, t) \tag{3.89}$$

(quasi-symmetry or also simply just symmetry of the Lagrangian). Then (by the same reasoning as above and by simply replacing 0 by $(d/dt)\mathcal{F}_\delta$ on the left-hand side of (3.86)) there is a corresponding conserved quantity, namely

$$P_\delta = p_a \delta_s q^a - \mathcal{F}_\delta \ . \tag{3.90}$$

Quasi-invariance arises e.g. when one considers the transformation of the free particle Lagrangian under Galilean boost transformations. Indeed, with $q^a \to x^i$ and $\dot{q}^a \to v^i = dx^i/dt$, the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2} m \vec{v}^2 = \tfrac{1}{2} m \delta_{ij} v^i v^j \tag{3.91}$$

Galilean boosts act as $\bar{x}^i = x^i - w^i t$, infinitesimally

$$\delta_s x^i = -\omega^i t \quad \Rightarrow \quad \delta_s v^i = -\omega^i \ . \tag{3.92}$$

Under these transformations, the Lagrangian is evidently not strictly invariant, but its variation is a total time derivative,

$$\delta_s \mathcal{L} = -m \delta_{ij} v^i \omega^j \equiv -m \omega_i v^i = \frac{d}{dt}(-m \omega_i x^i) \ . \tag{3.93}$$

Thus the associated conserved quantity is

$$P_\delta = p_i \delta_s x^i + m \omega_i x^i = (m x^i - p^i t)\omega_i \equiv G^i \omega_i \ . \tag{3.94}$$

As we will see below, the situation is much simpler for the relativistic particle, as the Lagrangian is strictly invariant under all Poincaré transformations, and thus one does not ever need to invoke this quasi-invariance variant of Noether's theorem.

Remarks:

1. Note that in the above we considered only variations of the paths $q^a(t)$, not variations of the independent variable t. This does not mean that we cannot deal with symmetries associated with transformations of t. What it means is that we should reinterpret them as transformations acting on the $q^a(t)$ alone. This avoids many completely unnecessary complications and pitfalls that invariably arise when one tries to formulate the Noether theorem directly for symmetries that involve explicit transformations of t. It is therefore surprising, that most textbook treatments of Noether's theorem actually take this latter, more complicated, approach.

2. In this more traditional approach, one considers infinitesimal transformations

$$\bar{t} = t + \epsilon X(q^a, t) \quad , \quad \bar{q}^a = q^a + \epsilon Y^a(q^a, t) \ . \tag{3.95}$$

which are such that under the substitution

$$t \to \bar{t} \quad , \quad q^a(t) \to \bar{q}^a(\bar{t}) \tag{3.96}$$

the action $\int dt L$ is invariant to order ϵ (and up to boundary terms). However, one can think of this combined transformation as defining a true variation (only $q^a(t)$ is varied, not t) via (retaining only the linear term in ϵ)

$$\delta q^a(t) = \bar{q}^a(t) - q^a(t) = \epsilon \left(Y^a(q(t), t) - X(q(t), t) \dot{q}^a(t) \right) \ . \tag{3.97}$$

Then the above invariance condition for the *action* (including the transformation of the integration measure dt) under (3.95) is completely equivalent to quasi-invariance of the *Lagrangian* under this variation (3.97). It is much more convenient to phrase the Noether theorem in these terms (but these notes are not the place to do this in general). See also sections 6.3 and 7.3 for some further explanations and illustrations of this in the field theory context.

3. In particular, we can think of infinitesimal time-translations $t \to \bar{t} = t + \epsilon$ alternatively as defining new (translated) paths $\bar{q}^a(t)$ by

$$\bar{q}^a(t) = q^a(t - \epsilon) \ . \tag{3.98}$$

Taylor expanding this, we have

$$\bar{q}^a(t) = q^a(t) - \epsilon \dot{q}^a(t) + \dots \tag{3.99}$$

The difference between the left-hand side and the first term on the right-hand side is now an infinitesimal difference between two different paths at the same point, and therefore this defines a variation. We are free to define the variation with either sign. For consistency with what we will do in the case of field theories, where it appears to be more natural to keep the minus sign, we thus define

$$\delta q^a(t) = -\epsilon \dot{q}^a(t) \quad , \quad \delta \dot{q}^a(t) = -\epsilon \ddot{q}^a(t) \ . \tag{3.100}$$

This is just a special case of the variation (3.97) introduced above, with $X = 1, Y^a = 0$. Acting with this variation on the Lagrangian, one finds

$$\delta \mathcal{L} = -\frac{d}{dt}(\epsilon \mathcal{L}) + \frac{\partial}{\partial t}(\epsilon \mathcal{L}) \ , \tag{3.101}$$

so the Lagrangian is quasi-invariant if $\mathcal{L}$ does not depend explicitly on t, and we are now entitled to call this variation a quasi-symmetry $\delta_s q^a(t)$. The corresponding conserved quantity is then essentially the Hamiltonian function (energy) $\mathcal{H}$,

$$p_a \delta_s q^a - \mathcal{F}_\delta = -\epsilon(p_a \dot{q}^a - \mathcal{L}) = -\epsilon \mathcal{H} \ . \tag{3.102}$$

3.6 Noether Theorem for the Relativistic Particle

With $q^a(t) \to x^a(\lambda)$, we can now specialise and apply this to the Lagrangian

$$L_\lambda = -mc\sqrt{-\eta_{ab} x'^a x'^b} \ . \tag{3.103}$$

To simplify the discussion, we will choose $\lambda = \lambda(\tau)$ to be a Lorentz scalar, and we will of course assume that the map $\tau \to \lambda$ is invertible, $d\lambda/d\tau \neq 0$. Dealing with situations like $\lambda = t$ (which is of course not a Lorentz scalar) is possible but requires a bit more thought - I will come back to this at the end of this section. Then this Lagrangian is, by construction, manifestly and strictly invariant under Poincaré transformations, i.e. Lorentz transformations and translations.

Since L_λ depends only on x'^a, for any variation δx^a we have

$$\delta L_\lambda = \frac{\partial L_\lambda}{\partial x'^a}\delta x'^a = p_a \delta x'^a \quad . \tag{3.104}$$

Explicitly, for infinitesimal translations

$$\delta_s x^a = \epsilon^a \tag{3.105}$$

we therefore have

$$\delta_s x'^a = 0 \quad \Rightarrow \quad \delta_s L_\lambda = p_a \delta_s x'^a = 0 \ . \tag{3.106}$$

And for infinitesimal Lorentz transformations (2.65)

$$\delta_s x^a = \omega^a{}_b x^b \quad \text{with} \quad \omega_{ab} \equiv \eta_{ac}\omega^c{}_b = -\omega_{ba} \tag{3.107}$$

we have

$$\delta_s x'^a = \omega^a{}_b x'^b = (d\tau/d\lambda)\omega^a{}_b p^b/m \tag{3.108}$$

and therefore

$$\delta_s L_\lambda = (d\tau/d\lambda)p_a\omega^a{}_b p^b = (d\tau/d\lambda)\omega_{ab}p^a p^b/m = 0 \tag{3.109}$$

by anti-symmetry of ω_{ab}.

Thus the conserved quantities (Noether charges) associated to spacetime translations are just the momenta p^a,

$$P_\delta = p_a \delta_s x^a = \epsilon^a p_a \quad \Rightarrow \quad p^a \quad \text{conserved} \ , \tag{3.110}$$

and those associated to Lorentz transformations are the components of an anti-symmetric tensor $\mathcal{L}^{ab}$,

$$P_\delta = p_a \delta_s x^a = \omega_{ab}p^a x^b = \tfrac{1}{2}\omega_{ab}(p^a x^b - p^b x^a) \quad \Rightarrow \quad \mathcal{L}^{ab} \equiv p^a x^b - p^b x^a \quad \text{conserved} \ . \tag{3.111}$$

Remarks:

1. To recall, the statement that a quantity C is "conserved" means that

$$\frac{d}{d\lambda}C = 0 \qquad \text{for a solution to the equations of motion} \ . \tag{3.112}$$

   In the case at hand, and by invertiblity of the relation between λ and τ, concretely we have the (rather trivial) assertions

$$\frac{d^2x^a}{d\tau^2} = 0 \quad \Rightarrow \quad \frac{d}{d\tau}p^a = 0 \quad , \quad \frac{d}{d\tau}(p^a x^b - p^b x^a) = 0 \ . \tag{3.113}$$

2. In particular, we see that p^0 is the conserved quantity associated to invariance under time translations, providing yet another rationale for identifying $E = cp^0$ with the energy.

3. Since p^a is a 4-vector, energy $E = cp^0$ and the spatial components p^i of the momentum mix (transform into each other) Lorentz transformations. As a consequence, conservation of the spatial components p^i of the momentum in every inertial system (equivalently conservation of the spatial components p^i of the momentum and Lorentz invariance) implies energy conservation, since

$$\bar{p}^i = L^i{}_b p^b = L^i{}_k p^k + L^i{}_0 p^0 \ . \tag{3.114}$$

4. Since $\mathcal{L}^{ab} = -\mathcal{L}^{ba}$, the six independent components are $\mathcal{L}^{ik} = -\mathcal{L}^{ki}$ and $\mathcal{L}^{0k}$. The $\mathcal{L}^{ik}$ are evidently just the three components of the angular momentum $\vec{L} = \vec{x} \times \vec{p}$, the familiar conserved quantities associated to spatial rotations,

$$\mathcal{L}^{ik} = p^i x^k - p^k x^i \quad \Leftrightarrow \quad \vec{L} = \vec{x} \times \vec{p} \ . \tag{3.115}$$

We see from this that a three-component vector can be promoted to a Lorentz tensor in different ways: the momenta are the spatial components of the momentum 4-vector p^a, while the angular momenta are (half of the) components of an anti-symmetric tensor $\mathcal{L}^{ab}$.

5. For a single particle, the conserved quantity

$$\mathcal{L}^{0k} = p^0 x^k - p^k x^0 \tag{3.116}$$

(note the similarity to (3.94)) associated to boosts is rather tautological and boring. Indeed, plugging in the solution to the equations of motion for $x^k = x^k(t)$, say, namely

$$x^k(t) = x^k(0) + tv^k(0) = x^k(0) + tp^k/m\gamma(v) \tag{3.117}$$

with $p^k = p^k(0)$, one finds

$$\mathcal{L}^{0k} = (E/c)(x^k(0) + tp^k/m\gamma(v)) - p^k ct = (E/c)x^k(0) \ , \tag{3.118}$$

so that the conserved quantity is esentially the initial position of the particle. For a multi-particle system the conservation of $\mathcal{L}^{0k}$ expresses the "center of energy" theorem, that the center of energy (rather than mass in the Newtonian theory) moves with constant velocity.

6. Under Lorentz transformations $\mathcal{L}^{ik}$ and $\mathcal{L}^{0k}$ will mix (transform into each other),

$$\bar{\mathcal{L}}^{ab} = L^a_{\ c} L^b_{\ d} \mathcal{L}^{cd} \ , \tag{3.119}$$

just as in Newtonian nechanics applying a Galilean boost to angular momentum one generates a term involving the conserved quantity $\vec{G}$ (3.94) associated to Galilean boosts,

$$\vec{x} \to \vec{x} - \vec{w}t \quad \Rightarrow \quad \vec{L} = \vec{x} \times \vec{p} \to \vec{L} + \vec{w} \times (m\vec{x} - \vec{p}t) = \vec{L} + \vec{w} \times \vec{G} \ . \tag{3.120}$$

Therefore, in all cases (single or multi particle, Galilean or Lorentzian boosts), the associated conserved quantity can also be thought of not as a new and independent conserved quantity, but as a quantity whose conservation is implied by conservation of angular momentum in every inertial system. Depending on the context this may or may not be the most useful perspective.

In the above discussion, we used the Lagrangian L_λ based on some parameter $\lambda = \lambda(\tau)$ that was simply some function of the proper time τ. Then the Lorentz invariant action $S \sim \int d\tau$ led to the strictly Lorentz invariant Lagrangian L_λ. However, in section 3.4 we saw that, given an inertial system with coordinates (t, x^i), it can also be convenient to parametrise the paths in the traditional way by $x^i = x^i(t)$, leading to the Lagrangian L_t defined by (3.64)

$$S[x] = -mc^c \int d\tau(x) = \int dt \ L_t \quad \Rightarrow \quad L_t = -mc^2 \sqrt{1 - \vec{v}^2/c^2} \ , \tag{3.121}$$

where $v^i = dx^i(t)/dt$. Evidently one has (3.72)

$$-mc^2 d\tau = L_t dt = L_{\bar{t}} d\bar{t} \ . \tag{3.122}$$

but the Lagrangian itself is *not* Lorentz invariant, since t is not Lorentz invariant. Rather, the infinitesimal transformation that leaves the action invariant,

$$x^a \to \bar{x}^a = x^a + \omega^a{}_b x^b \quad \Rightarrow \quad \left\{ \begin{array}{rl} t \to & \bar{t} = t + \omega^0{}_b x^b/c = t + \omega^0{}_k x^k/c \\ x^i \to & \bar{x}^i = x^i + \omega^i{}_b x^b \end{array} \right. \tag{3.123}$$

is a transformation of the type (3.95), which translates into a true variation (3.97)

$$\delta x^i = \omega^i{}_b x^b - v^i(\omega^0{}_k x^k/c) \ . \tag{3.124}$$

It is a fun exercise to show that indeed L_t is quasi-invariant under this variation, and that this leads to the same conserved quantities as in the manifestly Lorentz-invariant case $\lambda = \lambda(\tau)$ discussed above.

4 Lorentz-Covariant Formulation of Maxwell Theory

4.1 Maxwell Equations (Review)

In the traditional (non-covariant, 3-vector calculus) formulation, the Maxwell equations are the

1. Homogeneous Equations

$$\begin{aligned} \vec{\nabla}.\vec{B} &= 0 \\ \vec{\nabla}\times\vec{E} + \partial_t\vec{B} &= 0 \end{aligned} \tag{4.1}$$

2. Inhomogeneous Equations

$$\begin{aligned} \vec{\nabla}.\vec{E} &= \rho/\epsilon_0 \\ \vec{\nabla}\times\vec{B} - \frac{1}{c^2}\partial_t\vec{E} &= \mu_0\vec{J} \end{aligned} \tag{4.2}$$

Here $\vec{E}$ and $\vec{B}$ are the electric and magnetic fields, and the sources of these fields are the electric charge density ρ and the current density $\vec{J}$. ϵ_0 and μ_0 are constants (whose names, let alone their values, I can never remember) which are related to the velocity of light by

$$\epsilon_0\mu_0 = c^{-2} \quad . \tag{4.3}$$

The inhomogeneous equations imply the

3. Continuity Equation

$$\partial_t\rho + \vec{\nabla}.\vec{J} = 0 \quad . \tag{4.4}$$

In the absence of sources, the homogeneous and inhomogeneous equations together imply the

4. Wave Equations for the Electric and Magnetic Fields

$$\rho = \vec{J} = 0 \quad \Rightarrow \quad \Box\vec{E} = 0 \quad , \quad \Box\vec{B} = 0 \quad . \tag{4.5}$$

In order to (locally) solve the homogeneous equations, and also for other purposes and reasons, it is useful to introduce the

5. Electric Potential ϕ and Magnetic Potential $\vec{A}$

$$\begin{aligned} \vec{B} = \vec{\nabla}\times\vec{A} \quad &\Rightarrow \quad \vec{\nabla}.\vec{B} = 0 \\ \vec{E} = -\vec{\nabla}\phi - \partial_t\vec{A} \quad &\Rightarrow \quad \vec{\nabla}\times\vec{E} + \partial_t\vec{B} = 0 \end{aligned} \tag{4.6}$$

Introduction of these potentials gives rise to the

6. Gauge Transformations / Gauge Invariance

$$\phi \to \phi - \partial_t\Psi \quad , \quad \vec{A} \to \vec{A} + \vec{\nabla}\Psi \quad \Rightarrow \quad \vec{E} \to \vec{E} \quad , \quad \vec{B} \to \vec{B} \quad . \tag{4.7}$$

Finally, in terms of the potentials, the (remaining) inhomogeneous equations are the

7. Equations of Motion for the Potentials

$$\begin{aligned} &\Box\vec{A} - \vec{\nabla}G = -\mu_0\vec{J} \\ &\Box(-\phi/c) - \frac{1}{c}\partial_t G = \mu_0\rho c \end{aligned} \tag{4.8}$$

with

$$G = \vec{\nabla}.\vec{A} + \frac{1}{c}\partial_t(\phi/c) \ . \tag{4.9}$$

This is all we will need.

4.2 Lorentz Invariance of the Maxwell Equations: Preliminary Remarks

At first sight, the presumed Lorentz invariance of the Maxwell equations, as presented above, and the possible Lorentz-tensorial structure of their building blocks are totally obscure. What we have are various 3-vectors (i.e. vectors under spatial rotations), such as $\vec{E}$ and $\vec{J}$, 3-vectorial differential operators like $\vec{\nabla}$, and 3-scalars (i.e. scalars under spatial rotations) like ϕ. So where do the Lorentz tensors hide?

The issue is particularly puzzling for the electric and magnetic fields $\vec{E}$ and $\vec{B}$: while the electromagnetic field of a charge at rest is purely electric, that of a charge moving with a constant velocity contains both electric and magnetic fields. This means that the decomposition of an electromagnetic field into electric and magnetic fields depends on the inertial system and that under Lorentz boosts electric and magnetic fields will "mix", i.e. transform into each other. How can one combine the 3 components of $\vec{E}$ and the 3 components of $\vec{B}$ into a Lorentz tensor?

However, looking a bit closer at these equations, one finds some suggestive and intriguing hints that these equations really want to be written in a much nicer four-dimensional Lorentz covariant way:

1. Our first clue comes from the continuity equation (4.4). We had already seen in section 2.11, that such an equation (2.184) is Lorentz invariant provided that ρ and $\vec{J}$ can be assembled into the components of a Lorentz 4-vector. This is indeed true in the case at hand and will be the starting point of our discussion below.

2. Our second clue will come from looking at the potentials: both the gauge transformations (4.7) and the wave equations (4.8) strongly suggest that ϕ and $\vec{A}$ should then also be collected into a Lorentz (co)vector.

3. Once we know how ϕ and $\vec{A}$ transform under Lorentz transformations, we can also determine how $\vec{E}$ and $\vec{B}$ transform under Lorentz transformations, i.e. how they are assembled into a Lorentz tensor (and, as we will see, the covariant formulation makes this particularly simple).

4.3 Electric 4-Current and Lorentz Invariance of the Continuity Equation

We recall from section 2.11 that, in terms of

$$J^a = (c\rho, \vec{J}) \ , \tag{4.10}$$

the continuity equation (4.4) can be written as (2.184)

$$\frac{\partial}{\partial t}\rho + \vec{\nabla}.\vec{j} = 0 \quad \Leftrightarrow \quad \partial_a J^a(x) = 0 \ . \tag{4.11}$$

and that this equation is Lorentz invariant if J^a is a Lorentz 4-vector.

In order to determine the transformation behaviour of the charge density ρ and current density $\vec{J}$ under Lorentz boost transformations, it is sufficient to consider charge densities moving at constant velocities. Our starting point and physical input will be the empirical fact that the (differential) charge dQ contained in a volume element dV is independent of its velocity. In the restframe of the charge distribution, say, one has

$$dQ = \rho_0 dV_0 \quad \text{and} \quad \vec{J}_0 = 0 \ . \tag{4.12}$$

Here ρ_0 is the rest charge density, and as such (tautologically) a scalar under Lorentz transformations, much like the rest mass of a particle. In an inertial system moving relative to the restframe at constant velocity v, one has a charge density ρ and a current density

$$\vec{J} = \rho\vec{v} \ . \tag{4.13}$$

Lorentz contraction

$$dV = \gamma(v)^{-1} dV_0 \tag{4.14}$$

and invariance of the charge,

$$dQ = \rho_0 dV_0 = \rho dV \tag{4.15}$$

imply

$$\rho = \gamma(v)\rho_0 \tag{4.16}$$

(this is intuitively obvious: smaller volume leads to larger charge density) and therefore

$$\vec{J} = \rho_0\gamma(v)\vec{v} \ . \tag{4.17}$$

Thus the components of J^a are

$$(J^a) = (c\rho, \vec{J}) = \rho_0(\gamma(v)c, \gamma(v)\vec{v}) \ . \tag{4.18}$$

Here we recognise the components (3.5) of the Lorentz vector 4-velocity u^a,

$$(u^a) = (\gamma(v)c, \gamma(v)\vec{v}) \ . \tag{4.19}$$

Since ρ_0 is a Lorentz scalar, we have established that

$$J^a = \rho_0 u^a \tag{4.20}$$

is indeed a Lorentz 4-vector, the electric 4-current (density) of Maxwell theory. In particular, therefore, the continuity equation is now manifestly Lorentz invariant.

Remarks:

1. The argument given above for the 4-vector character of the current can also be applied to (discrete or continuous) distributions of relativistic particles: also in that case, the number density of particles ρ is such that $\rho/\gamma(v) = \rho_0$ is independent of the inertial system, and therefore
$$(J^a) = (c\rho, \rho\vec{v}) = \rho_0(u^a) \tag{4.21}$$
is a 4-vector.

2. For later convenience, we will henceforth also absorb the annoying constant μ_0 (cf. (4.8)) into the definition of the 4-current, i.e. we redefine
$$J^a = \mu_0\rho_0 u^a \ , \tag{4.22}$$
with covariant components
$$(J_a) = (-\mu_0 c\rho, \mu_0\vec{J}) = (-\rho/(\epsilon_0 c), \mu_0\vec{J}) \ . \tag{4.23}$$

4.4 Inhomogeneous Maxwell Equations I: 4-Potential

Having identified ρ and $\vec{J}$ as components of a Lorentz 4-vector, looking back at the Maxwell equations (4.8) and gauge transformations (4.7) strongly suggests to also combine the electric and magnetic potentials ϕ and $\vec{A}$ into a 4-component object.

Indeed, let us set
$$(A_a) = (-\phi/c, \vec{A}) \ . \tag{4.24}$$
Then the first obervation is that the gauge transformations (4.7) can uniformly and elegantly be written as
$$\phi \to \phi - \partial_t\Psi \quad , \quad \vec{A} \to \vec{A} + \vec{\nabla}\Psi \quad \Leftrightarrow \quad A_a \to A_a + \partial_a\Psi \tag{4.25}$$
for an arbitrary function $\Psi = \Psi(x)$ on Minkowski space. We also see that the function G introduced in (4.9) can simply be written as
$$G = \vec{\nabla}.\vec{A} + \frac{1}{c}\partial_t(\phi/c) = \partial_a A^a \tag{4.26}$$
(note that $(A^a) = (+\phi/c, \vec{A})$). With this, and the definition of the current J_a (including the factor of μ_0) we can write the equations of motion for the potentials (4.8) collectively and simply as
$$\Box A_a - \partial_a(\partial_b A^b) = -J_a \ . \tag{4.27}$$
Now, since $\Box$ is a Lorentz scalar, and ∂_a and J_a are Lorentz covectors, this equation will be Lorentz invariant if and only if A_a transforms as a Lorentz covector (and thus $\partial_b A^b$ is a Lorentz scalar).

We have thus, with very little effort, managed to write the inhomogeneous Maxwell equations in a manifestly Lorentz invariant form.

Remarks:

1. The gauge transformation behaviour (4.25)

$$A_a \to A_a + \partial_a \Psi \tag{4.28}$$

shows that the 4-potential should naturally be thought of as a covector A_a rather than as a vector A^a.

2. The result (4.27) is manifestly Lorentz invariant. It is also gauge invariant, as it has to be: under $A_a \to A_a + \partial_a \Psi$ one has

$$\Box A_a - \partial_a(\partial_b A^b) \to \Box A_a + \Box \partial_a \Psi - \partial_a(\partial_b A^b) - \partial_a(\partial_b \partial^b \Psi) = \Box A_a - \partial_a(\partial_b A^b) \tag{4.29}$$

(because partial derivatives commute). However, gauge invariance is not yet manifest, and we will rectify this in the next section (after having introduced the Maxwell field strength tensor). This field strength tensor will then also allow us to immediately read off the transformation behaviour of the electric and magnetic fields under Lorentz transformations.

3. The term $G = \partial_b A^b$ by itself is evidently not gauge invariant. A convenient gauge condition is the so-called Lorenz gauge (without the "t", named after Ludwig Lorenz, not Hendrik Lorentz)

$$G = \partial_a A^a = 0 \ . \tag{4.30}$$

Not only do the Maxwell equations decouple in this gauge,

$$G = 0 \quad \Rightarrow \quad \Box A_a = -J_a \tag{4.31}$$

(so that the general solution can immediately be written down in terms of Greens functions for the wave operator $\Box$). This gauge condition is also the (essentially unique) gauge condition on A_a that perserves Lorentz invariance (other common gauge conditions like the Coulomb gauge, $\vec{\nabla}.\vec{A} = 0$, or axial gauges like $A_0 = 0$, are evidently not Lorentz invariant).

4.5 Inhomogeneous Maxwell Equations II: Maxwell Field Strength Tensor

We now want to find out how to express the gauge invariant fields $\vec{E}$ and $\vec{B}$ in a Lorentz tensorial way. To that end we start with the observation that

$$\vec{E} = -\vec{\nabla}\phi - \partial_t \vec{A} \quad , \quad \vec{B} = \vec{\nabla} \times \vec{A} \tag{4.32}$$

are precisely those linear combinations of the first partial derivatives of the potentials ϕ and $\vec{A}$ that are gauge invariant. Thus, as our first step we determine how the first derivatives $\partial_a A_b$ of A_b transform under gauge transformations:

$$A_b \to A_b + \partial_b \Psi \quad \Rightarrow \quad \partial_a A_b \to \partial_a A_b + \partial_a \partial_b \Psi \ . \tag{4.33}$$

We see that in general the partial derivatives of A_b are not gauge invariant, as expected. But the offending term

$$\partial_a \partial_b \Psi = \partial_b \partial_a \Psi \tag{4.34}$$

has the one characteristic property that it is symmetric (because partial derivatives commute ...). Therefore, we can eliminate it by taking the anti-symmetrised derivative of A_b,

$$A_b \to A_b + \partial_b \Psi \quad \Rightarrow \quad \partial_a A_b - \partial_b A_a \to \partial_a A_b - \partial_b A_a \ . \tag{4.35}$$

These are now precisely the gauge invariant linear combinations of the first derivatives of the potentials, and thus they must be expressible in terms of $\vec{E}$ and $\vec{B}$ (and we will verify this shortly). In any case, this motivates us to define and introduce the Maxwell field strength tensor

$$F_{ab} = \partial_a A_b - \partial_b A_a \ . \tag{4.36}$$

In addition to gauge invariance, F_{ab} has the following two important properties:

- F_{ab} is anti-symmetric, $F_{ab} = -F_{ba}$. Thus it has 6 independent components, precisely the right number to accommodate $\vec{E}$ and $\vec{B}$: this is how two 3-vectors can combine into a Lorentz tensor!
- F_{ab} is a Lorentz (0,2)-tensor, i.e. under Lorentz transformations $\bar{x}^a = L^a_{\ b} x^b$ it transforms as

$$\bar{F}_{ab}(\bar{x}) = \Lambda_a^{\ c} \Lambda_b^{\ d} F_{cd}(x) \ . \tag{4.37}$$

Combining these two facts, we see that once we have determined the relation between the components of F_{ab} and those of $\vec{E}$ and $\vec{B}$, the Lorentz transformation of $\vec{E}$ and $\vec{B}$ is determined (and reduces to simple matrix multiplication).

Thus let us now determine the relation between F_{ab} and $\vec{E}, \vec{B}$. To that end, we first write the defining relations (4.32) in components as

$$E_i = -\partial_i \phi - \partial_t A_i \quad , \quad B_i = \epsilon_{ijk} \partial_j A_k \ \Leftrightarrow \ \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B_k \tag{4.38}$$

(I am deliberately not careful with the positioning of the spatial indices here, summation over repeated indices is still understood). Now we turn to the components of F_{ab} in this inertial system. Since F_{ab} is anti-symmetric, with

$$(A_a) = (-\phi/c, \vec{A}) \tag{4.39}$$

the independent components are

$$\begin{aligned} F_{0i} &= \partial_0 A_i - \partial_i A_0 = -E_i/c = -F_{i0} \\ F_{ij} &= \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B_k \ . \end{aligned} \tag{4.40}$$

Thus, as expected, F_{ab} can be expressed entirely and easily in terms of the electric and magnetic fields. In matrix form, one can also write this as

$$(F_{ab}) = \begin{pmatrix} 0 & -E_1/c & -E_2/c & -E_3/c \\ +E_1/c & 0 & +B_3 & -B_2 \\ +E_2/c & -B_3 & 0 & +B_1 \\ +E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{4.41}$$

It will also be useful to know the contravariant components

$$F^{ab} = \eta^{ac}\eta^{bd}F_{cd} \ . \tag{4.42}$$

For these one has

$$F^{0i} = -F_{0i} \quad , \quad F^{ij} = F_{ij} \ , \tag{4.43}$$

and thus

$$(F^{ab}) = \begin{pmatrix} 0 & +E_1/c & +E_2/c & +E_3/c \\ -E_1/c & 0 & +B_3 & -B_2 \\ -E_2/c & -B_3 & 0 & +B_1 \\ -E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{4.44}$$

Next we want to write the inhomogeneous Maxwell equations (4.27)

$$\Box A_b - \partial_b(\partial_a A^a) = -J_b \tag{4.45}$$

in terms of F_{ab}. Since F_{ab} is constructed from the first derivatives of A_a, we need to look at first derivatives of F_{ab}, and the result should be a covector. There is really only one possibility, namely $\partial^a F_{ab}$. Working this out, one finds that on the nose

$$\partial^a F_{ab} = \partial^a\partial_a A_b - \partial^a\partial_b A_a = \Box A_b - \partial_b(\partial_a A^a) \ . \tag{4.46}$$

Thus we can write the Maxwell equations in the simple and beautiful form

$$\partial^a F_{ab} = -J_b \quad \Leftrightarrow \quad \partial_a F^{ab} = -J^b \ . \tag{4.47}$$

This is the sought-for manifestly Lorentz and gauge invariant formulation of the Maxwell equations.

Remarks:

1. Using the explicit expression for the components of F^{ab} given above, it is straightforward to also verify directly that these equations are equivalent to the inhomogeneous Maxwell equations (4.2),

$$\partial_a F^{ab} = -J^a \quad \Leftrightarrow \quad \vec{\nabla}.\vec{E} = \rho/\epsilon_0 \quad , \quad \vec{\nabla}\times\vec{B} - \frac{1}{c^2}\partial_t\vec{E} = \mu_0\vec{J} \ . \tag{4.48}$$

   For example,

$$\partial_a F^{a0} = \partial_i F^{i0} = -\partial_i E_i/c = -\rho/(\epsilon_0 c) = -\mu_0\rho c = -J^0 \tag{4.49}$$

   and likewise for the spatial components $\partial_a F^{aj}$.

2. The continuity equation $\partial_a J^a = 0$ follows trivially from (4.47):

$$\partial_b J^b = -\partial_b\partial_a F^{ab} = 0 \tag{4.50}$$

   beacuse $\partial_b\partial_a$ is symmetric (partial derivatives commute ...) and F^{ab} is anti-symmetric.

4.6 Homogeneous Maxwell Equations I: Bianchi Identities

Looking back at the Maxwell equations recalled in section 4.1, we see that the only equations that we have not yet cast into manifestly Lorentz-invariant form are the homogeneous equations (4.1). One way to approach the question how to do go about this is to note that these equations are identically satisfied once one has introduced the potentials. In the present context, we are thus asking the question what differential equations are identically satisifed by an F_{ab} of the form $F_{ab} = \partial_a A_b - \partial_b A_a$.

- As a warm-up exercise (with one index less), let us consider the question what sort of differential equations are identically satisfied by a covector $F_a = \partial_a A$. In that case the well-known answer is that its anti-symmetrised derivative is zero

$$F_a = \partial_a A \quad \Rightarrow \quad \partial_a F_b - \partial_b F_a = \partial_a \partial_b A - \partial_b \partial_a A = 0 \tag{4.51}$$

  (partial derivatives commute ...).

- The same strategy works for $F_{ab} = \partial_a A_b - \partial_b A_a$: since partial derivatives commute, the totally anti-symmetrised derivative of F_{ab} will be identically zero,

$$F_{ab} = \partial_a A_b - \partial_b A_a \quad \Rightarrow \quad \partial_a F_{bc} - \partial_b F_{ac} + \text{ 4 more terms } = 0 \ . \tag{4.52}$$

  In general, such identities, resulting from anti-symmetrisation of differential operators, are referred to as *Bianchi Identities*.

  Using the results and notation of section 2.8, in particular the identity (2.143),

$$T_{abc} = T_{a[bc]} \quad \Rightarrow \quad T_{[abc]} = \tfrac{1}{3}(T_{abc} + T_{cab} + T_{bca}) \ , \tag{4.53}$$

  we can write this as

$$F_{ab} = \partial_a A_b - \partial_b A_a \quad \Rightarrow \quad \partial_{[a} F_{bc]} = 0 \quad \Leftrightarrow \quad \partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab} = 0 \ . \tag{4.54}$$

  The fact that the equation on the left implies the equation on the right is also easily verified directly.

While these equations, with their 3 indices, look somewhat intransparent (and of course we will improve that below!), already now we can verify that these are precisely 4 independent equations, and that, with F_{ab} expressed in terms of $\vec{E}$ and $\vec{B}$, they reproduce precisely the homogeneous Maxwell equations,

$$\partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab} = 0 \quad \Leftrightarrow \quad \vec{\nabla} \times \vec{E} + \partial_t \vec{B} = 0 \ , \ \vec{\nabla}.\vec{B} = 0 \ . \tag{4.55}$$

We need to consider 3 different cases:

1. two indices are equal

   We first observe that the equations on the left-hand side are empty (trivially satisfied for any anti-symmetric F_{ab}) if any 2 indices are equal (since the left-hand side is totally anti-symmetric, this could hardly be otherwise). Indeed, if $a = b$, say, then we have

$$\partial_a F_{ac} + \partial_a F_{ca} + \partial_c F_{aa} = \partial_a F_{ac} - \partial_a F_{ac} + 0 = 0 \tag{4.56}$$

   identically, just by anti-symmetry of F_{ab}. Thus all 3 indices have to be different.

2. all indices are spatial, e.g. $(a = 1, b = 2, c = 3)$

   In this case one has

$$\partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = \vec{\nabla}.\vec{B} \ . \tag{4.57}$$

3. one index is temporal and the others are spatial, e.g. $(a = 0, b = 1, c = 2)$ (or essentially, up to signs and permutations, two more possibilities)

   In this case one has

$$\partial_0 F_{12} + \partial_1 F_{20} + \partial_2 F_{01} = c^{-1}(\partial_t \vec{B} + \nabla \times \vec{E})_3 \tag{4.58}$$

   (and likewise for the remaining components).

This establishes (4.55).

Thus we can neatly summarise basically all of Maxwell theory by

$$\text{Maxwell Equations:} \quad \begin{cases} \partial_a F^{ab} = -J^b \\ \partial_{[a} F_{bc]} = 0 \end{cases} \tag{4.59}$$

A famous consequence of the Maxwell equations is that, in source-free regions of space(-time) the electric and magnetic fields propagate as waves with velocity c,

$$\rho = \vec{J} = 0 \quad \Rightarrow \quad \Box\vec{E} = \Box\vec{B} = 0 \ . \tag{4.60}$$

The usual non-covariant 3-vector calculus derivation of this is somewhat roundabout, and requires the full set of eight (homogeneous and inhomogeneous) Maxwell equations and judicious use of various 3-vector calculus identities. Here is a 1-line proof of the statement

$$\partial^a F_{ab} = -J_b = 0 \quad \Rightarrow \quad \Box F_{ab} = 0 \tag{4.61}$$

in our formulation:

$$0 = \partial^c(\partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab}) = \partial_a \partial^c F_{bc} + \partial_b \partial^c F_{ca} + \Box F_{ab} = \Box F_{ab} \ . \tag{4.62}$$

When the 4-current is not equal to zero, one has instead

$$\Box F_{ab} = \partial_b J_a - \partial_a J_b \ . \tag{4.63}$$

4.7 Homogeneous Maxwell Equations II: Dual Field Strength Tensor

While the form of the homogeneous Maxwell equation given in (4.59) is nicely manifestly Lorentz- and gauge invariant, there is a different way of writing it which makes it more manifest that these are indeed only precisely four equations, and which brings out a nice analogy between the homogeneous and inhomgeneous equations.

Recall that already in ordinary 3-vector calculus, frequently, instead of anti-symmetrising explicitly, it is much more convenient to let the ϵ- (or Levi-Civita) symbol ϵ_{ijk} do the job, as in

$$\partial_j A_k - \partial_k A_j \to \epsilon_{ijk}\partial_j A_k \equiv B_i \ . \tag{4.64}$$

In particular, then the identity $\vec{\nabla}.\vec{B} = 0$ becomes manifest because (once again ...) partial derivatives commute,

$$\partial_i B_i = \epsilon_{ijk}\partial_i\partial_j A_k = 0 \ . \tag{4.65}$$

In this 3-dimensional case, all the components of ϵ_{ijk} are determined by total anti-symmetry and the choice (of orientation) $\epsilon_{123} = 1$,

$$\epsilon_{ijk} = \epsilon_{[ijk]} \quad , \quad \epsilon_{123} = 1 \ . \tag{4.66}$$

In our 4-dimensional case, we can analogously introduce a totally anti-symmetric spacetime ϵ-symbol ϵ_{abcd} by

$$\epsilon_{abcd} = \epsilon_{[abcd]} \quad , \quad \epsilon_{0123} = +1 \ . \tag{4.67}$$

To be compatible with our conventions for raising and lowering indices, we also define ϵ^{abcd} by

$$\epsilon^{abcd} = \epsilon^{[abcd]} \quad , \quad \epsilon^{0123} = -1 \ . \tag{4.68}$$

Then, letting ϵ^{abcd} taking care of the total anti-symmetrisation, we can write the homogeneous Maxwell equations as

$$\partial_{[a}F_{cd]} = 0 \quad \Leftrightarrow \quad \epsilon^{abcd}\partial_a F_{cd} = \partial_a(\epsilon^{abcd}F_{cd}) = 0 \ . \tag{4.69}$$

We are thus led to introduce the *dual Maxwell field strength tensor* $\tilde{F}^{ab}$ by (the factor of $1/2$ is a convenient convention)

$$\tilde{F}^{ab} = \tfrac{1}{2}\epsilon^{abcd}F_{cd} \ . \tag{4.70}$$

Then we have

$$\partial_{[a}F_{cd]} = 0 \quad \Leftrightarrow \quad \partial_a\tilde{F}^{ab} = 0 \ , \tag{4.71}$$

and it is now manifest that these are indeed precisely 4 equations.

Thus we can write the full set of Maxwell equations as

$$\text{Maxwell Equations:} \quad \begin{cases} \partial_a F^{ab} = -J^b \\ \partial_a \tilde{F}^{ab} = 0 \end{cases} \tag{4.72}$$

Remarks:

1. Note that the 3-dimensional ϵ-symbol ϵ_{ijk} has the *cyclic symmetry* $\epsilon_{ijk} = \epsilon_{kij}$, because ϵ_{kij} can be obtained from ϵ_{ijk} by an *even* number of permutations,

$$\epsilon_{kij} = -\epsilon_{ikj} = +\epsilon_{ijk} \ . \tag{4.73}$$

   By contrast, for the 4-dimensional ϵ-symbol ϵ_{abcd} one has the anti-cyclic property

$$\epsilon_{dabc} = -\epsilon_{adbc} = +\epsilon_{abdc} = -\epsilon_{abcd} \ . \tag{4.74}$$

2. The dual field strength tensor $\tilde{F}^{ab}$ is, i.e. transforms as, a tensor under rotations and boosts (the transformations that we usually call Lorentz transformations), but because a choice of orientation is involved in the definition of ϵ_{abcd}, it transforms additionally with a

sign $\det(L) = \pm 1$ under general Lorentz transformations. This is just like in 3-dimensional vector calculus, where the vector product, defined with the help of ϵ_{ijk} defines not a vector but what is known as a pseudo-vector (sensitive to the orientation: right-hand versus left-hand rule). For the time being, however, since we are not interested in space or time reflections, we can ignore this subtlety.

3. Explicitly, the components of $\tilde{F}^{ab}$ are related to those of F_{ab} e.g. by

$$\begin{aligned} \tilde{F}^{01} &= \tfrac{1}{2}\epsilon^{01cd}F_{cd} = \tfrac{1}{2}(\epsilon^{0123}F_{23} + \epsilon^{0132}F_{32}) = \epsilon^{0123}F_{23} = -F_{23} \\ \tilde{F}^{23} &= \tfrac{1}{2}\epsilon^{23cd}F_{cd} = \epsilon^{2301}F_{01} = \epsilon^{0123}F_{01} = -F_{01} \end{aligned} \tag{4.75}$$

etc. In terms of $\vec{E}$ and $\vec{B}$ this means

$$\tilde{F}^{01} = -B_1 \quad , \quad \tilde{F}^{23} = E_1/c \tag{4.76}$$

etc., so that we can write $\tilde{F}^{ab}$ in matrix form as

$$(\tilde{F}^{ab}) = \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ +B_1 & 0 & +E_3/c & -E_2/c \\ +B_2 & -E_3/c & 0 & +E_1/c \\ +B_3 & +E_2/c & -E_1/c & 0 \end{pmatrix} \tag{4.77}$$

4. One can now also verify directly that

$$\partial_a \tilde{F}^{ab} = 0 \quad \Leftrightarrow \quad \vec{\nabla} \times \vec{E} + \partial_t \vec{B} = 0 \ , \ \vec{\nabla}.\vec{B} = 0 \ . \tag{4.78}$$

E.g.

$$\partial_a \tilde{F}^{a0} = \partial_i \tilde{F}^{i0} = \partial_i B_i = \vec{\nabla}.\vec{B} \tag{4.79}$$

(and likewise for the other components).

5. Comparison with (F^{ab}) (4.44),

$$(F^{ab}) = \begin{pmatrix} 0 & +E_1/c & +E_2/c & +E_3/c \\ -E_1/c & 0 & +B_3 & -B_2 \\ -E_2/c & -B_3 & 0 & +B_1 \\ -E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{4.80}$$

shows that $\tilde{F}^{ab}$ is obtained from F^{ab} by sending

$$F^{ab} \to \tilde{F}^{ab} \quad \Leftrightarrow \quad \vec{E}/c \to -\vec{B} \quad \text{and} \quad \vec{B} \to \vec{E}/c \ . \tag{4.81}$$

Thus this exchanges the electric and magnetic fields.

6. In fact, this transformation is known as the electric-magnetic duality transformation of Maxwell theory. You may have noticed before the curious fact that the Maxwell equations (without electric sources) are invariant under this transformation, i.e. the homogeneous equations get mapped to the inhomogeneous equations (without sources) and vice versa: it is obvious that the transformation exchanges

$$\vec{\nabla}.\vec{E} = 0 \quad \leftrightarrow \quad \vec{\nabla}.\vec{B} = 0 \ , \tag{4.82}$$

but it is also true that it exchanges the remaining equations, since

$$\vec{\nabla} \times \vec{B} - \frac{1}{c}\partial_t(\vec{E}/c) \quad \leftrightarrow \quad (\partial_t \vec{B} + \vec{\nabla} \times \vec{E})/c \ . \tag{4.83}$$

7. In the present formulation, this duality symmetry of the vacuum equations could not be more obvious. In the absence of electric sources, the Maxwell equations read

$$J^a = 0 \quad \Rightarrow \quad \partial_a F^{ab} = 0 \quad , \quad \partial_a \tilde{F}^{ab} = 0 \ , \tag{4.84}$$

which are manifestly invariant under the exchange $F^{ab} \leftrightarrow \tilde{F}^{ab}$. Unfortunately, in the presence of sources, this nice and intriguing duality symmetry is broken by the (unexplained) absence of magnetic monopole charges and currents in the real world.

4.8 Maxwell Theory and Lorentz Transformations I: Lorentz Scalars

Now that we know how the Maxwell field strength tensor F_{ab} transforms under Lorentz transformations, namely as a (0,2)-tensor, and how the components of F_{ab} are related to those of $\vec{E}$ and $\vec{B}$, we can now easily determine the transformation behaviour of $\vec{E}$ and $\vec{B}$ under Lorentz transformations, and we will come back to this below.

However, as always, it is useful to first think about and look for and at Lorentz scalars, i.e. objects that are actually invariant under Lorentz transformations. With the building blocks A_a and F_{ab} at our disposal, one Lorentz scalar that we could construct is

$$A_a A^a = \eta^{ab} A_a A_b \ , \tag{4.85}$$

but while this is a Lorentz scalar, it is not invariant under gauge transformations, and therefore of no interest to us. If we require gauge invariance in addition to Lorentz invariance, then we need to work with F_{ab}. The most obvious strategy to construct a scalar out of a $(0,2)$-tensor is (cf. the discussion in section 2.8) to take its η-trace, but beacuse F_{ab} is anti-symmetric, this will vanish,

$$F^a_{\ a} \equiv \eta^{ab} F_{ab} = 0 \ . \tag{4.86}$$

Thus there are no gauge invariant Lorentz scalars that are linear functions of $\vec{E}$ and $\vec{B}$. However, it is easy to construct a scalar that is quadratic in F_{ab}, namely

$$I_1 = \tfrac{1}{4} F_{ab} F^{ab} = \tfrac{1}{4} \eta^{ac} \eta^{bd} F_{ab} F_{cd} \tag{4.87}$$

(the factor of 1/4 is just a convention). Expressed in terms of $\vec{E}$ and $\vec{B}$, this is

$$I_1 = \tfrac{1}{4}(F_{0k}F^{0k} + F_{k0}F^{k0} + F_{ik}F^{ik}) = \tfrac{1}{2}(\vec{B}^2 - \vec{E}^2/c^2) \ . \tag{4.88}$$

The fact that this is a Lorentz scalar has some immediate consequences. Namely, if there is one inertial system in which $I_1 > 0$ (or $I_1 = 0$ or $I_1 < 0$), then in all inertial systems $I_1 > 0$ (or $I_1 = 0$ or $I_1 < 0$).

For example, consider the electromagnetic field of a charge at rest in some inertial system. In that inertial system, $\vec{E} \neq 0$ but $\vec{B} = 0$. In particular, therefore, I_1 is negative, $I_1 < 0$. In some

other inertial system, it is clear that there will be both an electric and a magnetic field, but the additional information that the invariant I_1 provides us with, without any further calculation, is that the magnetic field cannot exceed the electric field in magnitude,

$$I_1 = \bar{I}_1 < 0 \quad \Rightarrow \quad |\overline{\vec{B}}| < |\overline{\vec{E}}|/c \ . \tag{4.89}$$

There is another invariant that we can construct, namely

$$I_2 = \tfrac{1}{4} F_{ab}\tilde{F}^{ab} \ . \tag{4.90}$$

This is a scalar under rotations and boosts (but, like $\tilde{F}^{ab}$, transforms with the sign $\det L$ under general more general Lorentz transformations). Expressed in terms of $\vec{E}$ and $\vec{B}$, this is

$$I_2 = \vec{B}.\vec{E}/c \ . \tag{4.91}$$

In particular this implies that if e.g. $\vec{B} = 0$ in some inertial system, then in any inertial system the electric field will be orthogonal to the magnetic field. As regards the above example of a moving charge, this provides us with the additional information that the magnetic field of a moving charge will be orthogonal to its electric field.

One property of I_2 that we will come back to later in our discussion of an action principle for Maxwell theory is the fact that when $F_{ab} = \partial_a A_b - \partial_b A_a$, the invariant I_2 can (unlike I_1) be written as a total derivative. Indeed, writing

$$F_{ab}\tilde{F}^{ab} = \tfrac{1}{2}\epsilon^{abcd}F_{ab}F_{cd} = \epsilon^{abcd}F_{ab}\partial_c A_d \ , \tag{4.92}$$

we see that this can be written as

$$F_{ab}\tilde{F}^{ab} = \partial_c(\epsilon^{abcd}F_{ab}A_d) - \epsilon^{abcd}(\partial_c F_{ab})A_d = \partial_c(\epsilon^{abcd}F_{ab}A_d) \ , \tag{4.93}$$

where in the last step we used the Bianchi identity satisfied by F_{ab}.

Are there any further (independent) invariants we can construct? The answer is no (and one can prove this using group theory, but we shall not do this here). Here are some examples to illustrate this claim:

1. The most obvious candidate for another invariant is perhaps the square of the dual field strength tensor $\tilde{F}^{ab}$, but it is easy to see that

$$\tilde{I}_1 \equiv \tfrac{1}{4}\tilde{F}_{ab}\tilde{F}^{ab} = -\tfrac{1}{4}F_{ab}F^{ab} = -I_1 \ . \tag{4.94}$$

2. Any scalar constructed from an odd number of F_{ab} and/or $\tilde{F}^{ab}$ is automatically zero (because it can be regarded as the trace of an odd number of anti-symmetric matrices, which is zero). For example,

$$I_3 = F^a_{\ b}F^b_{\ c}F^c_{\ a} = 0 \ . \tag{4.95}$$

3. Scalars constructed from an even nunmber of F_{ab} and/or $\tilde{F}^{ab}$ can be expressed in terms of polynomials of I_1 and I_2. For example, for

$$I_4 = F^{ab}F_{bc}F^{cd}F_{da} \tag{4.96}$$

one finds, after an uninspiring but straightforward calculation, something like

$$I_4 = 8(I_1)^2 + 4(I_2)^2 \ . \tag{4.97}$$

4. One can also construct gauge invariant Lorentz scalars from derivatives of the fields, like $F_{ab}\Box F^{ab}$. These play a role in quantum field theory, as higher derivative (quantum) corrections to the classical action, but will not play any role in these notes.

4.9 Maxwell Theory and Lorentz Transformations II: Transformation of $\vec{E}, \vec{B}$

Finally, we turn to the simple (and purely algebraic) task of determining the transformation behaviour of $\vec{E}$ and $\vec{B}$ under Lorentz transformations. In general we already know that F_{ab} transforms like a (0,2) tensor field, i.e.

$$\bar{x}^a = L^a_{\ b} x^b \quad \Rightarrow \quad \bar{F}_{ab}(\bar{x}) = \Lambda_a^{\ c}\Lambda_b^{\ d} F_{cd}(x) \ . \tag{4.98}$$

As they stand, the above equations express the new fields at $\bar{x}$ in terms of the old fields at x. In order to express the new fields as functions of $\bar{x}$, as one would presumably like, all one needs to do is to write the x^a as

$$x^a = (L^{-1})^a_{\ b}\bar{x}^b \ , \tag{4.99}$$

so that

$$\bar{F}_{ab}(\bar{x}) = \Lambda_a^{\ c}\Lambda_b^{\ d} F_{cd}(L^{-1}\bar{x}) \ . \tag{4.100}$$

Under spatial rotations, $\vec{E}$ and $\vec{B}$ transform in the familiar was as 3-vectors. Thus we only need to look at Lorentz boosts, and without loss of generality we consider a boost in the x^1-direction, which has the form (cf. section 2.4)

$$(L^a_{\ b}) = \begin{pmatrix} \cosh\alpha & -\sinh\alpha & 0 & 0 \\ -\sinh\alpha & \cosh\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.101}$$

with

$$\cosh\alpha(v) = \gamma(v) \quad , \quad \sinh\alpha(v) = \beta(v)\gamma(v) \ . \tag{4.102}$$

Therefore, $\Lambda = (L^T)^{-1}$ has the form

$$\left(\Lambda_a^{\ b}\right) = \begin{pmatrix} \cosh\alpha & \sinh\alpha & 0 & 0 \\ \sinh\alpha & \cosh\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.103}$$

It follows that e.g. (suppressing the argument x or $\bar{x}$ for simplicity and for the time being)

$$\begin{aligned} \bar{F}_{01} &= \Lambda_0^{\ c}\Lambda_1^{\ d}F_{cd} = (\Lambda_0^{\ 0}\Lambda_1^{\ 1} - \Lambda_0^{\ 1}\Lambda_1^{\ 0})F_{01} = F_{01} \\ \bar{F}_{02} &= \Lambda_0^{\ c}\Lambda_2^{\ d}F_{cd} = \Lambda_0^{\ c}F_{c2} = \cosh\alpha F_{02} + \sinh\alpha F_{12} \\ \bar{F}_{12} &= \Lambda_1^{\ c}\Lambda_2^{\ d}F_{cd} = \Lambda_1^{\ c}F_{c2} = \sinh\alpha F_{02} + \cosh\alpha F_{12} \end{aligned} \tag{4.104}$$

etc. In terms of the components of the electric and magnetic fields one thus has

$$\begin{aligned} \bar{E}_1 &= E_1 \quad , \quad \bar{E}_2 = \gamma(E_2 - \beta c B_3) \quad , \quad \bar{E}_3 = \gamma(E_3 + \beta c B_2) \\ \bar{B}_1 &= B_1 \quad , \quad \bar{B}_2 = \gamma(B_2 + \beta E_3/c) \quad , \quad \bar{B}_3 = \gamma(B_3 - \beta E_2/c) \end{aligned} \tag{4.105}$$

We see that the "longitudinal" components of the fields are not changed by a boost, while the transverse components are deformed.

If we want to reinstate the dependence of the fields on the coordinates, then we proceed as in (4.100) above. In the case at hand, since L is symmetric, the components of L^{-1} are just those of Λ.

When originally there is just an electric field, these equations simplify to

$$\vec{B} = 0 \quad \Rightarrow \quad \overline{\vec{E}} = (E_1, \gamma E_2, \gamma E_3) \quad , \quad \overline{\vec{B}} = (0, \beta\gamma E_3/c, -\beta\gamma E_2/c) \tag{4.106}$$

and one can explicitly check the assertions regarding the invariants I_1 and I_2 made in the previous section, e.g. the fact that the new magnetic field is orthogonal to the new electric field.

4.10 Example: The Field of a Moving Charge (Outline)

One can now use these methods to solve in a very simple way some standard problems of electrodynamics, e.g. to determine the electromagnetic field created by a charge or current moving with constant velocity. To that end,

- one first solves the problem in the rest frame of the charge or current (so in this case this is the simple electrostatics problem of determining the electric field of a static charge or a charged wire)
- and one then applies a Lorentz transformation to this solution to obtain the electromagnetic field of the moving charge or electric current.

The only thing one has to pay attention to is, as mentioned above, the correct assignment of the coordinates to the fields.

Concretely, assume that a point particle with charge q is at rest at the origin of the inertial system with coordinates $x^a = (ct, \vec{x})$. Then it has a purely electric and time-independent field given by the solution to $\vec{\nabla}.\vec{E} = \rho/\epsilon_0$, namely

$$\vec{E}(\vec{x}) = Q\frac{\vec{x}}{|\vec{x}|^3} \quad , \tag{4.107}$$

where I have introduced the abbreviation

$$Q = \frac{q}{4\pi\epsilon_0} \quad . \tag{4.108}$$

It follows from the above formulae that in the inertial system with coordinates $\bar{x}^a$ (with respect to which the charge moves with constant velocity $-v$ in the $\bar{x}^1$-direction, apologies for the minus

sign ...), the electric field is given by

$$\begin{aligned}\bar{E}_1(\bar{x}) &= E_1(x) = Q\frac{x^1}{|\vec{x}|^3} \\ \bar{E}_2(\bar{x}) &= \gamma E_2(x) = \gamma Q\frac{x^2}{|\vec{x}|^3} \\ \bar{E}_3(\bar{x}) &= \gamma E_3(x) = \gamma Q\frac{x^3}{|\vec{x}|^3} \ .\end{aligned} \tag{4.109}$$

Thus all that is left to do is to express the spatial coordinates x^i on the right-hand side in terms of the spacetime coordinates $\bar{x}^a$ via the inverse Lorentz transformation. One can of course do this in general but, in order to simplify the subsequent formulae, let us choose an observer at rest in the new inertial system at a point P with spatial coordinates

$$\bar{x}^i_P = (0, \bar{x}^2 = b, 0) \ . \tag{4.110}$$

In terms of the coordinates x^i, this observer has the coordinates

$$x^i_P = (\gamma(v)\beta(v)\bar{x}^0, b, 0) = (\gamma(v)v\bar{t}, b, 0) \ . \tag{4.111}$$

In particular,

$$|\vec{x}_P| = (\gamma^2 v^2 \bar{t}^2 + b^2)^{1/2} \ . \tag{4.112}$$

Putting everything together, we find that in the inertial system in which the observer is at rest (and the charge moves with constant velocity), the observer sees a time-dependent electric field given by

$$\begin{aligned}\bar{E}_1(\bar{x}^i_P, \bar{t}) &= Q\frac{\gamma(v)v\bar{t}}{(\gamma^2 v^2\bar{t}^2 + b^2)^{3/2}} \\ \bar{E}_2(\bar{x}^i_P, \bar{t}) &= Q\frac{\gamma(v)b}{(\gamma^2 v^2\bar{t}^2 + b^2)^{3/2}} \\ \bar{E}_3(\bar{x}^i_P, \bar{t}) &= 0 \ .\end{aligned} \tag{4.113}$$

We see that the transverse component $\bar{E}_2$ reaches its maxmimum at the time $\bar{t} = 0$ (the time when the distance between the charge and the observer takes on its minimal value), with

$$\bar{E}_2(\bar{x}^i_P, \bar{t} = 0) = \frac{Q\gamma(v)}{b^2} \tag{4.114}$$

proportional to $\gamma(v)$, and hence large for a rapidly moving charge. The longitudinal component $\bar{E}_1$, on the other hand, changes sign at $\bar{t} = 0$, and it has extrema at

$$\bar{t}_\pm = \pm b/\sqrt{2}v\gamma(v) \tag{4.115}$$

(so for large velocities this is a narrow time interval) with

$$\bar{E}_1(\bar{x}^i_P, \bar{t}_\pm) = \pm\frac{2Q}{3\sqrt{3}b^2} \tag{4.116}$$

(which is independent of $\vec{v}$).

For the magnetic field, one sees that $\bar{B}_1 = \bar{B}_2 = 0$, but that there is a non-zero component

$$\bar{B}_3 = -\beta\gamma E_2/c = -\beta\bar{E}_2/c \tag{4.117}$$

of the magnetic field in the x^3-direction orthogonal to both the electric field and the velocity of the charge. This reflects what is known as the Biot-Savart law of magnetostatics. For an arbitrary direction of the velocity $\vec{v}$ the result can be written as

$$\overline{\vec{B}} = (\vec{v} \times \overline{\vec{E}})/c^2 \ . \tag{4.118}$$

In a similar way one can determine the electromagnetic field produced by a steady (constant velocity) current from the simple electrostatic field of a charged wire. In particular, this means that the magnetic field generated by a current can be regarded as a relativistic effect. Even though the typical velocities in a current, of the order $v \sim \mathcal{O}(1mm/s) \ll c$, are very far from what one would usually call "relativistic velocities", this is a very visible and common effect (electric motors!), because of the large (Avogadro-ish) number of charge carriers in a current which all contribute to the magnetic field.

4.11 Covariant Formulation of the Lorentz Force Equation

The non-relativistic (better: Galilean relativistic) equation of motion for a massive charged particle with mass m and charge q in an electromagnetic field is

$$\frac{d}{dt}(m\vec{v}) = q(\vec{E} + \vec{v} \times \vec{B}) \ , \tag{4.119}$$

where the force term on the right-hand side is known as the Lorentz force. Taking the scalar product of this equation with $\vec{v}$, one finds

$$\frac{d}{dt}(m\vec{v}^2/2) = q\vec{E}.\vec{v} \ , \tag{4.120}$$

which describes the change in the kinetic energy of the particle due to the work done on it by the electric field.

We already know how to modify the left-hand side of (4.119) in order to obtain a Lorentz-tensorial expression: we replace the velocity $\vec{v}$ by the 4-velocity u^a and the derivative with respect to time by the derivative with respect to proper time,

$$\frac{d}{dt}(m\vec{v}) \to \frac{d}{dt}(m\gamma(v)\vec{v}) = \frac{d}{dt}\vec{p} \to \frac{d}{d\tau}p^a = \frac{d}{d\tau}(mu^a) \ . \tag{4.121}$$

What about the right-hand side? In order to reproduce this we evidently need to construct a 4-vector that is linear in F_{ab} and linear in u^a. There are not so many possiblilities for this. In fact, up to signs and factors the only possibility is $F^{ab}u_b$. Let us calculate the spatial components of this:

$$F^{ib}u_b = F^{i0}u_0 + F^{ij}u_j = (-E_i/c)(-\gamma(v)c) + \epsilon_{ijk}\gamma(v)v^jB^k = \gamma(v)(\vec{E} + \vec{v} \times \vec{B})_i \ . \tag{4.122}$$

We see that, up to the γ-factor, we find on the nose and very naturally the rather peculiar Lorentz force term. We can thus write down our candidate Lorentz invariant equation of motion for a charged particle in the Maxwell field, namely

$$\frac{d}{d\tau}p^a = qF^{ab}u_b \ . \tag{4.123}$$

In section 4.12 below, we will derive (4.123) from a Lorentz- and gauge invariant action principle for a charged particle coupled to the Maxwell field.

Remarks:

1. Using the fact that $\gamma(v)$ is the conversion factor between $d\tau$ and dt, we see that the spatial components of this equation can be written as

$$\gamma(v)\frac{d}{dt}\vec{p} = \gamma(v)q(\vec{E} + \vec{v} \times \vec{B}) \quad \Leftrightarrow \quad \frac{d}{dt}\vec{p} = q(\vec{E} + \vec{v} \times \vec{B}) \tag{4.124}$$

   This differs from the non-relativistic equation (4.119) only by the replacement $m\vec{v} \to \vec{p} = m\gamma(v)\vec{v}$ on the left-hand side, while the right-hand Maxwell sides of the two equations are identical. In particular, this equation has the correct non-relativistic limit.

2. We noted before, in section 3.3, that any candidate equation of the form

$$\frac{d}{d\tau}p^a = K^a \tag{4.125}$$

   requires the force to be orthogonal to the 4-velocity,

$$\frac{d}{d\tau}p^a = ma^a = K^a \quad \Rightarrow \quad K^a u_a = 0 \ . \tag{4.126}$$

   In the case at hand, this is indeed satisfied,

$$K^a = qF^{ab}u_b \quad \Rightarrow \quad K^a u_a = qF^{ab}u_a u_b = 0 \tag{4.127}$$

   by anti-symmetry of F^{ab} and symmetry of $u_a u_b$.

3. It remains to discuss the temporal component of (4.123). It can be written as

$$\frac{d}{dt}p^0 = qF^{0k}u_k/\gamma(v) = q\vec{E}.\vec{v}/c \quad \Leftrightarrow \quad \frac{d}{dt}E = q\vec{E}.\vec{v} \tag{4.128}$$

   where $E = m\gamma(v)c^2$, and can therefore, exactly as (4.120), be interpreted as the change in the energy E of the particle due to the work performed on the particle by the electric field.

4. Just as (4.120) was implied by (4.119), in the present case (and in general for any K^a), one has

$$\frac{d}{d\tau}p^i = K^i \quad \Rightarrow \quad \frac{d}{d\tau}p^0 = K^0 \ . \tag{4.129}$$

   This is best understood as a consequence of the fact that the 4 components of K^a are not independent,

$$K^a u_a = 0 \quad \Leftrightarrow \quad K^0 = -K^i u_i/u_0 \ . \tag{4.130}$$

   Indeed, using the spatial components of the equation of motion, one finds an equation which is independent of the K^a,

$$\frac{d}{d\tau}p^0 = -K^i u_i/u_0 = -(\frac{d}{d\tau}p^i)u_i/u_0 \quad \Leftrightarrow \quad u_a\frac{d}{d\tau}p^a = 0 \ , \tag{4.131}$$

   and which is of course just the identity that 4-velocity and 4-acceleration are orthogonal, $u_a a^a = 0$.

4.12 Action Principle for a Charged Particle coupled to the Maxwell Field

We now want to look at the Lorentz force equation from the point of view of an action principle. This is rather straightforward, and it is also very instructive as it teaches us how to introduce forces / interactions in a free (non-interacting) matter theory in a Lorentz invariant manner by coupling the matter (here particles) to gauge fields in a Lorentz and gauge invariant way.

As a reminder, the action for a free relativistic particle was (we now use the subscript 0 on S_0 to indicate that this is the free action)

$$S_0[x] = -mc^2 \int d\tau \ . \tag{4.132}$$

with

$$\delta S_0[x] = \int d\tau \left(-\frac{d}{d\tau} p_a\right) \delta x^a \quad \Rightarrow \quad \frac{d}{d\tau} p_a = 0 \ . \tag{4.133}$$

We also know from the previous section that the equation of motion for a charged particle in the Maxwell field is

$$\frac{d}{d\tau} p_a = q F_{ab} u^b = q F_{ab} \dot{x}^b \ . \tag{4.134}$$

It is evident that in order to derive this equation from an action principle, we need to couple the particle to the Maxwell field. The action will thus take the form

$$S[x; A] = S_0[x] + S_I[x; A] \ , \tag{4.135}$$

where the 2nd term $S_I[x; A]$ describes the coupling (interaction) between particle and field, and I use the notation $S[x; A]$ to indicate that the action should depend on the gauge field $A_a(x)$, but that A_a is not, at this point, a dynamical variable that is to be varied separately. So our aim is to determine $S_I[x; A]$.

The low-brow (and perhaps not very insightful) way to go about this is to remind oneself how this is done in the non-relativistic case, and to then continue from there. Thus the coupling to an electric field is simply described by adding to the Lagrangian minus the potential electrostatic energy, which is nothing other than

$$V = q\phi \tag{4.136}$$

with ϕ the eletric potential (it is no coincidence that potentials are called potentials!). To describe the coupling to the magnetic field, one needs to introduce a (from the point of view of classical non-relativistic mechanics) rather peculiar velocity-dependent potential as well,

$$V = q\phi - q\vec{A}.\vec{v} \ . \tag{4.137}$$

Then one can show that the Euler-Lagrange equations resulting from

$$S = \int dt \ \left(\frac{m}{2}\vec{v}^2 - V\right) \tag{4.138}$$

are indeed precisely the Lorentz force equations (4.119).

One could then observe that, with our definition of A_a, the 2 terms in V can be combined into

$$-V = q(A_0 c + A_i v^i) = q A_a \frac{dx^a}{dt} \ , \tag{4.139}$$

and one might then perhaps be led to guess that the correct relativistic interaction action is

$$S_I[x;A] \stackrel{?}{=} q\int d\tau\ A_a\dot{x}^a \quad . \tag{4.140}$$

While this guess turns out to be correct, it is much more instructive to think about this (and arrive at this result) in a very different way, which requires no prior non-relativistic knowledge.

Our building blocks are $x^a = x^a(\tau), \dot{x}^a$ etc. for the particle, and A_a, F_{ab} etc. for the Maxwell field, and our aim is to find the simplest action that gives rise to Lorentz and gauge invariant equations of motion (and "simplest" here means lowest number of derivatives, lowest degree polynomial etc.).

Perhaps the simplest candidate for the interaction Lagrangian is $A_a x^a$. This is evidently Lorentz invariant, but equally evidently it will give rise to a contribution $\sim A_a$ to the force, which is not gauge invariant, and hence we discard it.

The next simplest term is $A_a\dot{x}^a$. This is again evidently Lorentz invariant, but what about gauge invariance? Under a gauge transformation $A_a \to A_a + \partial_a\Psi$ we find

$$A_a\dot{x}^a \to A_a\dot{x}^a + (\partial_a\Psi)\dot{x}^a = A_a\dot{x}^a + \frac{d}{d\tau}\Psi \quad . \tag{4.141}$$

Thus, even though $A_a\dot{x}^a$ is not gauge invariant, very cooperatively $A_a\dot{x}^a$ is gauge invariant up to a total derivative. Therefore the action only changes by a boundary term, and since this has no impact on the equations of motion, this is sufficient to ensure gauge invariance of the equations ot motion.

Therefore we postulate the action

$$S_I[x;A] = q\int d\tau\ A_a\dot{x}^a \quad . \tag{4.142}$$

We see that this agrees with the guess (4.140).

It is now straightforward to derive that the Euler-Lagrange equations derived from the action $S_0[x] + S_I[x;A]$ are indeed precisely the relativistic Lorentz force equations (4.134). Let us do this first, and then I will add some more comments on this action.

Since we already know the variation of $S_0[x]$, we just need to determine that of $S_I[x;A]$. For that we use that the variation of the 4-velocity is

$$\delta\dot{x}^a = \frac{d}{d\tau}\delta x^a \quad , \tag{4.143}$$

and that the variation of $A_a(x)$ induced by a variation $x^a \to x^a + \delta x^a$ is

$$\delta A_a = (\partial_b A_a)\delta x^b \quad . \tag{4.144}$$

We will also use

$$\frac{d}{d\tau}A_a = (\partial_b A_a)\dot{x}^b \quad . \tag{4.145}$$

With this we can calculate (using integration by parts and, as usual, dropping the boundary term)

$$\begin{aligned}\delta \int d\tau\ A_a \dot{x}^a &= \int d\tau \left((\partial_b A_a) \delta x^b \dot{x}^a + A_a \delta \dot{x}^a \right) \\ &= \int d\tau \left((\partial_a A_b) \delta x^a \dot{x}^b - \delta x^a \frac{d}{d\tau} A_a \right) \\ &= \int d\tau \left((\partial_a A_b - \partial_b A_a) \delta x^a \dot{x}^b \right) \\ &= \int d\tau\ F_{ab} \delta x^a \dot{x}^b \ .\end{aligned} \tag{4.146}$$

Thus combining this with (4.133) we find

$$\delta(S_0[x] + S_I[x, A]) = \int d\tau \left(-\frac{d}{d\tau} p_a + q F_{ab} \dot{x}^b \right) \delta x^a \tag{4.147}$$

and therefore the Euler-Lagrange equations are precisely the Lorentz force equations (4.134).

Remarks:

1. The rationale for introducing the charge q in front of the action (4.142) is that it is the *coupling constant*, i.e. a measure of the strength of the interaction between the particle and the Maxwell field (in particular, for an uncharged particle, $q = 0$, there is no such interaction).

2. Note that the momenta p_a in the above discussion are the covariant conjugate momenta of the free particle, i.e. $p_a = m u_a$. Because of the velocity dependendence of the interaction Lagrangian, these are not the same as the covariant conjugate momenta P_a associated to the sum of the free and interaction Lagrangian,

$$L = L_0 + L_I \quad \Rightarrow \quad P_a = \frac{\partial L}{\partial \dot{x}^a} = p_a + q A_a \ . \tag{4.148}$$

   The modification of the spatial components is already familiar from non-relativistic mechanics. Thus the quantity of interest is the temporal component

$$P^0 = p^0 + q A^0 = (E + q\phi)/c = (m\gamma(v)c^2 + q\phi)/c \ . \tag{4.149}$$

   This is the total (relativistic kinetic plus electric potential) energy of the particle.

3. The interaction action can be written as just the line integral of $A = A_a dx^a$ over the worldline (curve) C of the particle,

$$S_I[x; A] = q \int d\tau\ A_a \dot{x}^a = q \int_C A_a dx^a \equiv q \int_C A \ . \tag{4.150}$$

   Since one can integrate $A = A_a dx^a$ in a natural way only over 1-dimensional spaces, this makes it clear that the elementary objects that carry electric charge and that A_a can couple to are objects with 1-dimensional worldlines, i.e. particles. For some comments on generalisations of this kind of reasoning to other, more exotic, situations see section 7.1.

4. At this point it is natural to wonder if one can derive not just the Lorentz force equation but also the Maxwell equations themselves from an action principle. This is (of course) indeed the case, but requires an extension of action principles and variational calculus to field theories. This will be the subject of section 5.

5 Classical Lagrangian Field Theory

5.1 Introduction

In mechanics, the dynamical variables are functions of one variable, e.g. the paths $q^a = q^a(t)$ or $x^a = x^a(\tau)$. Maxwell theory, with its electric and magnetic fields $\vec{E}(t,\vec{x})$ and $\vec{B}(t,\vec{x})$ or, more fundamentally, with its potential $A_a(x^b)$, is the prime example of a field theory, i.e. a theory in which the dynamical variables are fields, functions of several space(-time) coordinates.

The modern description of all fundamental interactions of nature is in terms of (quantum) field theories, and the modern approach to constructing such field theories in an efficient manner is via the action principle. Maxwell theory provides us with the prototype of this and teaches us how to describe and introduce interactions, mediated by fields, in a Lorentz invariant manner.

Motivated by this, the first (and modest) aim of this section is to extend the usual variational or Lagrangian formalism of machanics to fields, i.e. to functions of several variables. This turns out to be straightforward.

We will then look concretely at Poincaré invariant action principles for scalar fields, as well as for Maxwell theory, and some variants and combinations thereof. In particular, we will see how to derive the Maxwell equations form an action principle, and how the action, and thus the equations of motion, are essentially determined by gauge invariance and Lorentz invariance.

One significant advantage of the Lagrangian or action based formalism is the availability of Noether's theorem which allows one to explore the consequences of the symmeteries of an action in a systematic and simple way. In particular, we will see how translation invariance leads to the notion of a (conserved) energy-momentum tensor.

5.2 Variational Calculus and Action Principle for Fields

In order to extend the usual variational calculus to field, i.e. to dynamical variables depending on more than one coordinate, we simply make the replacement

$$q^a(t) \quad \to \quad \Phi^A(x^a) \ , \tag{5.1}$$

where the x^a are some space-(time) coordinates, and where the $\Phi^A(x)$ denote a collection of fields or functions, which could be scalar fields, or components of vector fields, or something else. For the time being, and for the purposes of this section, the dimension of space(-time), i.e. the number of independent coordinates, is arbitrary, and thus we consider D-dimensional Euclidean or Minkowski space. We also do not need to be more specific about the precise nature of the fields $\Phi^A(x)$. This will of course change, when we consider concretely Poincaré-invariant actions for fields in $(3+1)$-dimensional Minkowski space, in which case x^a with $a = 0, 1, 2, 3$ are inertial coordinates for Minkowski space, and we will choose the fields $\Phi^A(x)$ to be appropriate Lorentz tensor fields.

Because we now have more than one coordinate, the velocities (ordinary derivatives) of the paths

$q^a(t)$ will be replaced by partial derivatives of the fields,

$$\dot{q}^a(t) \quad \to \quad \partial_a \Phi^A(x) \tag{5.2}$$

etc. The entire replacement procedure is summarised in the table below.

	Mechanics	Field Theory
Independent Variables	time t	space(-time) coordinates x^a $a = 0, \ldots, D-1$ oder $a = 1, \ldots D$
Dynamical Variables	paths $q^i(t)$	fields $\Phi^A(x^a)$ Φ^A: scalar , vector, tensor fields etc.
Derivatives	ordinary derivative $\dot{q}^i(t)$	partial derivatives $\partial_a \Phi^A(x)$
Lagrangian	$L = L(q^i, \dot{q}^i; t)$	$L = L(\Phi^A, \partial_a \Phi^A; x^a)$
Action	$S[q] = \int dt\ L$	$S[\Phi] = \int d^D x\ L$
Variations	$q^i(t) \to q^i(t) + \delta q^i(t)$	$\Phi^A(x) \to \Phi^A(x) + \delta \Phi^A(x)$

In particular, the functionals (actions) that we seek to extremise are now functionals $S[\Phi]$ of the fields Φ^A,

$$S: \quad \{\Phi^A\} \mapsto S[\Phi] \in \mathbb{R} \ , \tag{5.3}$$

and we will only consider local functionals, where local refers to the fact that they are are given by an integral over space(-time) of a Lagrangian function

$$L = L(\Phi^A, \partial_a \Phi^A, \ldots; x^a) \tag{5.4}$$

that depends on the Φ^A and a finite number of derivatives of Φ^A, as well as perhaps also explicitly on the coordinates x^a. We will only consider the case that the Lagrangian depends on the fields and their first partial derivatives, and thus the actions that we consider have the form

$$S[\Phi] = \int d^D x\ L(\Phi^A, \partial_a \Phi^A; x^a) \ . \tag{5.5}$$

Just as in mechanics, in order to determine the extrema or critical points of this action, we consider infinitesimal variations of the fields, i.e.

$$\Phi^A(x) \to \Phi^A(x) + \delta \Phi^A(x) \tag{5.6}$$

with the characteristic property that

$$\delta(\partial_a \Phi^A(x)) = \partial_a(\delta \Phi^A(x)) \ . \tag{5.7}$$

Using only this rule, we can now easily derive the field theory analog of the *Variational Master Equation* (VME) (3.80) derived in section 3.5, and then we can immediately deduce from this the field theory Euler-Lagrangian equations whose solutions extremise the action. As in the case of mechanics, the VME will also provide us with a 1-line proof of the field theory version of the Noether theorem (and we will come back to this in section 6.1 below).

Performing the variation, one obtains

$$\begin{aligned}\delta L &= \frac{\partial L}{\partial \Phi^A(x)}\delta\Phi^A(x) + \frac{\partial L}{\partial(\partial_a\Phi^A(x))}\delta(\partial_a\Phi^A(x)) \\ &= \frac{\partial L}{\partial \Phi^A(x)}\delta\Phi^A(x) + \frac{\partial L}{\partial(\partial_a\Phi^A(x))}\partial_a(\delta\Phi^A(x)) \\ &= \left(\frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a\Phi^A(x))}\right)\delta\Phi^A(x) + \frac{d}{dx^a}\left(\frac{\partial L}{\partial(\partial_a\Phi^A(x))}\delta\Phi^A(x)\right)\end{aligned} \tag{5.8}$$

This is already the field theory VME.

The only thing that may require some explanation here is the meaning of the operator d/dx^a. Just as the total time derivative d/dt acts on both the explicit and implicit dependence of a function of t, as in

$$\frac{d}{dt}F(q(t);t) = \frac{\partial}{\partial t}F(q(t);t) + \dot{q}(t)\frac{\partial}{\partial q(t)}F(q(t);t) \ , \tag{5.9}$$

the total derivative d/dx^a acts on both the explicit and the implicit x-dependence, as in

$$\frac{d}{dx^a}F(\Phi(x);x) = \frac{\partial}{\partial x^a}F(\Phi(x),x) + (\partial_a\Phi(x))\frac{\partial}{\partial\Phi(x)}F(\Phi(x),x) \ . \tag{5.10}$$

At the same time, however, d/dx^a acts as a partial derivative in the sense that the other coordinates are to be held fixed. In equations this means that if we simply consider F as a function of x, say

$$F(\phi(x),x) = G(x) \ , \tag{5.11}$$

then

$$\frac{d}{dx^a}F(\Phi(x);x) = \frac{\partial}{\partial x^a}G(x) \ . \tag{5.12}$$

Either way we have, in particular,

$$\frac{d}{dx^a}\Phi(x) = \frac{\partial}{\partial x^a}\Phi(x) \equiv \partial_a\Phi(x) \tag{5.13}$$

(what else could it be?). We need this total derivative in the VME because it is only the total derivative (which sees the entire x-dependence) that gives us a boundary term upon integration. Often such an implicit identification $F = G$ is made, and then it is not necessary to distinguish notationally the partial and total derivatives (and I will also adopt that in situations where no confusion should arise about what is meant).

We now integrate the VME (5.8) over a D-dimensional domain or volume V with boundary ∂V, and require the variations to vanish on ∂V. Then we find

$$\delta\Phi^A|_{\partial V} = 0 \quad \Rightarrow \quad \delta S[\Phi] = \delta\int_V d^Dx L = \int_V d^Dx\left(\frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a\Phi^A(x))}\right)\delta\Phi^A(x) \tag{5.14}$$

and therefore we obtain the Euler-Lagrange equations (the conditions for a field configuration Φ to extremise the action $S[\Phi]$)

$$\delta S[\Phi] = 0 \quad \forall\, \delta\Phi^A \quad \Leftrightarrow \quad \frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a \Phi^A(x))} = 0 \ . \tag{5.15}$$

Remarks:

1. Sometimes the Euler-Lagrange equations are written as the "variational derivative" (also called the "Euler-Lagrange derivative") of the Lagrangian L with respect to the fields Φ^A, i.e.

$$\frac{\delta L}{\delta \Phi^A}(x) \equiv \frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a \Phi^A(x))} = 0 \ . \tag{5.16}$$

   While fundamentally this does not make too much sense (one can and should think of the Euler-Lagrange equations as the variational derivative of the *action*, when the boundary terms are zero, not of the Lagrangian), it is a common and legitimate abbreviation.

   Note that with this notation,

$$\delta L \neq \frac{\delta L}{\delta \Phi^A}\delta\Phi^A \ . \tag{5.17}$$

   Rather, the VME (5.8) takes the form

$$\delta L = \frac{\delta L}{\delta \Phi^A}\delta\Phi^A + \frac{d}{dx^a}\left(\frac{\partial L}{\partial(\partial_a \Phi^A)}\delta\Phi^A\right) \ . \tag{5.18}$$

2. Another immediate consequence of the VME or the above calculation is that the Euler-Lagrange equations are not changed when one adds a total derivative to the Lagrangian,

$$L(\Phi^A, \partial_a\Phi^A; x) \to L(\Phi^A, \partial_a\Phi^A; x) + \frac{d}{dx^a}W^a(\Phi^A; x) \ . \tag{5.19}$$

   From the point of view of the action principle this is evident because it only changes the action by a boundary term. One can also read this off directly from (5.8), because the total derivative term only contributes to the last term in that identity: since by construction / definition variations commute with total derivatives, one has

$$\delta\left(\frac{d}{dx^a}W^a(\Phi^A; x)\right) = \frac{d}{dx^a}\left(\delta W^a(\Phi^A; x)\right) \ . \tag{5.20}$$

   One can of course also check explicitly that the addition of such a term to the Lagrangian does not change the equations of motion, i.e. that the Euler-Lagrange equations for a Lagrangian that is a total derivative are identically satisfied,

$$L = \frac{d}{dx^a}W^a(\Phi^A; x) \quad \Rightarrow \quad \frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a \Phi^A(x))} = 0 \quad \text{identically} \ . \tag{5.21}$$

   It is left as an exercise to show this.

Examples:

1. Laplace Equation

Consider a function (real scalar field) $\Phi(\vec{x})$ on $\mathbb{R}^3$. The simplest Lagrangian that we can write down that involves derivatives of Φ (otherwise we are not going to obtain any non-trivial Euler-Lagrange equations), and that is invariant under the Euclidean group of rotations and translations is

$$L = \tfrac{1}{2}\vec{\nabla}\Phi.\vec{\nabla}\Phi = \tfrac{1}{2}\partial_i\Phi\partial_i\Phi \ . \tag{5.22}$$

The Euler-Lagrange equations reduce to

$$\frac{d}{dx^k}\frac{\partial L}{\partial(\partial_k\Phi)} = \partial_k\partial_k\Phi = \Delta\Phi = 0 \ , \tag{5.23}$$

i.e. the Laplace equation. In particular, thinking of L as the electrostatic energy-density of the electric field $\vec{E} = -\vec{\nabla}\phi$, we learn that the electrostatic energy is minimised by solutions to the Laplace equation.

2. Schrödinger Equation

Consider a complex scalar field $\Psi(t, \vec{x})$ on $\mathbb{R} \times \mathbb{R}^3$, and the action

$$S[\Psi] = \int dt \int d^3x \left(\frac{i\hbar}{2}(\Psi^*\dot{\Psi} - \Psi\dot{\Psi}^*)) - \frac{\hbar^2}{2m}\vec{\nabla}\Psi^*.\vec{\nabla}\Psi - V(\vec{x})\Psi^*\Psi \right) \tag{5.24}$$

where $\dot{\Psi} = \partial_t\Psi$. Then one finds that this action is extremised by solutions to the Schrödinger equation

$$i\hbar\partial_t\Psi(t, \vec{x}) = (-\frac{\hbar^2}{2m}\Delta + V(\vec{x}))\Psi(t, \vec{x}) \ . \tag{5.25}$$

This calculation is best done once one has understood how to deal with complex scalar fields Φ in an efficient manner (namely that, for variational purposes, one is allowed to pretend that one can vary them and their complex conjugates Φ^* independently). This is something that we will discuss in section 5.4 below. We will then briefly return to this example in the context of the Noether theorem in section 6.1.

5.3 Poincaré-invariant Actions for Real Scalar Fields

We now specialise to $(3 + 1)$-dimensional Minkowski space, with inertial coordinates x^a. Our aim is to construct Poincaré invariant actions for various tensor fields, in particular for real and complex scalar fields, and for the covector field $A_a(x)$ of Maxwell theory (and in the latter case we will of course also require gauge invariance). Since the integration measure d^4x is Lorentz invariant (cf. section 2.10),

$$d^4\bar{x} = |\det(L)|d^4x = d^4x \ , \tag{5.26}$$

an action

$$S[\Phi] = \int d^4x \ L(\Phi^A, \partial_a\Phi^A) \tag{5.27}$$

is Lorentz invariant, provided that the Lagrangian L is a Lorentz scalar, and it is moreover translation (and thus Poincaré) invariant if L does not depend explicitly on the coordinates

x^a. [For now we regard these statements as being obviously true, but we will state this more carefully in section 6.3.]

A remark on terminology: occasionally, what I have referred to simply as the Lagrangian L above is called the *Lagrangian density*, and then one obtains the Lagrangian $\mathcal{L}$ by integrating the Lagrangian density over space (as for any density),

$$\mathcal{L} = \int d^3x \; L \;\; . \tag{5.28}$$

Then, as in particle mechanics, the action is given by integrating the Lagrangian over time t (or x^0),

$$S = \int dt \; \mathcal{L} \stackrel{c=1}{=} \int d^4x \; L \;\; . \tag{5.29}$$

While this terminology is useful for certain purposes, I will not use $\mathcal{L}$ at all and will therefore continue to refer to L (rather than $\mathcal{L}$) as the Lagrangian. The reason for avoiding the use of $\mathcal{L}$ is that it evidently depends on a decomposition of space-time into space and time (a choice of inertial system) and is therefore not Lorentz invariant even if L and S are.

As a warm-up exercise, in this section we start with a single real scalar field $\phi(x)$.

1. Free Massless Real Scalar Field: Wave Equation

   The simplest Lagrangian that we can write down that depends on the derivatives of ϕ and that is a Lorentz scalar is

   $$L = -\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi \;\; . \tag{5.30}$$

   The sign and prefactor have been chosen in such a (conventional) way that the kinetic (time derivative) term enters with a positive sign and with the usual factor of $1/2$,

   $$L = \tfrac{1}{2}(\partial_0\phi)^2 - \tfrac{1}{2}(\vec{\nabla}\phi)^2 \;\; . \tag{5.31}$$

   We can obtain the equations of motion either from the Euler-Lagrange equations,

   $$\frac{\partial L}{\partial \phi} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a\phi)} = \frac{d}{dx^a}(\eta^{ab}\partial_b\phi) = \eta^{ab}\partial_a\partial_b\phi = \Box\phi \tag{5.32}$$

   or directly from variation of the action (dropping boundary terms),

   $$\begin{aligned} \delta S[\phi] &= \delta \int d^4x(-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi) = \int d^4x(-\eta^{ab}\partial_a\phi\partial_b\delta\phi) \\ &= \int d^4x(\eta^{ab}\partial_b\partial_a\phi)\delta\phi = \int d^4x(\Box\phi)\delta\phi \end{aligned} \tag{5.33}$$

   Either way we find that the Euler-Lagrange derivative of L is

   $$\frac{\delta L}{\delta \phi} = \Box\phi \;\; . \tag{5.34}$$

   leading to the wave equation

   $$\Box\phi = 0 \;\; , \tag{5.35}$$

   This is referred to as the field equation for a free massless scalar field in Minkowski space.

Remarks:

(a) "Free" here refers to the fact that the equation is linear. Therefore the sum of two solutions is again a solution, which means that the field does not (self-)interact.

(b) The reason why it is called "massless" is because a basis of solutions of this equation is provided by the plane waves

$$\phi_p(x) = \mathrm{e}^{\,ip_a x^a/\hbar} = \mathrm{e}^{\,ik_a x^a} \quad \text{with} \quad k^a k_a = 0 \ , \tag{5.36}$$

appropriate for a massless particle with lightlike wave 4-vector k^a.

(c) In $(1+1)$ dimensions, one can introduce lightcone coordinates (2.89)

$$x^{\pm} = x^0 \pm x^1 \ . \tag{5.37}$$

In terms of these, the wave equation can be written and completely solved as

$$\Box\phi = 0 \quad \Leftrightarrow \quad \partial_+\partial_-\phi = 0 \quad \Leftrightarrow \quad \phi(x) = \phi^+(x^+) + \phi^-(x^-) \ , \tag{5.38}$$

with ϕ^+ (ϕ^-) corresponding to left (respectively right) moving waves.

2. Free Massive Real Scalar Field: Klein-Gordon Equation

The Klein-Gordon equation is the equation

$$(\Box - m^2)\phi = 0 \ . \tag{5.39}$$

This is still a linear equation, but now it contains what is known as a "mass term" $m^2\phi$ (the rationale for this terminology will be explained below), and hence this equation describes a free massive scalar field. It is easy to see that this can be derived from the action

$$S[\phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi - \tfrac{1}{2}m^2\phi^2\right) \tag{5.40}$$

(just like a linear harmonic oscillator force requires a quadratic potential).

Remarks:

(a) In writing the Klein-Gordon equation, I have adopted the particle physics convention to work in units where $\hbar = c = 1$. To make this equation dimensionally correct, with m a mass, one should replace

$$m^2 \to \frac{m^2c^2}{\hbar^2} \ . \tag{5.41}$$

(b) With this replacement, it is easy to see that a plane wave

$$\phi_p(x) = \mathrm{e}^{\,-iEt/\hbar + i\vec{p}.\vec{x}/\hbar} = \mathrm{e}^{\,ip_a x^a/\hbar} \tag{5.42}$$

will solve the Klein-Gordon equation when

$$E^2 = m^2c^4 + \vec{p}^2c^2 \tag{5.43}$$

which is precisely the mass shell condition (3.24) for a massive relativistic particle

$$p^a p_a = -m^2c^2 \ . \tag{5.44}$$

(c) Conversely, the Klein-Gordon operator $\Box - m^2c^2/\hbar^2$ can formally be obtained by "quantising" the mass shell relation, i.e. by replacing

$$(E \to i\hbar\partial_t, \vec{p} \to -i\hbar\vec{\nabla}) \quad \Leftrightarrow \quad p_a \to -i\hbar\partial_a \ . \tag{5.45}$$

Indeed, with this replacement

$$p^a p_a + m^2c^2 \to -\hbar^2(\Box - m^2c^2/\hbar^2) \ . \tag{5.46}$$

This may give the (mistaken!) impression that somehow the Klein-Gordon field ϕ is a quantum wave function of a massive relativistic particle. This is not true, but has historically caused quite some confusion. In a course on quantum field theory (QFT), one of the first things you will learn is how to correctly think of the Klein-Gordon field (namely as a classical field that itself needs to be promoted to an operator, among other things).

(d) If elsewhere you encounter the Klein-Gordon equation with the opposite relative sign between $\Box$ and m^2, then don't worry, it does not mean imaginary masses: it will simply be due the opposite sign convention $(\eta_{ab}) = \mathrm{diag}(+1, -1, -1, -1)$ for the Minkowski metric that is being used there (and most particle physics and quantum field theory practitioners use that convention).

3. Real Scalar Field with Self-Interaction

It is now also obvious how to include self-interactions of the scalar field: to that end one should add a potential that is not just a quadratic function of ϕ but e.g. a higher degree polynomial,

$$S[\phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi - V(\phi)\right) \ . \tag{5.47}$$

In order to deduce the equations of motion, we can either observe that

$$\delta V(\phi) = \frac{\partial V}{\partial\phi}\delta\phi \equiv V'(\phi)\delta\phi \ , \tag{5.48}$$

or we use

$$\frac{\partial L}{\partial\phi} = -\frac{\partial V}{\partial\phi} = -V'(\phi) \tag{5.49}$$

to conclude that the field equation is

$$\Box\phi = V'(\phi) \ . \tag{5.50}$$

Remarks:

(a) In particular, for $V(\phi) = m^2\phi^2/2$ one reproduces the Klein-Gordon equation.

(b) One interesting and non-trivial example is the quartic potential

$$V(\phi) = \frac{\lambda}{2}(\phi^2 - a^2)^2 \geq 0 \ , \tag{5.51}$$

depending on two real parameters λ and a. This potential is even, i.e. invariant under $\phi \to -\phi$. Since the derivative term in the Lagrangian also evidently has this symmetry, the entire Lagrangian has the discrete $\mathbb{Z}_2$ reflection symmetry $\phi \to -\phi$.

The two lowest energy solutions (ground states or vacua in QFT terminology) are the constant solutions

$$\phi^{\pm} = \pm a \ . \tag{5.52}$$

These are not invariant under (but exchanged by) the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$. This is a simple example of the phenomenon of *spontaneous symmetry breaking* (the ground state does not have all the symmetries of the theory).

(c) A famous and much studied non-linear equation in $(1+1)$-dimensions is the equation resulting from the potential

$$V(\phi) = m^2(1 - \cos\phi) \geq 0 \ . \tag{5.53}$$

Since

$$V(\phi) = \frac{m^2}{2}\phi^2 - \frac{m^2}{4!}\phi^4 + \dots \ , \tag{5.54}$$

this describes a massive sccalar field with self-interactions. The field equation is

$$\Box\phi = m^2 \sin\phi \tag{5.55}$$

and therefore this equation is commonly and unfortunately (physicists seem to love puns but are generally not very good at them) known as the *Sine-Gordon Equation*. Evidently the ground states of this theory are the constant solutions with $V(\phi) = 0$, i.e.

$$\phi = 0 \quad , \quad \phi = 2\pi \quad , \dots \tag{5.56}$$

Much more interesting is the fact that there are also so-called *solitonic* solutions to these equations which interpolate between different (but adjacent) vacua at $x^1 = \pm\infty$. A particular example is the time-independent solution

$$\phi(x) = 4\arctan\left(\mathrm{e}^{mx^1}\right) \ , \tag{5.57}$$

which (for a particular branch of the inverse tangent) interpolates between $\phi = 0$ at $x^1 = -\infty$ and $\phi = 4(\pi/2) = 2\pi$ at $x^1 = +\infty$. It is fun to verify explicitly that this is indeed a solution of (5.55).

Since the theory is Lorentz invariant, there also exist time-dependent solutions moving with constant velocity v, which are obtained by applying a boost to the above equation,

$$\phi(x) = 4\arctan\left(\mathrm{e}^{m\gamma(v)(x^1 - \beta(v)x^0)}\right) \ . \tag{5.58}$$

Things get really interesting when it comes to multi-soliton solutions, which show that solitons behave much like particles with elastic collisions, but this is not something I will get into here (I have already led us too far astray with these remarks).

4. Mulitple Real Scalar Fields

All of this is of course easily generalised to the case of multiple scalar fields $\phi^A(x)$, e.g. with action

$$S[\phi] = \int d^4x \left(-\tfrac{1}{2}\sum_A \eta^{ab}\partial_a\phi^A\partial_b\phi^A - V(\phi^A) \right) \ , \tag{5.59}$$

(but terms of the form $\eta^{ab}\partial_a\phi^A\partial_b\phi^B$ with $A \neq B$ are also Lorentz invariant and would hence also be allowed). The equations of motion are now evidently (varying independently the fields ϕ^A)

$$\Box\phi^A = \frac{\partial V}{\partial \phi^A} \quad . \tag{5.60}$$

5.4 Actions and Variations for Complex Scalar Fields

We now briefly consider a complex (i.e. complex valued) scalar field $\Phi(x)$. Since one can decompose such a complex scalar field into its real and imaginary parts,

$$\Phi(x) = \phi_1(x) + i\phi_2(x) \quad , \quad \Phi^*(x) = \phi_1(x) - i\phi_2(x) \quad , \tag{5.61}$$

with ϕ_1, ϕ_2 two real scalar fields, in principle we already know how to deal with this situation. Nevertheless, it is useful to know how to deal directly with the complex fields, without having to invoke the above decomposition.

Even though we have a complex scalar field, we want our action to be real. Thus for the derivative term in the Lagrangian we choose

$$L = -\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* + \dots \tag{5.62}$$

and we simply add a real potential $W(\Phi, \Phi^*)$ to arrive at the action

$$S[\Phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* - W(\Phi, \Phi^*)\right) \quad . \tag{5.63}$$

In order to determine the equations of motion, we first use the decomposition into real and imaginary parts, and then at the end reassemble the results into equations for the complex field Φ and Φ^*. We will then see that there is a shortcut to the result, which does not require this decomposition. I will phrase this procedure as an annotated exercise - you should fill in the missing details.

1. First of all, when writing the action in terms of ϕ_1, ϕ_2, we write the potential as

$$W(\Phi, \Phi^*) = V(\phi_1, \phi_2) \equiv V(\phi_A) \quad . \tag{5.64}$$

   Then the action becomes

$$S[\Phi] = \int d^4x \left(-\tfrac{1}{2}\sum_{A=1}^{2}\eta^{ab}\partial_a\phi_A\partial_b\phi_A - V(\phi_A)\right) \quad . \tag{5.65}$$

2. By the results of the previous section, the equations of motion for the ϕ_A are

$$\Box\phi_A = \frac{\partial V}{\partial \phi_A} \quad . \tag{5.66}$$

   Using identities like

$$\frac{\partial V}{\partial \phi_1} = \frac{\partial W}{\partial \Phi}\frac{\partial \Phi}{\partial \phi_1} + \frac{\partial W}{\partial \Phi^*}\frac{\partial \Phi^*}{\partial \phi_1} = \frac{\partial W}{\partial \Phi} + \frac{\partial W}{\partial \Phi^*} \tag{5.67}$$

   these equations can equivalently be written as

$$\Box\Phi = 2\frac{\partial W}{\partial \Phi^*} \quad , \quad \Box\Phi^* = 2\frac{\partial W}{\partial \Phi} \tag{5.68}$$

3. We now see that these equations also follow directly from the original action (5.63) if we formally treat the variations $\delta\Phi$ and $\delta\Phi^*$ as independent (rather than complex conjugate) variations. For example, if we only vary Φ^* in (5.63), we get (upon the standard integration by parts etc.)

$$\begin{aligned}\delta S[\Phi] &= \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\delta\Phi^* - \frac{\partial W}{\partial\Phi^*}\delta\Phi^* \right) \\ &= \int d^4x \left(\tfrac{1}{2}\Box\Phi - \frac{\partial W}{\partial\Phi^*} \right) \delta\Phi^*\end{aligned} \tag{5.69}$$

and we directly obtain the first of the equations in (5.68). Analogously for variations $\delta\Phi$.

Remarks:

1. Using this shortcut procedure, it is now also straightforward to see that the action (5.24) gives rise to the Schrödinger equation (5.25).

2. Instead of decomposing the complex scalar field into real and imaginary parts, one can also perform a polar decomposition

$$\Phi(x) = \rho(x)\mathrm{e}^{i\varphi(x)} \tag{5.70}$$

with ρ and φ real and φ defined modulo 2π. In terms of these fields, the kinetic term takes the (polar coordinate) form

$$-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* = -\tfrac{1}{2}((\partial\rho)^2 + \rho^2(\partial\varphi)^2) \tag{5.71}$$

where $(\partial\rho)^2$ is short for

$$(\partial\rho)^2 = \partial^a\rho\partial_a\rho = \eta^{ab}\partial_a\rho\partial_b\rho \tag{5.72}$$

etc.

3. When the potential is of the special form

$$W(\Phi,\Phi^*) = W(\Phi\Phi^*) \ , \tag{5.73}$$

the entire Lagrangian is manifestly invariant under the phase transformation

$$\Phi(x) \to \mathrm{e}^{i\theta}\Phi(x) \quad , \quad \Phi^*(x) \to \mathrm{e}^{-i\theta}\Phi^*(x) \ , \tag{5.74}$$

where θ is a constant real parameter. We will come back to this later, in our discussion of the Noether theorem (section 6.1) and in the context of gauging this symmetry and what is known as minimal coupling (section 6.2).

4. Moreover, when W is of this special form, in terms of the polar decomposition (5.70) the potential depends only on ρ and not on φ since

$$\Phi^*\Phi = \rho^2 \ . \tag{5.75}$$

5. One example of such a potential is a mass term,

$$W = \tfrac{1}{2}m^2\Phi^*\Phi \ , \tag{5.76}$$

leading to the Klein-Gordon equation for a complex scalar,

$$(\Box - m^2)\Phi = 0 \quad , \quad (\Box - m^2)\Phi^* = 0 \ . \tag{5.77}$$

6. Another prominent and important example is the complex version of the quartic potential (5.51), namely

$$W = \frac{\lambda}{2}(\Phi^*\Phi - a^2)^2 \geq 0 \ . \tag{5.78}$$

In this case, the ground states are the constant fields with $|\Phi| = a$. There is thus a 1-parameter family of them, labelled by a constant angle α,

$$\Phi_\alpha = a\mathrm{e}^{\,i\alpha} \ . \tag{5.79}$$

These are mapped into each other by the phase transformation (5.74),

$$\Phi_\alpha \to \Phi_{\alpha+\theta} \ , \tag{5.80}$$

but every ground state individually "spontaneouly" completely breaks this phase transformation symmetry.

5.5 Action for Maxwell Theory

We now come to the heart of the matter, namely the construction of an action principle for Maxwell theory. We will at first consider the case that there is no electric 4-current, $J^a = 0$, so that the Maxwell equations are simply $\partial_a F^{ab} = 0$.

Our Lorentz tensorial building blocks are A_a and F_{ab}, and we want to construct a gauge and Lorentz invariant Lagrangian

$$L = L(A_a, \partial_b A_a) \ . \tag{5.81}$$

We have already essentially solved this problem in section 4.8. The unique solution depending at most on first derivatives of A_a is a linear combination of the two invariants I_1 and I_2,

$$L = a_1 I_1 + a_2 I_2 = \frac{a_1}{4}F_{ab}F^{ab} + \frac{a_2}{4}F_{ab}\tilde{F}^{ab} \ . \tag{5.82}$$

Moreover, we had seen in that section that I_2 is actually a total derivative, so its variation would give no contribution to the equations of motion. Thus for the purposes of obtaining the classical equations of motion, we may as well set $a_2 = 0$, and thus we are left with $a_1 I_1$. [In the quantum theory, not just the equations of motion but the value of the action matters (think of the path integral), and therefore in that case the choice of a_2 can (and does!) play a role.]

The conventional normalisation for the Lagrangian corresponds to $a_1 = -1$,

$$L = -\tfrac{1}{4}F_{ab}F^{ab} \tag{5.83}$$

as this gives the same normalisation for the kinetic (time derivative) term as for a scalar field, namely

$$L = -\tfrac{1}{2}(\vec{B}^2 - \vec{E}^2/c^2) = \tfrac{1}{2}(\partial_0 \vec{A})^2 + \dots \tag{5.84}$$

Thus our candidate action for Maxwell theory is

$$S_0[A] = \int d^4x \left(-\tfrac{1}{4} F_{ab}F^{ab}\right) \ . \tag{5.85}$$

Does this give the Maxwell equations? Indeed it does. When we vary A_a in $F_{ab}F^{ab}$, a priori we get 4 terms, from the 4 appearances of A_a in

$$F_{ab}F^{ab} = \eta^{ac}\eta^{bd}(\partial_a A_b - \partial_b A_a)(\partial_c A_d - \partial_d A_c) \ , \tag{5.86}$$

but it is easy to see that all 4 terms are identical, and therefore

$$\delta(F_{ab}F^{ab}) = 4\eta^{ac}\eta^{bd}(\partial_a \delta A_b)(\partial_c A_d - \partial_d A_c) = 4(\partial_a \delta A_b)F^{ab} \ . \tag{5.87}$$

Therefore (with the usual integration by parts)

$$\delta S_0[A] = \int d^4x(-\partial_a \delta A_b)F^{ab} = \int d^4x(\partial_a F^{ab})\delta A_b \ . \tag{5.88}$$

and hence we obtain the vacuum Maxwell equations

$$\delta S_0[A] = 0 \quad \forall\, \delta A \quad \Rightarrow \quad \partial_a F^{ab} = 0 \ . \tag{5.89}$$

This would have been the modern and efficient way to "discover" the Maxwell equations, if we had not already known them: given the fields $A_a(x)$ and the requirements of gauge invariance and Lorentz invariance, the simplest possible action that satisfies these criteria gives rise to the Maxwell equations,

$$\text{Gauge Invariance} \quad \oplus \quad \text{Lorentz Invariance} \quad \Rightarrow \quad \text{Maxwell Theory} \tag{5.90}$$

Now let us include the current J^a. It should by now be evident that such a contribution to the equations of motion

$$\partial_a F^{ab} + J^b = 0 \tag{5.91}$$

will result from the (Lorentz invariant) coupling $A_b J^b$ of the gauge field to the 4-current,

$$\delta(S_0[A] + \int d^4x A_b J^b) = \int d^4x(\partial_a F^{ab} + J^b)\delta A_b \quad \Rightarrow \quad \partial_a F^{ab} + J^b = 0 \ . \tag{5.92}$$

Remarks:

1. In the same spirit as in our discussion of an action principle for the Lorentz force in section 4.12, we can think of this additional contribution to the action as the interaction term

$$S_I[A; J] = \int d^4x A_a J^a \tag{5.93}$$

which describes the coupling between the gauge fields and the electric 4-current. This is the generalisation of the interaction term (4.142)

$$S_I[x;A] = q\int d\tau\ A_a\dot{x}^a \quad , \tag{5.94}$$

for a particle coupled to the Maxwell field, to which it reduces when the current is simply the 1-dimensional (δ-function supported) current produced by a charged particle along its worldline.

2. As in the case of a charged particle, it remains to analyse the gauge invariance of $S_I[A,J]$ (the action $S_0[A]$ is manifestly gauge invariant). Under a gauge transformation $A_a \to A_a + \partial_a\Psi$ one has

$$\int d^4x A_a J^a \to \int d^4x A_a J^a + \int d^4x (\partial_a\Psi) J^a \quad . \tag{5.95}$$

We can write the second term as

$$\int d^4x (\partial_a\Psi) J^a = \int d^4x\ \partial_a(\Psi J^a) - \int d^4x\ \Psi\partial_a J^a \quad . \tag{5.96}$$

The first term on the right-hand side is a total derivative and hence a boundary term. Depending on the boundary conditions one imposes on J^a or Ψ, this term may or may not be zero, but regardless of this this term is no obstacle to the gauge invariance of the equations of motion.

However, a priori the second term on the right-hand side (which is not a boundary term) appears to be an obstacle to gauge invariance, and if this term is to vanish for all Ψ we need

$$\text{Gauge Invariance (up to boundary terms)} \quad \Rightarrow \quad \partial_a J^a \stackrel{!}{=} 0 \quad . \tag{5.97}$$

Of course we already know that the Maxwell equations imply this 4-current conservation law anyway,

$$\partial_a F^{ab} = -J^b \quad \Rightarrow \quad \partial_b J^b = 0 \quad . \tag{5.98}$$

However, here we have arrived at a somewhat stronger statement because we have derived this condition without using the Maxwell equations, just from the requirement of gauge invariance: a non-conserved current cannot be coupled in a gauge invariant way to a gauge field $A_a(x)$.

3. For the time being, the current (source) has been introduced purely phenomenologically. Whatever microscopic matter the electric current is actually built from, one would expect such a current to be conserved only by virtue of the matter equations of motion. We therefore need to introduce dynamics for the matter fields and couple them in a suitable way to the Maxwell field.

How this is accomplished, and how the coupling of matter to Maxwell theory is related to gauge invariance of the matter theory will be explained in section 6.2 below. [While thematically it would make sense to do this right here and now, both conceptually and calculationally it turns out to be slightly more convenient to do this after having explored

the consequences of global (phase) symmetries via Noether's theorem (section 6.1).] This will also allow us to sharpen somewhat (and make more precise) the statement made above regarding the relation between gauge invariance and charge (current) conservation.

6 Symmetries and Lagrangian Field Theories

6.1 Noether's 1st Theorem: Global Symmetries and Conserved Currents

We now return to the general setting of section 5.2, in particular to the VME (5.8)

$$\delta L = \left(\frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a} \frac{\partial L}{\partial(\partial_a \Phi^A(x))} \right) \delta\Phi^A(x) + \frac{d}{dx^a} \left(\frac{\partial L}{\partial(\partial_a \Phi^A(x))} \delta\Phi^A(x) \right) \tag{6.1}$$

and we proceed as in section 3.5 to deduce from this Noether's 1st theorem for Lagrangian field theories. Thus let $\Delta\Phi^A$ be a variation of the fields, that leaves the Lagrangian invariant up to a total derivative,

$$\Delta L = \frac{d}{dx^a} F^a_\Delta(\Phi^A, x) \tag{6.2}$$

(in the context of mechanics, we denoted such a variation by δ_s, but let us use the notation Δ here to slightly unburden the notation). Then evidently the current

$$J^a_\Delta = \frac{\partial L}{\partial(\partial_a \Phi^A(x))} \Delta\Phi^A(x) - F^a_\Delta \tag{6.3}$$

is conserved for any solution to the Euler-Lagrange field equations,

$$\frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a} \frac{\partial L}{\partial(\partial_a \Phi^A(x))} = 0 \quad \Rightarrow \quad \frac{d}{dx^a} J^a_\Delta = 0 \ . \tag{6.4}$$

This is already Noether's 1st theorem for field theories.

Remarks:

1. Note that nowhere in the above did we ever consider variations of the coordinates x^a, only variations of the fields $\Phi^A(x)$. This is unsurprising for certain kinds of symmetries, e.g. the phase invariance (5.74) of the action of a complex scalar field (with a suitable potential). Such symmetry transformations which are not related to any transformations of the spacetime coordinates are usually referred to as *internal symmetries*.

   However, there are of course also symmetries related to transformations of the spacetime coordinates, and so far we have thought of such *spacetime symmetries* like translations or Lorentz transformations as being associated with explicit transformations of the coordinates. However, this is neither necessary nor useful in the context of the Noether theorem, and I will explain in section 6.3 below how we will deal with such spacetime symmetries.

2. We just derived that in the field theory case Noether's theorem gives us not (or not directly) conserved charges but conserved currents. However, if we now specialise to Minkowski space, we can of course in the standard way (and with suitable asymptotic conditions) construct a conserved charge from a conserved current. In the following we drop the subscript Δ, i.e. we simply write $J^a_\Delta = J^a$, both for simplicity and because these considerations apply to an arbitrary conserved current, Noether or not.

   Thus we define the charge at time t to be the integral of the (charge) density

$$\rho \equiv J^0/c \tag{6.5}$$

over the 3-dimensional hypersurface Σ_t of constant t,

$$Q(t) = \int_{\Sigma_t} d^3x\ \rho\ . \tag{6.6}$$

Here are two proofs that the $Q(t)$ defined in this way is actually independent of t, and thus "conserved", provided that the spatial currents vanish at spatial infinity.

(a) The non-covariant argument (familiar from first year undergraduate physics: how to get integral conservation laws from differential conservation laws) uses the conservation law

$$\partial_a J^a = 0 \quad \Leftrightarrow \quad \partial_t \rho + \vec{\nabla}.\vec{J} = 0 \tag{6.7}$$

and Gauss' theorem to conclude that

$$\partial_t Q(t) = \int_{\Sigma_t} d^3x\ \partial_t \rho = -\int_{\Sigma_t} d^3x\ \vec{\nabla}.\vec{J} = -\oint_{S^2_\infty} d^2x\ \vec{n}.\vec{J} = 0\ . \tag{6.8}$$

Here S^2_∞ is the two-sphere "at infinity", $\vec{n}$ its normalised normal vector, and hence we get a conserved charge provided that there is no normal component of the current there.

(b) For the covariant version of this argument, we integrate $\partial_a J^a$ over a 4-dimensional volume V bounded by 2 spacelike hypersurfaces Σ_t at $t = t_1$ and $t = t_2$, and a timelike surface S "at infinity". Since $\partial_a J^a$ is a total derivative, its integral will be a boundary term, i.e. an integral over the boundary ∂V of V. Taking into account the opposite orientation of the 2 spacelike hypersurfaces (if the normal vector is inward pointing at $t = t_1 < t_2$, say, then with the same orientation at $t = t_2$ it would be outward pointing there, this boundary is

$$\partial V = \Sigma_{t_2} \cup (-\Sigma_{t_1}) \cup S\ . \tag{6.9}$$

Therefore we conclude from $\partial_a J^a = 0$ that

$$0 = \int d^4x\ \partial_a J^a = \int_{\Sigma_{t_2}} d^3x J^0 - \int_{\Sigma_{t_1}} d^3x J^0 + \quad \text{contributions from } S\ . \tag{6.10}$$

If there are no contributions from S, we conclude

$$\partial_a J^a = 0 \quad \Rightarrow \quad Q(t_2) = Q(t_1)\ , \tag{6.11}$$

which is another way of saying that Q is conserved.

3. There is an inherent ambiguity in extracting the conserved current J^a_Δ from the Noether theorem, not just regarding its sign and overall normalisation, as one can always add an identically conserved term $I^a(x)$ (constructed from the fields $\Phi^A(x)$ and their derivatives) to J^a_Δ. By identically conserved I mean that it satisfies

$$\partial_a I^a(x) = 0 \quad \text{identically}\ , \tag{6.12}$$

without use of the equations of motion. A simple way to construct such identically conserved terms is

$$I^a(x) = \partial_b U^{ab}(x) \quad \text{with} \quad U^{ab}(x) = -U^{ba}(x) \quad \Rightarrow \quad \partial_a I^a(x) = 0 \quad \text{identically}\ . \tag{6.13}$$

Then one has

$$\frac{d}{dx^a}(J^a_\Delta + I^a) = \frac{d}{dx^a}(J^a_\Delta) \quad , \tag{6.14}$$

which now vanishes for a solution to the equations of motion. While this changes the current in what appears to be a quite arbitrary way, the charge density only changes by a spatial total derivative,

$$J^0_\Delta \to J^0_\Delta + I^0 = J^0_\Delta + \partial_i U^{0i} \quad . \tag{6.15}$$

Therefore, while this arbitrariness in the definition changes what one means by the local charge density, it has no influence on the total charge provided that U^{0i} is chosen to fall off sufficiently fast at spatial infinity. In many situations, additional physical criteria (symmetries and gauge invariance etc.) can be used to select a preferred definition of the Noether current. We will see an example of this in the context of the Maxwell energy-momentum tensor in section 6.6.

EXAMPLES:

1. Complex Relativistic Scalar Field with a Phase-invariant Potential

   For our first example, we return to the complex scalar field action (5.63) with a potential of the form $W(\Phi\Phi^*)$ (5.73),

   $$S[\Phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* - W(\Phi\Phi^*)\right) \quad . \tag{6.16}$$

   This action is invariant under the phase transformations (5.74)

   $$\Phi(x) \to \mathrm{e}^{\,i\theta}\Phi(x) \quad , \quad \Phi^*(x) \to \mathrm{e}^{\,-i\theta}\Phi^*(x) \quad , \tag{6.17}$$

   where θ is a constant real parameter. Infinitesimally this is the statement

   $$\Delta\Phi = i\alpha\Phi \quad , \quad \Delta\Phi^* = -i\alpha\Phi^* \quad \Rightarrow \quad \Delta L = 0 \quad , \tag{6.18}$$

   with α infinitesimal. We are thus in a position to apply the Noether theorem, and we can now construct the Noether current and check explicitly that it is indeed conserved. Varying Φ and Φ^* independently, one finds

   $$J^a_\Delta = \frac{\partial L}{\partial(\partial_a\Phi)}\Delta\Phi + \frac{\partial L}{\partial(\partial_a\Phi^*)}\Delta\Phi^* = -(i\alpha/2)(\Phi\partial^a\Phi^* - \Phi^*\partial^a\Phi) \tag{6.19}$$

   Calculating its divergence, one finds (ignoring the irrelevant constant prefactor)

   $$\begin{aligned}\partial_a(\Phi\partial^a\Phi^* - \Phi^*\partial^a\Phi) &= \partial_a\Phi\partial^a\Phi^* + \Phi\Box\Phi^* - \partial_a\Phi^*\partial^a\Phi - \Phi^*\Box\Phi \\ &= \Phi\Box\Phi^* - \Phi^*\Box\Phi = 2(\Phi\partial W/\partial\Phi - \Phi^*\partial W/\partial\Phi^*)\end{aligned} \tag{6.20}$$

   where we already used the equations of motion (5.68),

   $$\Box\Phi = 2\frac{\partial W}{\partial\Phi^*} \quad , \quad \Box\Phi^* = 2\frac{\partial W}{\partial\Phi} \quad . \tag{6.21}$$

   This is not (and should not be) zero in general, but it is zero precisely when W has the special form that makes the action invariant under phase transformations, namely $W = W(\Phi^*\Phi)$. Indeed, in that case one has

   $$\partial W(\Phi^*\Phi)/\partial\Phi = W'(\Phi^*\Phi)\Phi^* \quad , \quad \partial W(\Phi^*\Phi)/\partial\Phi^* = W'(\Phi^*\Phi)\Phi \quad , \tag{6.22}$$

and therefore

$$\Phi\partial W/\partial\Phi - \Phi^*\partial W/\partial\Phi^* = W'(\Phi^*\Phi)\,(\Phi\Phi^* - \Phi^*\Phi) = 0 \quad . \tag{6.23}$$

2. Schrödinger Action

   The Schrödinger action (5.24)

$$S[\Psi] = \int dt \int d^3x \left(\frac{i\hbar}{2}(\Psi^*\dot{\Psi} - \Psi\dot{\Psi}^*)) - \frac{\hbar^2}{2m}\vec{\nabla}\Psi^*.\vec{\nabla}\Psi - V(\vec{x})\Psi^*\Psi \right) \tag{6.24}$$

   is also manifestly invariant under phase transformations

$$\Psi(t,\vec{x}) \to \mathrm{e}^{\,i\theta}\Psi(t,\vec{x}) \tag{6.25}$$

   (in agreement with the fact that these are physically equivalent states of a quantum system). On the other hand, it is also well known that in quantum mechanics there is a probability current with

$$\rho = \Psi^*\Psi \quad , \quad \vec{J} = \frac{\hbar}{2mi}(\Psi^*\vec{\nabla}\Psi - \Psi\vec{\nabla}\Psi^*) \tag{6.26}$$

   which is conerved for a solution to the Schrödinger equation,

$$i\hbar\partial_t\Psi(t,\vec{x}) = (-\frac{\hbar^2}{2m}\Delta + V(\vec{x}))\Psi(t,\vec{x}) \quad \Rightarrow \quad \partial_t\rho + \vec{\nabla}.\vec{J} = 0 \quad . \tag{6.27}$$

   The Noether theorem provides a charming link betweeen these two facts, since the Noether current associated to the invariance of the action under phase transformations is precisely the probablility current (as is readily verified).

6.2 Gauge Invariance and Minimal Coupling

As we saw, the complex scalar field action (6.16) with a potential of the form $W(\Phi\Phi^*)$ (5.73),

$$S[\Phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* - W(\Phi\Phi^*)\right) \quad . \tag{6.28}$$

is invariant under the phase transformations (6.17)

$$\Phi(x) \to \mathrm{e}^{\,i\theta}\Phi(x) \quad , \quad \Phi^*(x) \to \mathrm{e}^{\,-i\theta}\Phi^*(x) \quad , \tag{6.29}$$

where θ is a constant real parameter. Moreover, in section 6.1 we looked at this from the point of view of the Noether theorem and determined the corresponding conserved Noether current.

These phase transformations form an Abelian group,

$$\mathrm{e}^{\,i\theta_1}\mathrm{e}^{\,i\theta_2} = \mathrm{e}^{\,i(\theta_1+\theta_2)} = \mathrm{e}^{\,i\theta_2}\mathrm{e}^{\,i\theta_1} \quad . \tag{6.30}$$

While you can of course think of this as the group of 2-dimensional rotations, in the present (complex) context it is better to think of it as the group $U(1)$ of 1-dimensional unitary transformations. Thus we can say that the model we are considering has a global $U(1)$-symmetry, where "global" refers to the fact that the parameter θ is constant, i.e. independent of x.

The potential is also invariant under *local* (i.e. x-dependent) phase transformations

$$\Phi(x) \to \mathrm{e}^{\,i\theta(x)}\Phi(x) \quad , \quad \Phi^*(x) \to \mathrm{e}^{\,-i\theta(x)}\Phi^*(x) \ , \tag{6.31}$$

but the kinetic (derivative) term is not, because the partial derivatives do now not just transform with a phase, but also involve $\partial_a\theta$,

$$\partial_a\Phi \to \partial_a(\mathrm{e}^{\,i\theta}\Phi) = \mathrm{e}^{\,i\theta}(\partial_a\Phi + i(\partial_a\theta)\Phi) \ . \tag{6.32}$$

If, for whatever reasons, one wants to construct a theory that is invariant under local $U(1)$ transformations, in order to compensate the second term one needs to introduce a new field whose transformation behaviour under these transformations cancels this term. I.e. we need a field that transforms with $\partial_a\theta$ under such transformations. But we already know a field that has such a characteristic and unusual transformation behaviour under x-dependent transformations, namely the Maxwell gauge field $A_a(x)$,

$$A_a(x) \to A_a(x) + \partial_a\theta(x) \ . \tag{6.33}$$

Under the simultaneous transformations (6.31) and (6.33), the linear combination $\partial_a\Phi - iA_a\Phi$ transforms as

$$\partial_a\Phi - iA_a\Phi \to \mathrm{e}^{\,i\theta}(\partial_a\Phi + i(\partial_a\theta)\Phi) - i\mathrm{e}^{\,i\theta}(A_a + \partial_a\theta)\Phi = \mathrm{e}^{\,i\theta}(\partial_a\Phi - iA_a\Phi) \ . \tag{6.34}$$

We see that the derivative term $\partial_a\theta$ has indeed cancelled and that this particular linear combination transforms nicely (covariantly) under these local $U(1)$ transformations. We are thus led to introduce the (gauge) *covariant* derivative of Φ through

$$D_a\Phi = \partial_a\Phi - iA_a\Phi \quad , \quad D_a\Phi^* = (D_a\Phi)^* = \partial_a\Phi^* + iA_a\Phi^* \ . \tag{6.35}$$

Under the joint transformations of Φ and A,

$$\Phi(x) \to \mathrm{e}^{\,i\theta(x)}\Phi(x) \quad , \quad A_a(x) \to A_a(x) + \partial_a\theta(x) \ , \tag{6.36}$$

which we will now collectively refer to as the $U(1)$ gauge transformations of Φ and A, these transform covariantly, i.e. just like Φ and Φ^* themselves,

$$D_a\Phi \to \mathrm{e}^{\,i\theta}D_a\Phi \quad , \quad D_a\Phi^* \to \mathrm{e}^{\,-i\theta}D_a\Phi^* \ . \tag{6.37}$$

Therefore

$$\eta^{ab}D_a\Phi D_b\Phi^* \to \eta^{ab}D_a\Phi D_b\Phi^* \tag{6.38}$$

is gauge invariant, and we can write down a gauge invariant action

$$S[\Phi;A] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}D_a\Phi D_b\Phi^* - W(\Phi\Phi^*)\right) \ , \tag{6.39}$$

where gauge invariant means

$$S[\Phi;A_a] = S[\mathrm{e}^{\,i\theta}\Phi;A_a + \partial_a\theta] \tag{6.40}$$

for all $\theta(x)$.

Remarks:

1. We see that the introduction of a gauge field has allowed us to gauge (make local) the global $U(1)$-symmetry. This provides an answer to the question "what is a gauge field good for?" or "why do we need gauge fields?".

2. The requirement of gauge invariance has thus introduced a coupling of the scalar (matter) field to the Maxwell field. The way this gauge invariance and coupling is obtained is by the replacement
$$\partial_a \to D_a = \partial_a - iA_a \quad . \tag{6.41}$$
In a sense this is the simplest (minimal) way to achieve this goal, and therefore this prescription, in particular the replacement of ordinary by covariant derivatives, is known as *minimal coupling*.

3. In our world, the elementary electrically charged particles (electrons) are not described by a bosonic spin 0 scalar field, but by a fermionic spin 1/2 spinor field, but the principle (of minimal coupling etc.) is the same.

4. In the above action, the gauge field is not a dynamical field but simply a fixed background gauge field the scalar field is coupled to. However, we can easily rectify this, and provide the gauge field with its own dynamics, by simply adding the Maxwell action. Thus we consider
$$\begin{aligned} S_{tot}[\Phi, A] &= S_{\text{Maxwell}}[A] + S[\Phi; A] \\ &= \int d^4x \ \left(-\tfrac{1}{4}F_{ab}F^{ab} - \tfrac{1}{2}\eta^{ab}D_a\Phi D_b\Phi^* - W(\Phi\Phi^*)\right) \quad . \end{aligned} \tag{6.42}$$
In fact, even if we had not known Maxwell theory yet, and had only introduced A_a in order to gauge the global $U(1)$-symmetry, following the arguments in section 5.5, we would have now been led to Maxwell theory by the requirements of gauge and Lorentz invariance.

5. As Maxwell theory can thus be regarded as arising from the gauging of a global $U(1)$-symmetry, one can also think of Maxwell theory all by itself as an Abelian or $U(1)$ gauge theory.

6. This suggests the obvious and tempting possibililty to generalise all of this to the gauging of non-Abelian global symmetry groups and the construction of non-Abelian generalisations of Maxwell theory (known as *Yang-Mills theory*), but this is something that (for the time being at least) I will not address in these notes.

Returning to more elementary matters, we can now turn to the equations of motion for Φ and A. For Φ and Φ^* one finds (by varying Φ^* respectively Φ independently)
$$D_aD^a\Phi = 2\frac{\partial W}{\partial \Phi^*} \quad , \quad D_aD^a\Phi^* = 2\frac{\partial W}{\partial \Phi} \quad . \tag{6.43}$$
Using the explicit form of the potential, this can (as in Example 1 of the previous section) be written as
$$D_aD^a\Phi = 2W'(\Phi^*\Phi)\Phi \quad , \quad D_aD^a\Phi^* = 2W'(\Phi^*\Phi)\Phi^* \quad . \tag{6.44}$$

These equations of motion are gauge invariant, as they should be, as under gauge transformations both sides of the equations transform in the same way.

Variation of the action with respect to the gauge fields leads to

$$\delta S_{tot} = \int d^4x \ \left(\partial_a F^{ab} + J^b\right) \delta A_b \tag{6.45}$$

where the current is obtained from varying the minimally coupled matter action with respect to A. Since A_b appears once in the form $-iA_b\Phi D^b\Phi^*$, and once in the form $D^b\Phi(+iA_b)\Phi^*$, both with an overall factor of $-1/2$, this current is

$$J^b = (i/2)(\Phi D^b\Phi^* - \Phi^* D^b\Phi) \ . \tag{6.46}$$

This current is also gauge invariant, as it should be, since Φ and $D^b\Phi^*$ transform inversely to each other under gauge transformations (and likewise for the second contribution to the current).

Note also that this current, which looks like the covariantised (minimally coupled) version of the Noether current (6.19) of the ungauged theory, is actually also the Noether current associated to the invariance of the gauged theory (invariant under *local* gauge transformations) under *global* (constant) gauge transformations. I will come back to this below.

The equations of motion $\partial_a F^{ab} + J^b = 0$ imply (and therefore require) that $\partial_b J^b = 0$. We will now show that this is indeed satisfied as a consequence of the equations of motion for Φ. Ignoring the constant prefactor, we start with

$$\partial_b(\Phi D^b\Phi^*) = (\partial_b\Phi)D^b\Phi^* + \Phi\partial_b D^b\Phi^* \ . \tag{6.47}$$

Subtracting and adding $iA_b\Phi(D^b\Phi^*)$, we can write this in the nicer form

$$\partial_b(\Phi D^b\Phi^*) = (D_b\Phi)D^b\Phi^* + \Phi D_b D^b\Phi^* \ . \tag{6.48}$$

In section 7.4 I will give a more conceptual explanation below for why such identities are true. In any case, repeating the calculation for the second contribution to J^b we find

$$\begin{aligned} \partial_b(\Phi D^b\Phi^* - \Phi^* D^b\Phi) &= (D_b\Phi)D^b\Phi^* + \Phi D_b D^b\Phi^* - (D_b\Phi^*)D^b\Phi - \Phi^* D_b D^b\Phi \\ &= \Phi D_b D^b\Phi^* - \Phi^* D_b D^b\Phi \ . \end{aligned} \tag{6.49}$$

Now using the matter equations of motion one finds (exactly as in the case of the Noether current (6.19) of the ungauged theory) that these two terms cancel for a potential of the form $W = W(\Phi^*\Phi)$, and thus

$$D_b D^b\Phi = 2W'(\Phi^*\Phi)\Phi \quad , \quad D_b D^b\Phi^* = 2W'(\Phi^*\Phi)\Phi^* \quad \Rightarrow \quad \partial_b J^b = 0 \ . \tag{6.50}$$

Remarks:

1. This illustrates the remark made at the end of section 5.5, that the electric current source for the Maxwell equations obtained by coupling the matter fields to the Maxwell field (in a gauge invariant way) will be conserved as a consequence of the equations of motion of the matter fields, as required by gauge invariance.

2. We can now also understand more precisely, in which sense current (or charge) conservation is associated with (and a consequence of) a symmetry of the action, in the spirit of the Noether theorem. Indeed, the total action (6.42) is, in particular, invariant under constant gauge transformations,

$$\delta\Phi = i\alpha\Phi \quad , \quad \delta A_b = \partial_b \alpha = 0 \ , \tag{6.51}$$

leading to the Noether current

$$\begin{aligned} J^a_{\text{Noether}} &= \frac{\partial L}{\partial(\partial_a \Phi)}\delta\Phi + \frac{\partial L}{\partial(\partial_a \Phi^*)}\delta\Phi^* + \frac{\partial L}{\partial(\partial_a A_b)}\delta A_b \\ &= (-i\alpha/2)(\Phi D^a \Phi^* - \Phi^* D^a \Phi) \ . \end{aligned} \tag{6.52}$$

We see that, up to a constant factor, this is equal to the source current J^a (6.46) of the theory,

$$\alpha \quad \text{constant} \quad \Rightarrow \quad J^a_{\text{Noether}} = -\alpha J^a \ . \tag{6.53}$$

In particular, therefore, invariance of the gauged action under global gauge transformation implies charge conservation.

3. Since the theory is invariant not only under global gauge transformations, but under the infinity of local gauge transformations, naively one might perhaps expect the Noether theorem to provide one with a corresponding infinity of conserved currents or charges. However, this is *not* the case.

   Performing the same calculation as above, but now for local (x-dependent) transformations, one finds that the corresponding Noether current is (now the Maxwell term contributes to the current)

$$\alpha = \alpha(x) \quad \Rightarrow \quad J^a_{\text{Noether}} = -\alpha J^a - F^{ab}\partial_b \alpha \ . \tag{6.54}$$

   Upon closer inspection, this can be written as

$$J^a_{\text{Noether}} = -\alpha(\partial_b F^{ba} + J^a) - \partial_b(\alpha F^{ab}) \ . \tag{6.55}$$

   This current is trivial in the sense that it is a linear combination of a term (the first one) that is identically zero for a solution to the equations of motion, and another term (the second one) that is identically conserved (independently of any equations of motion), by anti-symmetry of F^{ab}, $\partial_a\partial_b(\alpha F^{ab}) \equiv 0$.

   This foreshadows and anticipates a general feature of theories with local symmetries (Noether's 2nd theorem), and various aspects of this will be explored in more detail in section 8.

6.3 Spacetime Symmetries and Variations I: Translations

As stressed in section 6.1, in our simple 1-line derivation of Noether's theorem we have only considered variations of the fields, not in addition possible variations of the coordinates. This raises the question if and/or how one can deal with spacetime symmetries, i.e. transformations

of the fields that are associated with transformations of the coordinates, like translations or Lorentz transformations.

For some reason, at this point most textbooks dealing with this issue opt to generalise the Noether theorem to situations where one also considers and allows explicit variations of the coordinates. However, this leads to all kinds of unnecessary complications, for instance the transformation of the integration volume element $d^D x$ and the integration domain. All these problems, and other issues related to disentangling true from false variations, are absent when one reformulates the action of spacetime transformations rightaway as transformations that act on the fields alone, not on the coordinates.

I have already briefly mentioned how to go about this in the context of the Noether theorem in mechanics in section 3.5, and I will explain this in some more detail in the field theory context here.[1]

Let me start with translations of the spacetime coordinates. Infinitesimally these take the form

$$x^a \to \bar{x}^a = x^a + \epsilon^a \quad . \tag{6.56}$$

Under such translations, not just Lorentz scalars but all the Lorentz tensor fields that we have discussed transform as scalars, i.e. one has

$$\bar{\phi}(\bar{x}) = \phi(x) \quad , \quad \bar{A}_a(\bar{x}) = A_a(x) \tag{6.57}$$

etc. While this is true and simple, it really does not tell us much about how fields transform under infinitesimal translations. The statement that e.g. $\bar{\phi}(\bar{x}) - \phi(x) = 0$ does not mean that the field does not change: after all we are comparing two fields not at the same point but two fields at two different points. Variations, on the other hand, are obviously always differences between two fields at the same point,

$$\delta\Phi^A(x) = (\Phi^A(x) + \delta\Phi^A(x)) - \Phi^A(x) \quad , \tag{6.58}$$

and it is this fact that ensures the crucial property of a variation that variations and partial derivatives "commute".

The way to translate infinitesimal translations into true variations is to think of such infinitesimal translations as defining new translated fields $\bar{\phi}(x)$ via

$$\bar{\phi}(x) = \phi(x - \epsilon) \quad . \tag{6.59}$$

Taylor expanding this to first order, one finds

$$\bar{\phi}(x) = \phi(x) - \epsilon^a \partial_a \phi(x) \quad . \tag{6.60}$$

The difference between the left-hand side and the first term on the right-hand side is now a difference between two fields at the same point, and this therefore defines a variation. We can thus define the translational variation $\delta_T \phi$ of ϕ by

$$\delta_T \phi(x) = -\epsilon^a \partial_a \phi(x) \quad , \tag{6.61}$$

[1] For a nice treatment, which has also helped me to inprove my presentation of the subject, see M. Banados, I. Reyes, *A short review on Noether's theorems, gauge symmetries and boundary terms*, `https://arxiv.org/abs/1601.03616`.

and likewise for any other tensor field, e.g.

$$\delta_T A_b(x) = -\epsilon^a \partial_a A_b(x) \ , \tag{6.62}$$

and in general

$$\delta_T \Phi^A(x) = -\epsilon^a \partial_a \Phi^A(x) \quad , \quad \delta_T(\partial_b \Phi^A(x)) = -\partial_b(\epsilon^a \partial_a \Phi(x)) = -\epsilon^a \partial_a \partial_b \Phi(x) \ . \tag{6.63}$$

Acting with δ_T on any Lagrangian $L(\Phi^A, \partial_b \Phi^A; x)$ one finds

$$\begin{aligned} \delta_T L &= \frac{\partial L}{\partial \Phi^A(x)} \delta_T \Phi^A(x) + \frac{\partial L}{\partial(\partial_b \Phi^A(x))} \delta_T(\partial_b \Phi^A(x)) \\ &= -\frac{\partial L}{\partial \Phi^A(x)} \epsilon^a \partial_a \Phi^A(x) - \frac{\partial L}{\partial(\partial_a \Phi^A(x))} \epsilon^a \partial_a \partial_b \Phi(x) \\ &= -\frac{d}{dx^a}(\epsilon^a L) + \frac{\partial}{\partial x^a}(\epsilon^a L) \ . \end{aligned} \tag{6.64}$$

Thus we see that the variation is a total derivative (and hence the infinitesimal translations are infinitesimal symmetries) if the Lagrangian does not depend explicitly on the coordinates x^a,

$$\frac{\partial L}{\partial x^a} = 0 \quad \Rightarrow \quad \delta_T L = \frac{d}{dx^a}(-\epsilon^a L) \tag{6.65}$$

(note that in the derivation of this result it is clearly necessary to carefully distinguish the partial and the total derivative).

While this is certainly the expected result, anticipated already in our construction of Poincaré invariant actions in section 5, we have now derived this from the point of view of variations and the Noether theorem. The conserved currents associated with this translation invariance will be explored in section 6.4 below.

Remarks:

1. The minus signs in the above equations may seem to be a nuisance, and we could simply have defined the variations with the opposite sign. However, the analogous considerations for Lorentz transformations in section 7.3 will show that it is more natural to keep the minus sign where it is.

2. We can also decompose δ_T into the variations along the different directions, as

   $$\delta_T = \epsilon^a \delta_{(a)} \ . \tag{6.66}$$

   say. With that notation one can write

   $$\delta_{(a)} \Phi^A = -\partial_a \Phi^A \quad , \quad \delta_{(a)} \partial_b \Phi^A = -\partial_a \partial_b \Phi^A \ . \tag{6.67}$$

3. This above discussion of translations also teaches us how to deal with Lorentz transformations, which are of course also associated with spacetime transformations, but under which additionally Lorentz tensor fields transform in a non-trivial (namely tensorial) way. As this is something we do not really need in the course, a discussion of this will be deferred to section 7.3.

Suffice it so say here, that the result one finds precisely mirrors that we have found for translations. I.e. if we denote the infinitesimal generator of a Lorentz transformation by ω^a,

$$x^a \to \bar{x}^a = x^a + \omega^a_{\ b} x^b \equiv x^a + \omega^a \quad , \tag{6.68}$$

then under Lorentz variations δ_L a Lorentz scalar Lagrangian L transforms as

$$\delta_L L = \frac{d}{dx^a}(-\omega^a L) \quad . \tag{6.69}$$

Thus it is in this sense that a Lorentz invariant Lagrangian gives rise to a Lorentz symmetry in the (variational) sense that appears in the Noether theorem.

6.4 Spacetime Translation Invariance and the Energy-Momentum Tensor

After this preparation, we can now immediately deduce that we obtain 4 conserved currents $J^a_{(b)}$ associated to spacetime translation invariance provided that the Lagrangian does not depend explicitly on the spacetime coordinates x^b,

$$\frac{\partial L}{\partial x^b} = 0 \quad \Rightarrow \quad \exists \quad \text{conserved currents} \quad J^a_T = \epsilon^b J^a_{(b)} \quad . \tag{6.70}$$

We know that the conserved currents are only defined up to overall factors, signs, and the addition of identically conserved terms, but for now we just take them as they come out of the Noether theorem directly (and we will then make a consistency check on the choice of sign).

Combining (6.65) with the general expresssion (6.3) for the Noether current, we find the Noether current associated to the translational symmetry $\Delta = \delta_T$ to be

$$J^a_T = \frac{\partial L}{\partial(\partial_a \Phi^A)} \delta_T \Phi^A + \epsilon^a L = \epsilon^b \left(-\frac{\partial L}{\partial(\partial_a \Phi^A)} \partial_b \Phi^A + \delta^a_{\ b} L \right) \quad . \tag{6.71}$$

With the decomposition $J^a_T = \epsilon^b J^a_{(b)}$ this results in 4 currents

$$J^a_{(b)} = -\frac{\partial L}{\partial(\partial_a \Phi^A)} \partial_b \Phi^A + \delta^a_{\ b} L \tag{6.72}$$

indexed by (b), associated to the 4 spacetime translations $x^b \to x^b + \epsilon^b$. By construction, these are conserved for any solution to the Euler-Lagrange equations, provided that the Lagrangian does not depend explicitly on the x^b,

$$\frac{\partial L}{\partial x^b} = 0 \quad \Rightarrow \quad \frac{d}{dx^a} J^a_{(b)} = 0 \quad \text{(on solutions)} \quad . \tag{6.73}$$

Since everything in (6.72) is tensorial, the 4 currents actually nicely combine into a Lorentz (1,1)-tensor, known as the *Noether Energy-Momentum Tensor*, or as the

$$\text{Canonical Energy-Momentum Tensor:} \quad \Theta^a_{\ b} = -\frac{\partial L}{\partial(\partial_a \Phi^A)} \partial_b \Phi^A + \delta^a_{\ b} L \quad . \tag{6.74}$$

We will also define

$$\Theta_{ab} = \eta_{ac} \Theta^c_{\ b} = -\frac{\partial L}{\partial(\partial^a \Phi^A)} \partial_b \Phi^A + \eta_{ab} L \quad . \tag{6.75}$$

By construction, and from the Noether theorem, we have

$$\frac{\partial L}{\partial x^b} = 0 \quad \Rightarrow \quad \frac{d}{dx^a}\Theta^a_{\ b} = 0 \quad \text{(on solutions)} \ . \tag{6.76}$$

While we deduced this result from the general Noether theorem, it is also straightforward to verify it directly and explicitly by simply computing the divergence of $\Theta^a_{\ b}$,

$$\frac{d}{dx^a}\Theta^a_{\ b} = \ldots = \frac{\delta L}{\delta \Phi^A}\partial_b \Phi^A + \frac{\partial L}{\partial x^b} \ . \tag{6.77}$$

Here I used the shorthand notation (5.16)

$$\frac{\delta L}{\delta \Phi^A(x)} \equiv \frac{\partial L}{\partial \Phi^A(x)} - \frac{d}{dx^a}\frac{\partial L}{\partial(\partial_a \Phi^A(x))} \tag{6.78}$$

for the Euler-Lagrange equations. Please make sure that you know how to derive this, backwards and forwards.

We now turn to the physical interpretation of the components of $\Theta^a_{\ b}$ and Θ_{ab}. In the remainder of this section I will (finally) work in natural units in which the velocity of light $c = 1$. This permits us to not have to worry about the distinction betweeen matter and energy densities, and which factors of c we should perhaps have included in either the definition of the Lagrangian or that of Θ_{ab}, say.

We begin with the conserved charges

$$P_b = \int d^3x \ J^0_{(b)} = \int d^3x \ \Theta^0_{\ b} \ . \tag{6.79}$$

I have called these P_b because they are the conserved charges associated to spacetime translations, and therefore are what we usually call momenta and energy or 4-momenta.

More specifically, in mechanics we had $p_0 = -E$ (in units with $c = 1$), and therefore, in order to agree with this, the definition of the current $J_{(0)}$ should be such that its zero-component is *minus* the *energy density* ϵ,

$$J^0_{(0)} = \Theta^0_{\ 0} = -\epsilon \quad \Rightarrow \quad P_0 = -\int d^3x \ \epsilon = -E \ . \tag{6.80}$$

Also note that this implies that

$$\Theta_{00} = +\epsilon \ . \tag{6.81}$$

It turns out that with the choice of sign for the Noether currents we made at the beginning of this section this comes out correctly. In fact, this is already very plausible from the explicit expression for $\Theta^0_{\ 0}$,

$$\Theta^0_{\ 0} = -\frac{\partial L}{\partial(\partial_0 \Phi^A)}\partial_0 \Phi^A + L \ , \tag{6.82}$$

which is exactly *minus* the Legendre transform of the Lagrangian, and hence *minus* what one might like to call the Hamiltonian density or energy density.

Likewise, the zero-components of the Noether currents associated to spatial translation invariance must have the interpretation of *momentum densities* π_k,

$$J^0_{(k)} = \Theta^0_{\ k} = \pi_k \quad \Rightarrow \quad \int d^3x \ \pi_k = P_k \ . \tag{6.83}$$

The conservation laws then provide us with the interpretation of the remaining components. For example, comparison of the standard formula

$$\partial_t \rho + \partial_i J^i = 0 \tag{6.84}$$

with

$$\partial_a \Theta^a{}_0 = \partial_0 \Theta^0{}_0 + \partial_i \Theta^i{}_0 = -\partial_0 \epsilon + \partial_i \Theta^i{}_0 \tag{6.85}$$

tells us that the $\Theta^i{}_0$ are (minus) *energy current densities*. Likewise, from

$$\partial_a \Theta^a{}_k = \partial_0 \Theta^0{}_k + \partial_i \Theta^i{}_k = \partial_0 \pi_k + \partial_i \Theta^i{}_k \tag{6.86}$$

we deduce that the $\Theta^i{}_k$ are what one might call *momentum current densities*. However, momentum currents lead to pressure and stresses, and therefore the $\Theta^i{}_k$ are more commonly referred to as *stress tensor densities*. Note that these are indeed the components of a spatial 3-tensor (under rotations), and this is the way stresses and pressures are e.g. described in elasticity theory.

In terms of Θ_{ab}, we have

$$\begin{aligned} \Theta_{00}: &\quad \text{energy density } \epsilon \\ \Theta_{i0}: &\quad \text{(minus) energy current density} \\ \Theta_{0k}: &\quad \text{(minus) momentum density } -\pi_k \\ \Theta_{ik}: &\quad \text{stress tensor density} \end{aligned} \tag{6.87}$$

6.5 Energy-Momentum Tensor for a Scalar Field

As our first example (where everything works nicely), we look at the energy-momentum tensor of a (real, interacting) scalar field described by the action (5.47)

$$S[\phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi - V(\phi)\right) \equiv \int d^4x \left(-\tfrac{1}{2}(\partial\phi)^2 - V(\phi)\right) \tag{6.88}$$

with

$$(\partial\phi)^2 = \eta^{ab}\partial_a\phi\partial_b\phi = -\dot{\phi}^2 + (\vec{\nabla}\phi)^2 \ . \tag{6.89}$$

The energy-momentum tensor is (6.74)

$$\Theta^a{}_b = -\frac{\partial L}{\partial(\partial_a\phi)}\partial_b\phi + \delta^a{}_b L = \partial^a\phi\ \partial_b\phi - \delta^a{}_b \left(\tfrac{1}{2}(\partial\phi)^2 + V(\phi)\right) \tag{6.90}$$

or

$$\Theta_{ab} = \partial_a\phi\ \partial_b\phi - \eta_{ab}\left(\tfrac{1}{2}(\partial\phi)^2 + V(\phi)\right) \ . \tag{6.91}$$

This energy-momentum tensor has the following properties:

1. As expected, and by construction, Θ_{ab} is conserved for a solution to the equations of motion. Since we know Θ_{ab} as an explicit function of ϕ and its derivatives, on which we can act with the partial derivatives, and because there is no explicit x-dependence anywhere, we do not need to invoke the total derivative d/dx^a and can simply write this assertion as

   $$\Box\phi = V'(\phi) \quad \Rightarrow \quad \partial_a\Theta^a{}_b = \partial^a\Theta_{ab} = 0 \ . \tag{6.92}$$

   This is a simple calculation you should do (and should be able to do) yourself.

2. The (00)-component Θ_{00} is

$$\Theta_{00} = \dot{\phi}^2 - \eta_{00}\left(\tfrac{1}{2}(\partial\phi)^2 + V(\phi)\right) = \tfrac{1}{2}\left(\dot{\phi}^2 + (\vec{\nabla}\phi)^2\right) + V(\phi) \ . \tag{6.93}$$

This is the correct energy density $\epsilon = \Theta_{00}$ (6.81) of a scalar field, in particular with the correct sign, namely non-negative for a non-negative potential $V(\phi)$,

$$V(\phi) \geq 0 \quad \Rightarrow \quad \epsilon = \Theta_{00} \geq 0 \ . \tag{6.94}$$

This confirms the sign choice made at the beginning of the previous section 6.4.

Applied to the interacting examples (quartic potential or the sine-Gordon model) of section 5.3, we can now also see that a constant solution $\phi(x) = \phi_0$ at a minimum $V(\phi_0) = 0$ of the potential is indeed a lowest (zero) energy solution, $\epsilon = 0$.

3. Θ_{ab} is manifestly symmetric,

$$\Theta_{ab} = \Theta_{ba} \ . \tag{6.95}$$

This is true for any Lorentz invariant theory of scalar fields, but is not true in general (as we will see in the case of Maxwell theory in section 6.6 below).

4. One implication of the symmetry of Θ_{ab} is that we can construct conserved currents $J^{[ab]}$ for each anti-symmetric pair of indices $[ab]$, with components

$$J^{[ab]c} = x^b\Theta^{ca} - x^a\Theta^{cb} \ . \tag{6.96}$$

Indeed, calculating the divergence, we find

$$\begin{aligned} \partial_c J^{[ab]c} &= \partial_c(x^b\Theta^{ca} - x^a\Theta^{cb}) \\ &= \delta^b_{\ c}\Theta^{ca} + x^b\partial_c\Theta^{ca} - \delta^a_{\ c}\Theta^{cb} - x^a\partial_c\Theta^{cb} \\ &= \Theta^{ba} - \Theta^{ab} + x^b\partial_c\Theta^{ca} - x^a\partial_c\Theta^{cb} \ . \end{aligned} \tag{6.97}$$

The first two terms cancel by symmetry of Θ_{ab} and the other terms vanish for a solution to the equations of motion. Thus we conclude

$$\partial_c J^{[ab]c} = 0 \quad \text{(on solutions)} \tag{6.98}$$

This conclusion holds for any symmetric and conserved tensor Θ^{ab}.

5. To understand the physical significance or interpretation of these conserved currents, one can look at the corresponding charge densities

$$J^{[ab]0} = x^b\Theta^{0a} - x^a\Theta^{0b} \ , \tag{6.99}$$

in particular

$$J^{[ik]0} = x^k\Theta^{0i} - x^i\Theta^{0k} = x^k\pi^i - x^i\pi^k \ . \tag{6.100}$$

These resemble the conserved charges $L^{ab} \sim x^a p^b - x^b p^a$ (3.111) (in particular the angular momentum) associated to Lorentz invariance in relativistic mechanics, and this suggests that we have just constructed the conserved currents associated to Lorentz invariance of the scalar field action (in fact, what else could they be?).

6. That these currents are indeed precisely the Noether currents associated to the Lorentz invariance of the action and the infinitesimal anti-symmetric Lorentz transformation parameters ω_{ab}, can be seen by using the result (6.69)

$$\delta_L L = \frac{d}{dx^a}(-\omega^a L) \ , \tag{6.101}$$

valid for any Lorentz scalar, in particular therefore also for ϕ itself, and derived in section 7.3. Since this variation is a total derivative, there is a corresponding conserved current which we can write as

$$J^c_L = \frac{\partial L}{\partial(\partial_c\phi)}\delta_L\phi + \omega^c L = \frac{\partial L}{\partial(\partial_c\phi)}(-\omega^a\partial_a\phi) + \delta^c{}_a\omega^a L \ . \tag{6.102}$$

Comparing with the definition of the energy-momentum tensor, we learn that

$$J^c_L = \omega^a\Theta^c{}_a = \omega^a{}_b x^b\Theta^c{}_a = \omega_{ab}x^b\Theta^{ca} \ . \tag{6.103}$$

Since ω_{ab} is anti-symmetric, we anti-symmetrise the other contribution to deduce

$$J^c_L = \tfrac{1}{2}\omega_{ab}(x^b\Theta^{ca} - x^a\Theta^{cb}) = \tfrac{1}{2}\omega_{ab}J^{[ab]c} \ . \tag{6.104}$$

This establishes the claim. It is also clear from the above derivation that for higher rank Lorentz tensor fields there will be additional contributions to the currents arising from the non-trivial transformation behaviour of Lorentz tensors under Lorentz transformations.

Because of all these desirable properties, there is no reason to modify the definition of the energy-momentum tensor for a scalar field in any way, and we do not need to make a notational distinction between the Noether or canonical energy-momentum tensor Θ_{ab} and the symbol that is usually used for the energy-momentum tensor, namely T_{ab}. Thus for a scalar field we have

$$T_{ab} = \Theta_{ab} = \partial_a\phi\,\partial_b\phi - \eta_{ab}\left(\tfrac{1}{2}(\partial\phi)^2 + V(\phi)\right) \ . \tag{6.105}$$

All of this also generalises in a straightforward way to actions for multiple real or complex scalar fields. Something different, however, happens in the case of actions for higher rank Lorentz tensor fields, and we will take a closer look at this in the case of Maxwell theory below.

6.6 Energy-Momentum Tensor for Maxwell Theory

We now turn our attention to pure Maxwell gauge theory (i.e. without sources, $J^a = 0$). Thus the Lagrangian is

$$L = -\tfrac{1}{4}F_{ab}F^{ab} \ , \tag{6.106}$$

and the translational variation of the gauge field is

$$\delta_T A_c = -\epsilon^b\partial_b A_c \ . \tag{6.107}$$

Because the Maxwell Lagrangian does not depend explicitly on the coordinates x^a, under this variation it transforms as (6.65)

$$\delta_T L = \frac{d}{dx^a}(-\epsilon^a L) \ . \tag{6.108}$$

Therefore the conserved Noether energy-momentum tensor Θ_{ab} (6.75) is

$$\Theta_{ab} = -\frac{\partial L}{\partial(\partial^a A_c)}\partial_b A_c + \eta_{ab} L = F_a{}^c \partial_b A_c - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} \ . \tag{6.109}$$

This energy-momentum tensor has the following properties (bugs and features):

1. Feature: By construction, it is conserved for a solution to the equations of motion,

$$\partial^a \Theta_{ab} = 0 \quad \text{(on solutions)} \ . \tag{6.110}$$

   Note that both sets of Maxwell equations are required to derive this, i.e.

$$\partial_a F^{ab} = 0 \quad \text{and} \quad \partial_{[a} F_{bc]} = 0 \quad \Rightarrow \quad \partial^a \Theta_{ab} = 0 \ . \tag{6.111}$$

2. Bug: Θ_{ab} is evidently not gauge invariant. In particular, the expression for the energy-density is not gauge-invariant and does not agree with the standard expression (I continue to units in which $c = 1$)

$$\Theta_{00} \neq \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{6.112}$$

   Therefore Θ_{ab} cannot be the physically correct answer.

3. Fact: Θ_{ab} is not symmetric. In particular, therefore, the candidate angular momentum current (6.96) is not conserved,

$$\partial_c(x^b \Theta^{ca} - x^a \Theta^{cb}) \neq 0 \tag{6.113}$$

   (even though Maxwell theory is Lorentz invariant). This should not come as a surprise, given that we already noted above that for higher rank Lorentz tensor fields (6.104) cannot be the whole story.

This situation can be improved by first of all manipulating Θ_{ab} as

$$\begin{aligned} \Theta_{ab} &= F_a{}^c(\partial_b A_c - \partial_c A_b) - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} + F_{ac}\partial^c A_b \\ &= F_a{}^c F_{bc} - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} + F_{ac}\partial^c A_b \ . \end{aligned} \tag{6.114}$$

Here the first two terms are already nice and gauge invariant. The last term can be written as a sum of two terms,

$$F_{ac}\partial^c A_b = \partial^c(F_{ac} A_b) - (\partial^c F_{ac}) A_b \ . \tag{6.115}$$

The first of these is identically conserved because of $F_{ac} = -F_{ca}$.

$$\partial^a \partial^c (F_{ac} A_b) = 0 \quad \text{identically} \ . \tag{6.116}$$

We are thus in the situation discussed in section 6.1: we can modify Noether currents by identically conserved terms, and we are therefore led to define

$$\hat{\Theta}_{ab} = \Theta_{ab} - \partial^c(F_{ac} A_b) \tag{6.117}$$

By construction, this energy-momentum tensor is still conserved on solutions,

$$\partial^a \hat{\Theta}_{ab} = 0 \quad \text{(on solutions)} \ . \tag{6.118}$$

Moreover, the second term in (6.115) actually vanishes on solutions,

$$(\partial^c F_{ac})A_b = 0 \quad \text{(on solutions)}. \tag{6.119}$$

and therefore this new $\hat{\Theta}_{ab}$ is now also gauge invariant on solutions,

$$\begin{aligned} \hat{\Theta}_{ab} &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} - (\partial^c F_{ac})A_b \\ &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \quad \text{(on solutions)} \ . \end{aligned} \tag{6.120}$$

Since we are only interested in the energy-momentum tensor for solutions to the equations of motion, there is no point in carrying around a term that is zero for solutions. Therefore one can define a new (and vastly improved) energy-momentum tensor T_{ab} by

$$T_{ab} = F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \ . \tag{6.121}$$

This T_{ab} now has the following features (and no bugs!):

1. T_{ab} is still on-shell conserved,

   $$\partial^a T_{ab} = 0 \quad \text{(on solutions)} \tag{6.122}$$

   (again both sets of Maxwell equations are required to establish this). With an external source,

   $$\partial_{[a}F_{bc]} = 0 \quad , \quad \partial_a F^{ab} = -J^b \tag{6.123}$$

   one has the non-conservation law

   $$\partial^a T_{ab} = J^a F_{ab} \ , \tag{6.124}$$

   where the term on the right-hand side (a generalised Lorentz force) describes the exchange of energy between the electromagnetic field and the matter fields.

   If done correctly, the proof of (6.124) is quite simple. It does, however, require the ability to manipulate Lorentz tensorial equations (relabelling of indices, anti-symmetrisation etc.) in an accident-free and intelligent manner, so this is a good exercise for you to test your understanding of the formalism.

   Proof of (6.124):

   - From (6.121) we find

     $$\partial^a T_{ab} = (\partial^a F_{ac})F_b{}^c + F_{ac}\partial^a F_b{}^c - \tfrac{1}{2}(\partial_b F_{cd})F^{cd} \ . \tag{6.125}$$

   - Using the inhomogeneous Maxwell equations, the first term on the right-hand side already gives us the right-hand side of (6.124),

     $$(\partial^a F_{ac})F_b{}^c = -J_c F_b{}^c = J^a F_{ab} \ . \tag{6.126}$$

   - In order to be able to combine the remaining terms, we relabel and raise/lower the indices such that

     $$F_{ac}\partial^a F_b{}^c - \tfrac{1}{2}(\partial_b F_{cd})F^{cd} = F^{ac}\partial_a F_{bc} - \tfrac{1}{2}(\partial_b F_{ac})F^{ac} = F^{ac}(\partial_a F_{bc} - \tfrac{1}{2}\partial_b F_{ac}) \ . \tag{6.127}$$

- Since $F^{ac} = -F^{ca}$, only the anti-symmetric part of $\partial_a F_{bc}$ contributes, and therefore we anti-symmetrise explicitly, to find

$$F^{ac}(\partial_a F_{bc} - \tfrac{1}{2}\partial_b F_{ac}) = \tfrac{1}{2}F^{ac}(\partial_a F_{bc} - \partial_c F_{ba} - \partial_b F_{ac}) \tag{6.128}$$

- Finally, by the homogeneous Maxwell equations, the term in brackets is zero,

$$\partial_a F_{bc} - \partial_c F_{ba} - \partial_b F_{ac} = \partial_a F_{bc} + \partial_c F_{ab} + \partial_b F_{ca} = 0 \ . \tag{6.129}$$

2. T_{ab} is gauge-invariant and correctly gives the gauge-invariant and positive-definite energy-density as

$$T_{00} = F_{0c}F_0{}^c - \tfrac{1}{4}\eta_{00}F_{cd}F^{cd} = \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{6.130}$$

   This follows from $F_{0k} = -E_k$ and (4.88).

3. Moreover, the components of T_{i0} are exactly minus the components of the Poynting vector,

$$\vec{S} = \vec{E} \times \vec{B} \ , \tag{6.131}$$

   which is known to describe the energy flux of the electromagnetic field,

$$T_{i0} = F_{ic}F_0{}^c = F_{ij}F_0{}^j = -\epsilon_{ijk}B_k E_j = -S_i \ , \tag{6.132}$$

   in complete agreement (signs and all) with the identifications in (6.87).

4. Finally, the spatial components T_{ik} agree with the components of what is known as the Maxwell stress tensor (but we will not look at these in detail here).

5. T_{ab} is symmetric,

$$T_{ab} = T_{ba} \ . \tag{6.133}$$

6. As a consequence, also the currents

$$J^{[ab]c} = x^b T^{ca} - x^a T^{cb} \tag{6.134}$$

   are conserved,

$$\partial_c J^{[ab]c} = 0 \quad \text{(on solutions)} \ , \tag{6.135}$$

   and are the Noether currents associated with the Lorentz invariance of Maxwell theory (modulo identically conserved terms and terms that vanish on solutions).

T_{ab} is therefore clearly *the* correct energy-momentum tensor of Maxwell theory.

While the result that we have obtained is clearly very satisfactory, equally clearly the way that we have arrived at it is not. Are there not perhaps (and should there not be) better, more systematic and conceptually clearer, shorter, less ad-hoc and round-about ways of arriving at the result? Indeed there are, and I will mention three of them.

1. The Elegant and Elementary Way: Gauge-Invariant Translations

   This is the only approach I will describe in detail, because it is really nice and easy to understand (and it is therefore also the only one I expect you to know and understand).

   Our starting point is the obervation that the source of the lack of gauge invariance of the Maxwell Noether energy-momentum tensor Θ_{ab} (6.109) is the lack of gauge invariance of the translational variation (6.107)

$$\delta_T A_c = -\epsilon^b \partial_b A_c \ . \tag{6.136}$$

   Let us see what we can do with that. We write this as

$$\delta_T A_c = -\epsilon^b (\partial_b A_c - \partial_c A_b) - \epsilon^b \partial_c A_b = -\epsilon^b F_{bc} - \partial_c (\epsilon^b A_b) \ . \tag{6.137}$$

   Here the first term is nicely gauge invariant, and the second term is just a gauge transformation of A_c, with parameter

$$\Psi = \epsilon^b A_b \ . \tag{6.138}$$

   But since the Lagrangian of Maxwell theory is gauge invariant, it does not matter whether we act on it with a translational variation or with a translational variation plus a gauge transformation. Therefore we define a new (gauge invariant) translational variation by

$$\Delta_T A_c = \delta_T A_c + \partial_c (\epsilon^b A_b) = -\epsilon^b F_{bc} \ , \tag{6.139}$$

   or

$$\Delta_{(b)} A_c = -F_{bc} \ . \tag{6.140}$$

   Acting on any gauge invariant object, Δ_T reduces to δ_T. You can (and should) check this explicitly e.g. for the field strength tensor $F_{cd} = \partial_c A_d - \partial_d A_c$:

$$\Delta_T F_{cd} = \partial_c \Delta_T A_d - \partial_c \Delta_T A_d = \ldots = -\epsilon^b \partial_b F_{cd} = \delta_T F_{cd} \tag{6.141}$$

   (fill in the dots!). In particular, for the gauge invariant Lagrangian L of Maxwell theory one still has (6.108)

$$\Delta_T L = \delta_T L = \frac{d}{dx^a}(-\epsilon^a L) \ . \tag{6.142}$$

   But now, instead of (6.109), the energy-momentum tensor is

$$\frac{\partial L}{\partial(\partial^a A_c)} \Delta_{(b)} A_c + \eta_{ab} L = F_a{}^c F_{bc} - \tfrac{1}{4} \eta_{ab} F_{cd} F^{cd} = T_{ab} \ . \tag{6.143}$$

   Thus in this way we obtain directly and on the nose the correct gauge invariant energy-momentum tensor (6.121) of Maxwell theory, without having to play any silly games.

   This construction can also be used to define gauge invariant Lorentz variations (or gauge invariant general coordinate transformation variations) of gauge fields, and also works for non-Abelian gauge fields. It is a very simple, clever and elegant way to avoid ever having to deal with non-gauge invariant objects when performing variations of gauge fields.

2. The Time-Honoured Way: Belinfante Improvement Procedure

   The procedure (6.114)-(6.121) to obtain a symmetric and conserved T_{ab} from the canonical Noether energy-momentum tensor Θ_{ab} of a Poincaré-invariant field theory, illustrated above in the case of Maxwell theory, can be understood in a more general and systematic, but still somewhat round-about way by appealing to the Lorentz-invariance of the action and taking into account the non-trivial transformation behaviour of higher rank Lorentz tensor fields under Lorentz transformations. This (time-honoured) recipe is known as the *Belinfante improvement procedure*.

   Here one reverse-engineers the above construction leading one eventually to (6.135), i.e. one starts from the conserved Noether currents for Lorentz transformations, and then tries to put them into the form (6.134) for some symmetric T_{ab}, by adding/removing identically conserved terms or terms that are zero on solutions, in order to then deduce from the conservation of these currents that the T_{ab} that arises in that way is a conserved tensor which one then identifies as a suitable candidate for the energy-momentum tensor. This procedure is explained in many places, with wildly varying degree of comprehensibility (or comprehension).[2]

   However, I am not going to get into this here, beacuse I believe that, at least for current purposes, this procedure misses the point entirely. The main problem with Θ_{ab} for Maxwell theory is not, that it is not symmetric and that therefore the currents (6.113) are not conserved (all that means is that the conserved Lorentz currents are something else). The glaring problem with Θ_{ab} for Maxwell theory is that it is not gauge invariant. This has nothing to do with Lorentz invariance. After all, a gauge invariant theory should have a gauge invariant energy-momentum tensor even when it is not Lorentz invariant.

3. The Cool and Fundamental (General Relativity) Way: T_{ab} is the Source of Gravity

   Here one asks the question: how should one fundamentally, independently of any symmetries or conservation laws, define the energy-momentum tensor? General Relativity, Einstein's relativistic theory of gravity, provides the answer to that: it is well known that mass or energy density, what we have called T_{00}, can create gravitational fields, i.e. can act as a source for gravitational fields. But in a tensorial theory, if T_{00} appears as a source, then all the T_{ab} must appear and be able to act as sources of the gravitational field. Turning this around, and appealing to the universality of gravity, one simply *defines* T_{ab} to be the source of the gravitational field.

   To see how that helps one to actually define the energy-momentum tensor for a Lagrangian field theory, it is useful to first think about the analogous question how to define the source (current J^a) of the electromagnetic field. The answer is very simple, as we have seen in our discussion of minimal coupling in section 6.2:

   - First we couple the matter Lagrangian to the electromagnetic field A_a (e.g. by the minimal coupling replacement $\partial_a \to D_a$),

   $$S[\Phi] \to S[\Phi; A] \ . \tag{6.144}$$

[2] For a detailed and comprehensible explanation, see e.g. section 2 of T. Ortin, *Gravity and Strings*.

- Then by construction the current J^a (source term in the field equations for A_a) is obtained from the variation of the minimally coupled action with respect to the gauge field A_a, symbolically
$$J^a \sim \frac{\delta S[\Phi, A]}{\delta A_a} \quad . \tag{6.145}$$
Note that this can be deduced without knowing (or specifying) the action for the gauge field A_a itself.

The construction for gravity proceeds analogously. It turns out (and this is one of the fundamental insights of Einstein) that the dynamical variable in gravity is the spacetime metric itself, i.e. the gravitational field is a symmetric $(0,2)$-tensor g_{ab} which defines a line element $ds^2 = g_{ab}(x)dx^a dx^b$ (here the x^a are arbitrary, not inertial, coordinates). At this point one can repeat the two steps above:

- First we couple the matter (or Maxwell) Lagrangian to the gravitational field g_{ab} (e.g. by some minimal coupling replacement $\partial_a \to D_a$ - this also works for gravity, with some minor additional decorations),
$$S[\Phi] \to S[\Phi; g] \quad . \tag{6.146}$$
- Then by construction the source term T_{ab} in the field equations for g_{ab} is obtained from the variation of the minimally coupled action with respect to the gravitational field g_{ab}, symbolically
$$T^{ab} \sim \frac{\delta S[\Phi, g]}{\delta g_{ab}} \quad . \tag{6.147}$$
Note that this can be deduced without knowing (or specifying) the action for the gravitational field g_{ab} itself, even without knowing the field equations (the Einstein equations for the gravitational field).

Specialising now $g_{ab} \to \eta_{ab}$, one obtains the candidate energy-momentum tensor in Minkowsi space. By construction, the T_{ab} obtained in this way will always have the following properties:

- T_{ab} is conserved on solutions
[This is not obvious from what I have said but is implied by general covariance, the invariance under general coordinate transformations, just like gauge invariance implies $\partial_a J^a = 0$]
- T_{ab} is symmetric
- T_{ab} will automatically inherit all the local and global symmetries of the Minkowski space matter Lagrangian.

In particular, if one applies this prescription to the minimal coupling of Maxwell theory to the gravitational field, it is a 1-line calculation to show that one obtains directly and on the nose the correct and gauge invariant Maxwell energy momentum tensor (6.121), without having to invoke any kind of voodoo improvement procedure.

Of course, I cannot explain this in more detail here, and I refer you to my course and Lecture Notes on General Relativity for a detailed discussion of everything that is required to understand the above paragraphs (and much more ...).

7 Symmetries and Gauge Theories: Selected Advanced Topics

7.1 Higher Dimensional and Higher Rank Generalisations of Maxwell Theory

As an aside, and as a sequel to our discussion of electric-magnetic duality in section 4.7 and the coupling of particles to the electromagnetic field in section 4.12, here are some comments on two generalisations of Maxwell theory in $(3+1)$-dimensions, namely higher dimensional generalisations, and generalisations to higher rank gauge fields.

Starting with the former, as they stand the Maxwell equations in the form given in (4.59),

$$\text{Maxwell Equations:} \quad \begin{cases} \partial_a F^{ab} = -J^b \\ \partial_{[a} F_{bc]} = 0 \end{cases} \tag{7.1}$$

make sense in any number of spacetime dimensions, and can be used to define the gauge theory of a gauge field $A_a(x)$. However, in passing to the formulation given in (4.72) in terms of the dual field strength tensor $\tilde{F}^{ab}$, we explicitly used the 4-dimensional ϵ-symbol to define (4.70)

$$\tilde{F}^{ab} = \tfrac{1}{2}\epsilon^{abcd} F_{cd} \ . \tag{7.2}$$

What would happen in other dimensions? Well, if we are in $D = d+1$ spacetime dimensions, then we can define and construct a D-dimensional ϵ-symbol by

$$\epsilon_{a_1 \ldots a_D} = \epsilon_{[a_1 \ldots a_D]} \quad , \quad \epsilon_{01\ldots d} = +1 \ . \tag{7.3}$$

Then we have

$$\partial_{[a} F_{cd]} = 0 \quad \Leftrightarrow \quad \partial_{a_1} \tilde{F}^{a_1 \ldots a_{D-2}} = 0 \ , \tag{7.4}$$

where the dual field strength tensor is the totally anti-symmetric $(D-2,0)$-tensor

$$\tilde{F}^{a_1 \ldots a_{D-2}} = \tfrac{1}{2}\epsilon^{a_1 \ldots a_{D-2} cd} F_{cd} \ . \tag{7.5}$$

Thus we see that it is a special feature of 4=3+1 dimensions that the dual of the field strength tensor is again a rank-2 tensor. Moreover, as we will see below, this implies that only in 4 dimensions the hypothetical magnetic dual of an electrically charged particle would again be a particle.

As an aside (of an aside), let me point out that e.g. in 5 dimensions one can construct an identically conserved current

$$J_I^a = \epsilon^{abcde} F_{bc} F_{de} \quad \Rightarrow \quad \frac{d}{dx^a} J_I^a = 0 \quad \text{identically} \tag{7.6}$$

(the subscript "I" is for "Instanton", for reasons that I will not explain here), whose charge density is essentially the $D = 4$ invariant I_2 of section 4.8. Apart from things like this, however, the structure of Maxwell theory in $D \neq 4$ dimensions is pretty much the same as that of Maxwell theory in $D = 4$ dimensions.

These considerations also lead one to contemplate a different generalisation of Maxwell theory, namely to higher rank gauge fields, in which $\tilde{F}_{a_1 \ldots a_{D-2}}$ would arise as the field strength tensor of a rank $(D-3)$ "gauge field".

It is indeed possible, and of independent interest, to generalise Maxwell theory in such a way, namely to gauge theories of higher rank (totally anti-symmetric) gauge fields. The simplest case to consider is that of a rank-2 gauge field $B_{ab} = B_{[ab]}$. In this case the field strength could be defined by

$$H_{abc} \sim \partial_{[a} B_{bc]} \ , \tag{7.7}$$

and this would be invariant under gauge transformations

$$B_{ab} \to B_{ab} + \partial_a \Psi_b - \partial_b \Psi_a \tag{7.8}$$

(because second partial derivatives commute ... in case you missed me saying this for a while). In this case, the Bianchi identity takes the form

$$\partial_{[a} H_{bcd]} = 0 \ , \tag{7.9}$$

and a candidate gauge invariant equation of motion would be something like

$$\partial_a H^{abc} = J^{bc} \quad \Rightarrow \quad \partial_b J^{bc} = 0 \ , \tag{7.10}$$

with a conserved source $J^{bc} = -J^{cb}$.

What sort of objects could be "charged" under such a gauge field, i.e. what are the objects that one can couple to B_{ab} or that could give rise to a source J^{ab}? Well, following the logic in section 4.12, the B_{ab} are objects that can naturally be integrated over 2-dimensional spaces (surfaces) S. Indeed, if that space has coordinates τ and σ, say, then one could construct something like

$$\int d\sigma d\tau \ B_{ab}(\dot{x}^a x'^b - \dot{x}^b x'^a) \equiv \int_S B \tag{7.11}$$

where $x^a = x^a(\tau, \sigma)$ and

$$\dot{x}^a = \frac{\partial x^a}{\partial \tau} \quad , \quad x'^a = \frac{\partial x^a}{\partial \sigma} \ . \tag{7.12}$$

Objects whose "worldlines" (better "worldvolumes") are $(1+1)$-dimensional are themselves 1-dimensional, *strings*! And indeed such a field B_{ab} appears and plays a fundamental role in string theory, where it is known as the *Kalb-Ramond field*, or just as *the B-field*.

Likewise rank-3 totally anti-symmetric gauge fields C_{abc} can couple naturally to (and therefore appear in theories of) 2-dimensional membranes with $(2+1)$-dimensional woldvolumes etc.

Finally, combining the two observations in this section, we see that

- a (hypothetical) magnetic dual of an electrically charged particle in 4 dimensions would again be a particle,

$$D = 4: \quad \text{particle} \to A_a \to F_{ab} \to \tilde{F}_{ab} \to \tilde{A}_a \to \text{dual particle} \tag{7.13}$$

- while e.g. the (even more hypothetical) magnetic dual of an electrically charged particle in 5 dimensions would be a magnetically charged string,

$$D = 5: \quad \text{particle} \to A_a \to F_{ab} \to \tilde{F}_{abc} \to \tilde{A}_{ab} \to \text{dual string} \tag{7.14}$$

- while the dual of an electrically charged string in 6 dimensions would be a magnetically charged string ("string-string duality"),

$$D = 6: \quad \text{string} \to B_{ab} \to H_{abc} \to \tilde{H}_{abc} \to \tilde{B}_{ab} \to \text{dual string} \tag{7.15}$$

- etc. etc.

7.2 Abelian Chern-Simons Gauge Theory

We have seen in our discussion of Lorentz scalars in Maxwell theory (section 4.8) and an action principle for Maxwell theory (section 5.5) that essentially the unique choice for a gauge theory Lagrangian (depending at most on 1st derivatives of the gauge field A_a) in any dimension is the Maxwell Lagrangian $L \sim F^2$. However, there is one exception to this, in 3 dimensions. This is the (Abelian) Chern-Simons Lagrangian

$$L_{CS} = \tfrac{1}{2}\epsilon^{abc} A_a F_{bc} = \epsilon^{abc} A_a \partial_b A_c \ , \tag{7.16}$$

with action

$$S_{CS}[A] = \int d^3x \ \tfrac{1}{2}\epsilon^{abc} A_a F_{bc} \ . \tag{7.17}$$

Here the indices $a, b, \ldots$ can take either the values $(0, 1, 2)$ (then we are in a (2+1)-dimensional spacetime), or the values $(1, 2, 3)$ (so then we are dealing with a 3-dimensional space). Note that this Lagrangian, unlike that of Maxwell theory, is linear (rather than quadratic) in the 1st derivatives of the fields.

We will need to discuss the issues of gauge invariance and Lorentz invariance of this action:

1. Gauge Invariance

   Admittedly, at first sight L_{CS} does not look like a great candidate for a gauge theory Lagrangian, because it does not look particularly gauge invariant. At second sight, however, we see that under

   $$\delta_\theta A_a = \partial_a \theta \tag{7.18}$$

   we have, by the Bianchi idenitity for F_{bc},

   $$\delta_\theta L_{CS} = \tfrac{1}{2}\epsilon^{abc}(\partial_a \theta) F_{bc} = \frac{d}{dx^a}\left(\tfrac{1}{2}\epsilon^{abc}\theta F_{bc}\right) \ . \tag{7.19}$$

   Thus the Lagrangian is invariant up to a total derivative, the action only changes by a boundary term, and therefore the equations of motion must be gauge invariant, and indeed they are, as we will verify below.

2. Lorentz Invariance

   The Lagrangian is clearly invariant under (2+1)-dimensional rotations and boosts (or 3-dimensional rotations). However, because of the appearance of the ϵ-symbol, which requires a choice of orientation, the Lagrangian is not invariant under reflections. This, however, is more a feature than a bug of Chern-Simons theory.

Now let us turn to the equations of motion. Varying the action

$$S_{CS}[A] = \int d^3x\ \epsilon^{abc} A_a \partial_b A_c \tag{7.20}$$

one finds

$$\begin{aligned} \delta S_{CS}[A] &= \int d^3x\ \epsilon^{abc}\left[(\delta A_a)\partial_b A_c + A_a \partial_b(\delta A_c)\right] \\ &= \int d^3x\ \epsilon^{abc}\left[(\delta A_a)\partial_b A_c - (\partial_b A_a)(\delta A_c)\right] \\ &= \int d^3x\ \epsilon^{abc}\left[(\delta A_a)\partial_b A_c - (\partial_c A_b)(\delta A_a)\right] \\ &= \int d^3x\ \epsilon^{abc}(\delta A_a) F_{bc}\ . \end{aligned} \tag{7.21}$$

and therefore

$$\delta S_{CS}[A] = 0 \quad \forall\ \delta A \quad \Rightarrow \quad F_{bc} = 0\ , \tag{7.22}$$

which is indeed as gauge invariant as it gets.

Nevertheless, you may have the impression that this "Chern-Simons theory" cannot possibly be particularly interesting, and I agree with you: as it stands, the Abelian Chern-Simons action, all by itself, in Minkowski space or Euclidean space, is not particularly interesting. In particular, in these circumstances one can solve the equations of motion by

$$F_{bc} = 0 \quad \Rightarrow \quad A_b = \partial_b \theta\ , \tag{7.23}$$

so that, modulo gauge transformations, the unique solution of the equations of motion is $A_b = 0$.

Things become more interesting, however, if any one of the above conditions is relaxed, and we will now look at one instance of this, namely when one adds the Abelian Chern-Simons Lagrangian to the Maxwell Lagrangian. Thus we consider the Lagrangian

$$L = L_{\text{Maxwell}} + kL_{CS} = -\tfrac{1}{4} F_{ab}F^{ab} + \tfrac{1}{2} k \epsilon^{abc} A_a F_{bc}\ . \tag{7.24}$$

Here I have introduced a relative constant between the two terms, the Chern-Simons "level" k, which is an a priori arbitrary real constant parameter. The equations of motion resulting from this Lagrangian are evidently

$$\partial_a F^{ab} + k\epsilon^{bcd} F_{cd} = 0\ . \tag{7.25}$$

In terms of the dual field strength

$$G^b \equiv \tilde{F}^b = \tfrac{1}{2}\epsilon^{bcd} F_{cd} \tag{7.26}$$

the Bianchi idenitity for F_{ab} can, as in (7.4), be written as

$$\partial_b G^b = 0\ . \tag{7.27}$$

Moreover, after some ϵ-symbol gymnastics, the equation of motion can equivalently be written as

$$\partial_a F^{ab} + k\epsilon^{bcd} F_{cd} = 0 \quad \Leftrightarrow \quad \partial_a G_b - \partial_b G_a = 2k\epsilon_{abc} G^c\ . \tag{7.28}$$

Acting on this equation with ∂^a, and using the Bianchi identity and the equations of motion, one finds (now in Minkowski signature with $\epsilon_{abc}\epsilon^{abd} = -2\delta^d_c$)

$$\Box G_b = 2k\epsilon_{abc}\partial^a G^c = 2k^2\epsilon_{abc}\epsilon^{acd}G_d = 4k^2 G_b \ . \tag{7.29}$$

Therefore the Chern-Simons term generates a mass term for G_b or F_{ab}, with

$$(m_G)^2 = 4k^2 \ . \tag{7.30}$$

For this reason (and because the Chern-Simons theory by itself is in some suitable sense "topological"), Maxwell-Chern-Simons theory is also known as "topologically massive" Maxwell theory. Note that the naive way to introduce a mass term for the gauge field, by adding $m^2 A_a A^a$ to the Lagrangian, would not have been compatible with gauge invariance, while the Chern-Simons term provides a gauge inviariant way to give a mass to the gauge field. Unfortunately, there is no obvious and simple generalisation of this simple mechanism to higher dimensions. For a different mechanism, which plays a crucial role in the Standard Model of Particle Physics (the Higgs mechanism), see section 7.5.

Chern-Simons theory becomes much more interesting for non-Abelian gauge groups, with connections and applications to all kinds of branches of physics and mathematics (from condensed matter physics, integrable models and gravity in (2+1) dimensions to knot theory and the topology of 3-manifolds), but this shall suffice as a teaser or appetiser.

7.3 Spacetime Symmetries and Variations II: Lorentz Transformations

In section 6.3 we had already discussed how to reformulate infinitesimal spacetime translations on arbitary tensor fields as variations (which one can then use e.g. in the Noether theorem). Here I sketch how the same procedure can be applied to Lorentz transformations.

We begin with a Lorentz scalar field $\phi(x)$. Under Lorentz transformations,

$$\bar{x}^a = L^a_{\ b} x^b \ , \tag{7.31}$$

such a Lorentz scalar field transforms as

$$\bar{\phi}(\bar{x}) = \phi(x) \ . \tag{7.32}$$

As in the case of translations in section 6.3, we think of this as defining new (Lorentz rotated) fields at x, this time via

$$\bar{\phi}(x) = \phi(L^{-1}x) \ . \tag{7.33}$$

For an infinitesimal Lorentz transformation, we have

$$L^a_{\ b} = \delta^a_{\ b} + \omega^a_{\ b} \quad \Rightarrow \quad x^a \to \bar{x}^a = x^a + \omega^a_{\ b} x^b \equiv x^a + \omega^a \ , \tag{7.34}$$

with $\omega^a = \omega^a_{\ b} x^b$ the (x-dependent) infinitesimal generator of Lorentz transformations. We thus have

$$\bar{\phi}(x) = \phi(x^a - \omega^a) = \phi(x) - \omega^a \partial_a \phi(x) \ , \tag{7.35}$$

and we can define the Lorentz variation by

$$\delta_L \phi(x) = -\omega^a \partial_a \phi(x) \ . \tag{7.36}$$

Note that we can write this as

$$\delta_L \phi = -\partial_a(\omega^a \phi) = \frac{d}{dx^a}(-\omega^a \phi) \tag{7.37}$$

because

$$\partial_a \omega^a = \partial_a(\omega^a_{\ b} x^b) = \omega^a_{\ b} \delta^b_{\ a} = \omega^a_{\ a} = 0 \ , \tag{7.38}$$

by anti-symmetry of ω_{ab}.

Since $\delta_L \phi$ is a variation, for the derivative we have

$$\delta_L(\partial_a \phi) = -\partial_a(\omega^b \partial_b \phi) = -\omega^b_{\ a} \partial_b \phi - \omega^b \partial_b \partial_a \phi \ . \tag{7.39}$$

Note that here the new $\omega^b_{\ a}$-term arises automatically, reflecting the fact that $\partial_b \phi$ is a covector. More succinctly, from (7.37), we can also write this as

$$\delta_L(\partial_b \phi) = \partial_b \partial_a(-\omega^a \phi) \tag{7.40}$$

These are now the variations one can use e.g. in order to investigate the invariance of an action of a scalar field under Lorentz transformations.

More generally, however, this shows that if we have a Lorentz scalar Lagrangian L (constructed from arbitrary Lorentz tensor fields), under Lorentz variations it will transform by a total derivative,

$$\delta_L L = \frac{d}{dx^a}(-\omega^a L) \ . \tag{7.41}$$

Thus a Lorentz scalar Lagrangian L indeed also has a Lorentz symmetry in the sense of the Noether theorem. If one is slightly skeptical about the above reasoning, one can also explicitly calculate the variation of a Lagrangian in terms of the Lorentz variations of the (scalar or other) fields it is built from,

$$\delta_L L = \frac{\partial L}{\partial \Phi^A} \delta_L \Phi^A + \ldots , \tag{7.42}$$

but the result will not change.

As our next example, we consider a vector field $V^a(x)$. Under a Lorentz transformation one has

$$\bar{V}^a(\bar{x}) = L^a_{\ b} V^b(x) \ . \tag{7.43}$$

For an infinitesimal Lorentz transformation, one thus has

$$\bar{V}^a(\bar{x}) = V^a(x) + \omega^a_{\ b} V^b(x) \ . \tag{7.44}$$

One might therefore be tempted to regard the second term on the right-hand side as a variation,

$$\delta(?) V^a(x) = \bar{V}^a(\bar{x}) - V^a(x) = \omega^a_{\ b} V^b(x) \ . \tag{7.45}$$

But even though a Lorentz vector does transform in such a way infinitesimally, for a Lorentz vector field this is not a variation because it is the difference between two fields at two distinct

points. To rectify this, we proceed as above. We think of an (infinitesimal) Lorentz transformation as defining a new (Lorentz rotated) field via

$$\bar{V}^a(x) = L^a_{\ b} V^b(L^{-1}x) \ . \tag{7.46}$$

Note that this is really just the same equation as (7.43) above, just evaluated at the point x instead of $\bar{x}$. For an infinitesimal Lorentz transformation, we can now write

$$\bar{V}^a(x) = (\delta^a_{\ b} + \omega^a_{\ b}) V^b(x^c - \omega^c) \tag{7.47}$$

and expand to first order in $\omega^a_{\ b}$ to find

$$\bar{V}^a(x) = V^a(x) + \omega^a_{\ b} V^b(x) - \omega^c \partial_c V^a(x) \ . \tag{7.48}$$

We can therefore define the variation as

$$\delta_L V^a(x) = \bar{V}^a(x) - V^a(x) = +\omega^a_{\ b} V^b(x) - \omega^c \partial_c V^a(x) \ . \tag{7.49}$$

We can now also (finally) understand why we have kept the minus sign in the part of the variation involving the derivative along ω^a (or along ϵ^a in the case of translations): with this choice, the Lorentz variation is really just the infinitesimal transformation of a vector under Lorentz transformations, namely $V^a \to V^a + \omega^a_{\ b} V^b$, plus a correction term that correctly takes into account the x-dependence and the fact that we are comparing the original and the transformed field at the same point x.

Entirely in terms of the generator ω^a, this result can also be written compactly as

$$\delta_L V^a = -\omega^b \partial_b V^a + V^b \partial_b \omega^a \ . \tag{7.50}$$

In this form, this relation also generalises to other (including higher rank) tensor fields, with a sign flip for covariant indices (because they transform inversely to contravariant indices). E.g. for a covector field one has

$$\delta_L A_a = -\omega^b \partial_b A_a - A_b \partial_a \omega^b = -\omega^b \partial_b A_a - \omega^b_{\ a} A_b \ . \tag{7.51}$$

For $A_a = \partial_a \phi$ this agrees precisely with the result (7.39) derived before.

For higher rank tensors, the result can be deduced from what we already know. There is a universal term $(-\omega^a \partial_a)$ acting on any tensor, and then each contravariant oder covariant index is treated like that in V^a or A_a.

In particular, for a $(0,2)$-tensor we have

$$\delta_L T_{ab} = -\omega^c \partial_c T_{ab} - (\partial_a \omega^c) T_{cb} - (\partial_b \omega^c) T_{ac} \ . \tag{7.52}$$

For $T_{ab} = \eta_{ab}$ the Minkowski metric we get

$$\delta_L \eta_{ab} = -\partial_a \omega_b - \partial_b \omega_a = -\omega_{ba} - \omega_{ab} = 0 \tag{7.53}$$

by anti-symmetry of ω_{ab}. This is how the invariance of the Minkowski metric under Lorentz transformations is encoded in, or emerges from, this way of writing things.

As a concluding remark I just want to mention that these formulae we have derived for the transformation of Lorentz tensors under Lorentz transformations are also true for the transformation of tensors under arbitrary coordinate transformations, with $\xi^a = \xi^a(x)$ the infinitesimal, but now arbitrary, generator,

$$\bar{x}^a = x^a + \xi^a(x) \ . \tag{7.54}$$

Note that now, due to the arbitrariness of $\xi^a(x)$, the new coordinates $\bar{x}^a$ are in general no longer inertial coordinates, but this does not prevent us from considering such coordinate transformations (e.g. the transformation to polar or spherical coordinates).

In this more general context the variation of a tensor is called the *Lie Derivative* of the tensor along (the vector field) $\xi^a(x)$,

$$\delta_\xi T^{a\ldots}_{b\ldots} = -L_\xi T^{a\ldots}_{b\ldots} \ , \tag{7.55}$$

with

$$L_\xi T^{a\ldots}_{b\ldots} = \xi^c \partial_c T^{a\ldots}_{b\ldots} \pm \ldots \tag{7.56}$$

Since general covariance (invariance under general coordinate transformations) is at the heart of the theory of General Relativity, Einstein's theory of gravity, the Lie derivative plays an important role in this context. For much more on this, see my Lecture Notes on General Relativity.

7.4 Some Properties of the Gauge Covariant Derivative

In this section we look at some further properties of the covariant derivative D_a introduced in section 6.2 to describe the minimal coupling of a complex scalar field to the Maxwell field. This is interesting in its own right and can also help to simplify and demystify certain calculations.

Let us say that a field $\Phi^{(q)}$ has charge q if under phase transformations it transforms as

$$\Phi \to \mathrm{e}^{i\theta}\Phi \quad \Rightarrow \quad \Phi^{(q)} \to \mathrm{e}^{iq\theta}\Phi^{(q)} \ . \tag{7.57}$$

Thus I have (arbitrarily) normalised the charge of the field Φ and its complex conjugate Φ^* to be ± 1. Examples of fields with integer charge $q = n > 0$ or $q = -m < 0$ are

$$\Phi^{(n)} = (\Phi)^n \quad , \quad \Phi^{(-m)} = (\Phi^*)^m \ . \tag{7.58}$$

The covariant derivative on a field of charge q should act as

$$D_a\Phi^{(q)} = \partial_a\Phi^{(q)} - iqA_a\Phi^{(q)} \ , \tag{7.59}$$

because this will ensure that the derivative indeed transforms covariantly, i.e. the same way as the charged field itself,

$$\Phi^{(q)} \to \mathrm{e}^{iq\theta}\Phi^{(q)} \quad \Rightarrow \quad D_a\Phi^{(q)} \to \mathrm{e}^{iq\theta}D_a\Phi^{(q)} \ . \tag{7.60}$$

One way of guaranteeing or enforcing this on fields built from products of Φ and Φ^* and their covariant derivatives is to require that the covariant derivative satisfies the product rule (or

Leibniz rule). For example, consider the field Φ^2. It has charge $q = 2$, and therefore its covariant derivative should be

$$D_a\Phi^2 = \partial_a\Phi^2 - 2iA_a\Phi^2 \ . \tag{7.61}$$

Evaluating this further, we find

$$\begin{aligned} D_a\Phi^2 &= (\partial_a\Phi)\Phi + \Phi\partial_a\Phi - 2iA_a\Phi^2 \\ &= (\partial_a\Phi - iA_a\Phi)\Phi + \Phi(\partial_a\Phi - iA_a\Phi) \\ &= (D_a\Phi)\Phi + \Phi(D_a\Phi) = 2\Phi(D_a\Phi) \ . \end{aligned} \tag{7.62}$$

Thus, conversely, the charge 2 covariant derivative arises automatically from the charge 1 covariant derivative of Φ if one requires the product rule. More generally,

$$D_a(\Phi^{(p)}\Phi^{(q)}) = (D_a\Phi^{(p)})\Phi^{(q)} + \Phi^{(p)}(D_a\Phi^{(q)}) \tag{7.63}$$

is satisfied, if the three covariant derivatives appearing in this identity are those appropriate for fields of charge $p+q, p, q$ respectively.

In particular, on a field of charge $q = 0$, one has

$$D_a\Phi^{(q=0)} = \partial_a\Phi^{(q=0)} \ . \tag{7.64}$$

A field of charge 0 means that it is invariant under phase transformations. An examples is $\Phi^*\Phi$, with (the A_a-terms cancel out)

$$D_a(\Phi^*\Phi) = (D_a\Phi^*)\Phi + \Phi^* D_a\Phi = (\partial_a\Phi^*)\Phi + \Phi^*\partial_a\Phi = \partial_a(\Phi^*\Phi) \ . \tag{7.65}$$

Another example, and this brings me back to the calculation we performed in (6.48), is the phase invariant (charge 0) combination $\Phi D^b\Phi^*$. For this we can now use the above rules to immediately deduce that

$$\partial_b(\Phi D^b\Phi^*) = D_b(\Phi D^b\Phi^*) = (D_b\Phi)D^b\Phi^* + \Phi D_b D^b\Phi^* \ , \tag{7.66}$$

without having to manually add and subtract terms involving A_a.

Thus the covariant derivative shares with the ordinary partial derivative the property that it satisfies the product rule. However, crucially and characteristically, one property that it does not share is the useful (and much used in these notes) fact that partical derivatives commute. In fact, it is easy to calculate the commutator of covariant derivatives on Φ. Using the fact that partial derivatives do commute and that also $A_aA_b = A_bA_a$, one finds

$$[D_a, D_b]\Phi = [\partial_a - iA_a, \partial_b - iA_b]\Phi = -i(\partial_aA_b - \partial_bA_a)\Phi = -iF_{ab}\Phi \ . \tag{7.67}$$

Thus the commutator of covariant derivatives gives us the field strength tensor! And this could have been an alternative way to find or define F_{ab}.

7.5 Spontaneously Broken Symmetries (Goldstone and Higgs): Toy Models

The aim of this section is to illustrate, in a very simple, classical and Abelian, toy model, two mechanisms / phenomena that are associated with the spontaneous breaking of global or

gauge symmetries, and that play a crucial and fundamental role in various fields of physics, in particular for the understanding of the properties of (elementary) particles within the framework of what is known as the Standard Model of Particle Physics. These are

- the *Goldstone Mechanism*, explaining the appearance of massless particles (Nambu-Goldstone bosons) as a consequence of the spontaneous breaking of a global symmetry, and
- the *Higgs Mechanism* (or Brout-Englert-Higgs-Guralnik-Hagen-Kibble mechanism), explaining the emergence of massive gauge bosons from (what looks like) the spontaneous breaking of a local (gauge) symmetry.

Of course the real mechanisms are statements about the spontaneous breaking of non-Abelian symmetries in interacting quantum field theories, and are much more subtle and harder to prove rigorously.

The model we will look at is that of a complex scalar field, with action (5.63),

$$S[\Phi] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab}\partial_a\Phi\partial_b\Phi^* - W(\Phi,\Phi^*)\right) \quad . \tag{7.68}$$

and with a specific choice of potential, namely the quartic potential (5.78)

$$W(\Phi,\Phi^*) = W(\Phi^*\Phi) = \frac{\lambda}{2}(\Phi^*\Phi - a^2)^2 \quad . \tag{7.69}$$

In particular, this theory has the global $U(1)$-symmetry (5.74)

$$\Phi(x) \to \mathrm{e}^{\,i\theta}\Phi(x) \quad , \quad \Phi^*(x) \to \mathrm{e}^{\,-i\theta}\Phi^*(x) \quad . \tag{7.70}$$

We will also (subsequently) look at the minimally coupled theory (cf. section 6.2), where this global $U(1)$-symmetry has been gauged, but for now we continue with the ungauged action.

As already mentioned in section 5.4, the lowest energy solutions (ground states, vacua) of this theory are the constant fields with $|\Phi| = a$, i.e.

$$\Phi = \Phi_\alpha = a\mathrm{e}^{\,i\alpha} \quad . \tag{7.71}$$

labelled by a constant angle α, and mapped into each other by the $U(1)$-symmetry.

$$\Phi_\alpha \to \Phi_{\alpha+\theta} \quad . \tag{7.72}$$

However, every ground state individually "spontaneouly" completely breaks this global symmetry, i.e. it is not invariant under any non-trivial $U(1)$-transformation.

To better understand the properties of this theory, and the consequences of this, it is convenient to use the polar decomposition (5.70)

$$\Phi(x) = \rho(x)\mathrm{e}^{\,i\varphi(x)} \quad \Rightarrow \quad \Phi^*\Phi = \rho^2 \quad , \tag{7.73}$$

in terms of which the Lagrangian takes the form

$$L = -\frac{1}{2}\left((\partial\rho)^2 + \rho^2(\partial\varphi)^2\right) - \frac{\lambda}{2}(\rho^2 - a^2)^2 \quad . \tag{7.74}$$

At first sight this does not look particularly enlightning. But we now proceed as one would in quantum field theory. In that setting, particles arise as excitations of the field above the vacuum. In our classical setting, this means that we should expand the field around one of its ground states, which we can without loss of generality take to be the field

$$\Phi_0 = a : \quad \rho = a \quad , \quad \varphi = 0 \ . \tag{7.75}$$

We therefore parametrise Φ as

$$\Phi = (a + \sigma)\mathrm{e}^{i\varphi} \tag{7.76}$$

with σ and φ "small", meaning that we will only keep terms to quadratic order in these fields (higher order terms corresponding to small couplings and interactions). In particular, for the potential we find

$$W(\rho = a + \sigma) = \frac{\lambda}{2}(2a\sigma + \sigma^2)^2 \approx \frac{1}{2}(4\lambda a^2)\sigma^2 + \dots \ , \tag{7.77}$$

so this is a mass term for σ, with mass

$$(m_\sigma)^2 = 4\lambda a^2 \ , \tag{7.78}$$

and no mass term (of course no potential whatsoever, as a consequence of the $U(1)$-symmetry of the potential) for φ,

$$(m_\varphi)^2 = 0 \ . \tag{7.79}$$

In the kinetic term, we can approximate

$$\rho^2(\partial\varphi)^2 \approx a^2(\partial\varphi)^2 \ , \tag{7.80}$$

so this now becomes a standard kinetic term for the field $a\varphi$, and thus to leading (quadratic) order the Lagrangian is

$$L = -\tfrac{1}{2}(\partial\sigma)^2 - \tfrac{1}{2}a^2(\partial\varphi)^2 - \tfrac{1}{2}(m_\sigma)^2\phi^2 \ . \tag{7.81}$$

The spectrum of the theory therefore consists of one massive particle σ with mass m_σ, and one massless particle φ.

The appearance of a massive particle in the spectrum is unsurprsing and completely generic: it arises whenever one expands around the minimum of a potential, even for just one real scalar field, say with $V(\phi_0) = 0$,

$$V(\phi) = V(\phi_0) + (\phi - \phi_0)V'(\phi_0) + \tfrac{1}{2}(\phi - \phi_0)^2V''(\phi_0) + \dots = \tfrac{1}{2}(\phi - \phi_0)^2V''(\phi_0) + \dots \tag{7.82}$$

This is a mass term for the field $\sigma = \phi - \phi_0$.

What is much more interesting is the appearance of a massless field φ in the spectrum. This field is associated with the phase of the complex field, and its appearance is strictly correlated with the fact that this global $U(1)$ phase symmetry has been spontaneously broken. One can loosely think of it as reflecting the ability of the field to fluctuate in that direction, i.e. along the minima of the potential, without any cost in energy.

In more generality, *Goldstone's theorem* states that whenever a global symmetry is spontaneously broken (down to some subgroup), one obtains a massless particle (a *Goldstone boson* or *Nambu-Goldstone boson*) for each generator of the global symmetry group that has been broken. This

mechanism finds applications in a wide variety of fields, from condensed matter and solid state physics ("phonons" and "magnons") to particle physics ("pions").

Now what happens, when the spontaneouly broken symmetry in question is not a global symmetry but a gauge symmetry? At first, this sounds dangerous: you do not really want to break a gauge symmetry (which is supposed to just represent a certain redundancy in our description of the physics, which is supposed to be invariant under gauge symmetry transformations). But maybe things are fine when the gauge symmetry in question is broken spontaneously? Actually, one can prove that in a quantum theory there is no such thing like a spontaneously broken gauge symmetry (this is known as *Elitzur's theorem*), but let us not worry about this here (at the rather imprecise classical and heuristic level at which we are working here, it is more an issue of terminology ...).

Thus to address this question, again in the framework of our classical Abelian toy model, we gauge the $U(1)$-symmetry by minimal coupling (section 6.2), and we therefore consider the action

$$S[\Phi;A] = \int d^4x \left(-\tfrac{1}{2}\eta^{ab} D_a\Phi D_b\Phi^* - W(\Phi,\Phi^*)\right) \quad , \tag{7.83}$$

with the same quartic potential as above (this is the action of what is known as the *Abelian Higgs Model*). In terms of the polar decomposition of Φ, gauge transformations now act as shifts of φ, while ρ is gauge invariant,

$$\Phi(x) = \rho(x)\mathrm{e}^{i\varphi(x)} \to \mathrm{e}^{i\theta(x)}\Phi(x) \quad \Rightarrow \quad \rho(x) \to \rho(x) \quad , \quad \varphi(x) \to \varphi(x) + \theta(x) \ . \tag{7.84}$$

In particular, the linear combination $A_a - \partial_a\varphi$ is gauge invariant,

$$B_a = A_a - \partial_a\varphi \to B_a \ . \tag{7.85}$$

For the covariant derivative we find

$$D_a\Phi = (\partial_a\rho + i\rho\partial_a\varphi - iA_a\rho)\mathrm{e}^{i\varphi} = (\partial_a\rho + i\rho(A_a - \partial_a\varphi))\mathrm{e}^{i\varphi} = (\partial_a\rho + i\rho B_a)\mathrm{e}^{i\varphi} \ . \tag{7.86}$$

We see that the gauge invariance of the theory, and the covariance of the covariant derivative under gauge transformations, are reflected in the fact that A_a only appears in the gauge invariant combination $B_a = A_a - \partial_a\varphi$. As a consequence, the field φ has also completely disappeared from the Lagrangian, which now reads

$$L = -\tfrac{1}{2}(\partial\rho)^2 - \tfrac{1}{2}\rho^2 B_a B^a - W(\rho^2) \ . \tag{7.87}$$

Again this theory (supplemented by the Maxwell action, say, which is invariant under $A_a \to B_a$) has the ground states Φ_α (7.71) (supplemented by $A_a = 0$), and we can again expand around one of them, say Φ_0, as above, with the result that instead of a massless particle φ we now get what looks like a mass term

$$-\tfrac{1}{2}a^2 B_a B^a = -\tfrac{1}{2}(m_B)^2 B_a B^a \tag{7.88}$$

for the gauge field B_a!

This is remarkable: clearly an explicit mass term in the action for the gauge field is not allowed by gauge invariance, but such a mass term can arise from (what looks like) the spontaneous

breaking of the gauge symmetry, arising e.g. from an appropriate complex scalar field and a suitable potential. This is the famous *Higgs Mechanism*!

Remarks:

1. One might worry about what happens to the degrees of freedom of the theory when the massless field φ just disappears from the spectrum. The resolution is that, while a massless gauge field in four dimensions has 2 degrees of freedom, a massive gauge field has 3. Particle physicists like to say that the gauge field has "eaten" the massless Goldstone boson to become massive (but you should not think of this as an explanation of anything).

2. A slightly more involved variant of this quartic potential, built from a doublet (Φ_1, Φ_2) of complex scalar fields, appears as the potential for the Higgs field in the Standard Model of particle physics. In this case the massive (and short range) gauge fields emerging from this mechanism are the $W^{\pm}$ and Z bosons (while the photon remains massless).

In concluding this section I want to stress once more that the purely classical picture and explanation given here of these effects is inadequate (and misleading in several respects), and a full quantum field theory treatment of these issues, with quite some care and mathematical rigour, is required.

8 General Structure of Theories with Local Symmetries: Noether's 2nd Theorem

8.1 Maxwell Theory Revisited

While we have already studied the general structure of Maxwell theory in quite some detail in previous sections, also from the point of view of gauge symmetries (e.g. the relation between gauge invariance and current conservation described in section 5.5), there are some other related properties of Maxwell theory that we have not yet discussed. These are not only interesting and instructive in their own right. They are also prototypical of the structure of theories with local (or gauge) symmetries in general. This general story is the content of Noether's remarkable and non-trivial 2nd Theorem, and a simplified version of it will be described in section 8.3 below.

The two aspects of Maxwell theory I want to highlight are, in turn,

- the issue of Noether currents and Noether charges for gauge symmetries, and
- the characteristic (constrained) structure of the field equations.

1. Noether's 1st Theorem and Gauge Symmetries

   We have seen that for any finite-dimensional symmetry group of an action, e.g. global $U(1)$ phase transformations, translations, Lorentz transformations, Noether's theorem provides us with conserved charges or currents, equal in number to the dimension of the symmetry group, i.e. the number of generators or independent constant parameters (one, or four, or six in the above examples).

   The gauge symmetry of Maxwell theory, however, depends on an arbitrary function we called $\Psi(x)$ or $\theta(x)$, and is therefore an ininite-dimensional symmetry group. Does this mean that Noether's theorem will provide us with an infinite number of non-trivial conserved currents for Maxwell theory? At first sight, that seems to be the logical, albeit perhaps somewhat unlikely, conclusion. Let us see what actually happens.

   We begin with the pure Maxwell Lagrangian $L = -F^2/4$ (and we will include the current later). This Lagrangian is strictly invariant under gauge transformations (I will continue to use the notation $\theta(x)$, as in section 6.2),

$$\delta_\theta A_b = \partial_b \theta \quad \Rightarrow \quad \delta_\theta L = 0 \ . \tag{8.1}$$

   Therefore, the Noether theorem tells us that the current

$$J^a_\theta = \frac{\partial L}{\partial(\partial_a A_b)} \delta_\theta A_b = -F^{ab} \partial_b \theta \tag{8.2}$$

   is conserved. This is of course indeed true, as we can check by calculating

$$\partial_a J^a_\theta = -(\partial_a F^{ab}) \partial_b \theta - F^{ab} \partial_a \partial_b \theta = 0 \ , \tag{8.3}$$

where the first term is zero by the Maxwell equations, and the second term because of the anti-symmetry of F^{ab}. However, does this actually contain any non-trivial information? No. To see this, write the current as

$$J^a_\theta = -\partial_b(F^{ab}\theta) + (\partial_b F^{ab})\theta \ . \tag{8.4}$$

The second term vanishes for any solution to the Maxwell equations, and what remains,

$$J^a_\theta = \partial_b(-F^{ab}\theta) \quad \text{(on solutions)} \ , \tag{8.5}$$

is precisely of the form (6.13)

$$I^a(x) = \partial_b U^{ab}(x) \quad \text{with} \quad U^{ab}(x) = -U^{ba}(x) \quad \Rightarrow \quad \partial_a I^a(x) = 0 \quad \text{identically} \tag{8.6}$$

of an identically conserved current, which we can always add to or subtract from any Noether current. In particular, the associated Noether charges (which we are only ever interested in for solutions) are all zero (provided that either the gauge fields or the gauge transformation parameter $\theta(x)$ vanish in an appropriate way at infinity),

$$Q_\theta = \int_\Sigma d^3x \ J^0_\theta = \int_\Sigma d^3x \ \partial_k(F^{k0}\theta) = 0 \ . \tag{8.7}$$

Thus our potentially infinite number of conserved charges for Maxwell theory have just been reduced to zero (in number and value).

In section 8.3 below I will give a very simple argument to show that this must be true for the Noether charges associated to any local symmetries. The more intricate 2nd theorem of Noether will then, among other things, provide us with more detailed information about how this comes about.

If we add an electric source current, which for clarity I will now denote by J^a_S,

$$L = -\tfrac{1}{4}F^2 + A_a J^a_S \quad \Rightarrow \quad \partial_a F^{ab} + J^b_S = 0 \ , \tag{8.8}$$

then from section 5.5 we know that

(a) this current has to be conserved (e.g. by the matter equations of motion of a matter theory minimally coupled to Maxwell theory, as in section 6.2),

(b) when this condition is satisfied, the Lagrangian is invariant under gauge transformations up to a total derivative,

$$\delta_\theta L = \frac{d}{dx^a}(J^a_S\theta) \ . \tag{8.9}$$

Therefore now Noether's 1st theorem gives us the conserved current

$$J^a_\theta = -F^{ab}\partial_b\theta - J^a_S\theta = -\partial_b(F^{ab}\theta) + (\partial_b F^{ab} - J^a_S)\theta \ . \tag{8.10}$$

Thus on solutions the Noether current reduces to the same identically conserved quantity as before, with the same conclusions.

We had also already found the same kind of result for the Noether current of the gauge invariant minimally coupled theory of a complex scalar field coupled to Maxwell theory in (6.55) of section 6.2:

(a) From the action (6.42)

$$S_{tot}[\Phi, A] = S_{\text{Maxwell}}[A] + S[\Phi; A] \tag{8.11}$$

we obtained the Maxwell equations of motion

$$\partial_a F^{ab} + J_S^b = 0 \ , \tag{8.12}$$

where the source current is obtained from varying the minimally coupled matter action with respect to A (6.46),

$$J_S^b = (i/2)(\Phi D^b \Phi^* - \Phi^* D^b \Phi) \ . \tag{8.13}$$

(b) This source current is (up to a constant factor) equal to the Noether current associated to the invariance of the *gauged* action under *global* (constant) gauge transformations,

$$\theta \quad \text{constant} \quad \Rightarrow \quad J_\theta^a = -\theta J_S^a \ . \tag{8.14}$$

In particular, therefore, invariance of the gauged action under global gauge transformation implies charge conservation.

(c) However, the Noether current associated to non-constant local gauge transformations can be written as (6.55)

$$J_\theta^a = -\theta(\partial_b F^{ba} + J_S^a) - \partial_b(\theta F^{ab}) \ , \tag{8.15}$$

precisely as in the example above, with a fixed (non-dynamical) external current J_S^a, and therefore is again trivial.

2. Constrained Structure of the Maxwell Field Equations

If we have a real scalar field satisfying an equation like $\Box\phi = 0$ (or one of its variants), then a solution $\phi(t, \vec{x})$ is uniquely determined everywhere by specifying suitable initial data on an initial spacelike hypersurface, e.g. "position" $\phi(0, \vec{x})$ and "momentum" $\dot{\phi}(0, \vec{x})$ at $t = 0$. Likewise, when we have N scalar fields satisfying such 2nd order differential equations, then their solutions are also uniquely determined by specifying suitable initial data for these N fields.

With this in mind, let us now look at the Maxwell equations (with $J^a = 0$ for simplicity),

$$\partial_a F^{ab} = 0 \ . \tag{8.16}$$

These are $N = 4$ 2nd order differential equations for the $N = 4$ components of the gauge field $A_a(x)$. At first sight, this looks like just the right number of equations to determine the $A_a(x)$ uniquely once suitable initial data have been specified at $t = 0$.

At second sight, however, this cannot possibly be correct: after all, the theory is gauge invariant, and the $A_a(x)$ can and should not be determined uniquely at later times, but

only up to gauge transformations. I.e. even if you specify initial data that are not gauge invariant (and specifying $A_a(t=0,\vec{x})$ cannot possibly be gauge invariant), you should still be able to perform gauge transformations at a later time, i.e. with some function $\theta(t,\vec{x})$ that vanishes for $t \leq 0$, say, and obtain a different solution for $A_a(t,\vec{x})$ from the same initial data. Therefore gauge invariance implies that the $N=4$ Maxwell equations should not determine the $N=4$ components of $A_a(x)$ uniquely. How does that come about?

The resolution is that the 4 Maxwell equations are not independent: there is one differential relation among them, namely

$$\partial_b(\partial_a F^{ab}) = 0 \quad . \tag{8.17}$$

As a consequence, only 3 of the 4 equations are independent differential equations, and this is precisely the right number to determine the 4 components of $A_a(x)$ up to gauge transformations, i.e. up to 1 function.

This may all sound a bit abstract, but we can also understand this very concretely. If all 4 equations were standard (2nd order in time) differential equations, then this would be like $N=4$ equations for $N=4$ scalar fields, and this would be in conflict with gauge invariance. But we know that among these 4 equations there is one, namely

$$\partial_a F^{a0} = \partial_k F^{k0} = 0 \quad \Leftrightarrow \quad \vec{\nabla}.\vec{E} = 0 \tag{8.18}$$

which only involves first time-derivatives of the gauge field. Therefore, this is not at all a standard evolution equation, but a *constraint* on the initial data at a given time: they can not be chosen arbitrarily. Rather, they need to be chosen such in such a way that they satisfy $\vec{\nabla}.\vec{E} = 0$.

There is another way of seeing or understanding that such a constraint equation has to exist, just as a consequence of the identity (8.17). Namely, let us write (8.17) as

$$\partial_0(\partial_k F^{k0}) = -\partial_k(\partial_a F^{ak}) \quad . \tag{8.19}$$

Since the Maxwell equations are 2nd order differential equations, the right-hand side contains at most 2nd time derivatives. This implies that $\partial_k F^{k0}$ can at most contain 1st time derivatives, and therefore the zero-component $\partial_k F^{k0} = 0$ of the Maxwell equation is not at all an evolution equation, but is rather a condition relating the fields and their time derivatives at any given time. In particular, this equation is a constraint on the allowed initial data!

The charm and power of Noether's 2nd Theorem, to be discussed below, is that it not only establishes results analogous to those discussed in the two items above in the previous section 8.1 in complete generality, for any theory with local symmetries, but that it moreover also provides a general direct link and strict relation between the two observations, namely

- identically conserved Noether currents, and
- the existence of differential relations among the equations of motion (Euler-Lagrange derivatives).

8.2 Noether Charges for Local Symmetries are Identically Zero

Before turning to this, let me give you a simple argument that in any theory with local symmetries, i.e. symmetries depending on a certain number of arbitrary functions of the spacetime coordinates, a conserved Noether charge associated to such a symmetry is necessarily identically zero. In this argument, we will not need to make any assumptions about the currents themselves, in particular whether or not they are identically conserved.

As in the previous section, let Q_θ be the candidate conserved Noether charge associated to some arbitrary function (or collection of functions) $\theta(x)$, i.e.

$$Q_\theta(t) = \int_{\Sigma_t} d^3x \; J^0_\theta \tag{8.20}$$

If $Q_\theta(t)$ is conserved, then this means that

$$Q_\theta(t_2) = Q_\theta(t_1) \ . \tag{8.21}$$

Now consider a different collection of functions $\vartheta(x)$, such that

$$\begin{aligned} &\vartheta(x) = \theta(x) \qquad \text{in a neighbourhood of} \quad \Sigma_{t_1} \\ &\vartheta(x) = 0 \qquad\quad \text{in a neighbourhood of} \quad \Sigma_{t_2} \end{aligned} \tag{8.22}$$

(the existence of such functions is guaranteed by the premise that we have local symmetries depending on arbitrary functions). Because of the first condition, we clearly have

$$Q_\vartheta(t_1) = Q_\theta(t_1) \ , \tag{8.23}$$

and because of the second condition we have

$$Q_\vartheta(t_2) = 0 \ . \tag{8.24}$$

But now, conservation of Q_ϑ means

$$Q_\vartheta(t_2) = 0 \quad \Rightarrow \quad Q_\vartheta(t_1) = 0 \quad \Rightarrow \quad Q_\theta(t_1) = 0 \quad \Rightarrow \quad Q_\theta(t) = 0 \quad \forall\, t \ . \tag{8.25}$$

Isn't this a nice and simple argument?

8.3 Noether's 2nd Theorem

Let us now turn to the non-trivial part of Emmy Noether's famous and fundamental work *Invariante Variationsprobleme* on symmetries and variational problems, which was actually prompted by questions of Hilbert regarding the apparent failure of what he referred to as the "energy theorem" in Einstein's theory of General Relativity.[3]

Noether considered the completely general case of a Lagrangian L

[3] See e.g. N. Byers, *E. Noether's Discovery of the Deep Connection Between Symmetries and Conservation Laws*, `https://arxiv.org/abs/physics/9807044` for some historical context, and the monograph *The Noether Theorems* by Y. Kosmann-Schwarzbach for a detailed and erudite account.

- depending on an arbitrary number N of functions of p variables and their first q derivatives,
- invariant (up to total derivatives) under local transformations that depend on r arbitrary functions and their first s derivatives.

Then, among other things, she showed that

- there are r identities among the N Euler-Lagrange derivatives of L and their derivatives up to order s (the so-called Noether identities);
- conversely, if there are such identities among the Euler-Lagrange derivatives, then there exist corresponding local symmetries;
- the associated (infinite number of) conserved currents are identically conserved.

In order to illustrate this theorem, I will consider the special case where $q = 1$ (the Lagrangian only depends on the N fields $\Phi^A(x)$ and their first derivatives) and $s = 1$ (the local transformations depend on r functions $\theta^I(x)$ and their first derivatives). I will also assume that the local infinitesimal symmetry variations depend linearly on the θ^I and their first derivatives (Noether shows that one can assume this without loss of generality). Finally, for notational simplicity I will assume that the Lagrangian L (and the local symmetry transformations) do not depend explicitly on x, but nothing substantial needs to be changed in the following argument if one drops that assumption.

Thus, concretely, we have N fields $\Phi^A(x)$ and r functions $\theta^I(x)$,

$$\Phi^A(x) \quad A = 1, \ldots, N \quad , \quad \theta^I(x) \quad I = 1, \ldots, r \ , \tag{8.26}$$

and we assume that we have a Lagrangian $L = L(\Phi^A, \partial_a \Phi^A)$ that transforms as

$$\delta_\theta L = \frac{d}{dx^a} F^a_\theta \tag{8.27}$$

under variations $\delta_\theta \Phi^A$ of the fields (note that, in line with our previous discussions e.g. in section 6.3, we are not considering explicit variations of the coordinates). Since by assumption $s = 1$, these variations can be expanded as

$$s = 1 \quad \Rightarrow \quad \delta_\theta \Phi^A(x) = \Delta^A_{\ I}(\Phi)\theta^I(x) + \Delta^{Ab}_{\ \ I}(\Phi)\partial_b \theta^I(x) \ . \tag{8.28}$$

I will also introduce the notation

$$\Pi^a_A = \frac{\partial L}{\partial(\partial_a \Phi^A)} \tag{8.29}$$

for the generalised momenta of the fields Φ^A. With this notation the Euler-Lagrange derivatives are

$$\frac{\delta L}{\delta \Phi^A} = \frac{\partial L}{\partial \Phi^A} - \frac{d}{dx^a}\Pi^a_A \ , \tag{8.30}$$

and the variational master equation (5.18) takes the form

$$\delta L = \frac{\delta L}{\delta \Phi^A}\delta\Phi^A + \frac{d}{dx^a}\left(\Pi^a_A \delta \Phi^A\right) \ . \tag{8.31}$$

The infinite number of conserved currents predicted by Noether's 1st theorem are the

$$J^a_\theta = \Pi^a_A \delta_\theta \Phi^A - F^a_\theta \ , \tag{8.32}$$

with

$$\frac{d}{dx^a} J^a_\theta = \frac{\delta L}{\delta \Phi^A} \delta_\theta \Phi^A \ , \tag{8.33}$$

and thus

$$\frac{\delta L}{\delta \Phi^A} = 0 \quad \Rightarrow \quad \frac{d}{dx^a} J^a_\theta = 0 \quad \forall\, \theta^I(x) \ . \tag{8.34}$$

We start with the Noether identities satisfied by the Euler-Lagrange derivatives, following directly the argument given by Noether herself (in more generality). To that end, we write (8.33) more explicitly as

$$\frac{d}{dx^a} J^a_\theta = \frac{\delta L}{\delta \Phi^A} (\Delta^A_{\ I} \theta^I + \Delta^{Ab}_{\ \ I} \partial_b \theta^I) \tag{8.35}$$

and "integrate by parts" the last term, to arrive at

$$\frac{d}{dx^a} \left(J^a_\theta - \frac{\delta L}{\delta \Phi^A} \Delta^{Aa}_{\ \ I} \theta^I \right) = \left(\frac{\delta L}{\delta \Phi^A} \Delta^A_{\ I} - \partial_b (\Delta^{Ab}_{\ \ I} \frac{\delta L}{\delta \Phi^A}) \right) \theta^I \tag{8.36}$$

Now this is true for arbitrary θ^I, and so we can now integrate this over arbitrary domains, with functions that are arbitrary in the interior of the domain but which are required to be zero on the boundary together with their 1st derivatives (in general, with vanishing derivatives on the boundary up to the order with which they appear in the term in brackets on the left-hand side). Then we will always get zero on the left-hand side, and this in turn implies that the function on the right-hand side has to be identically zero. Therefore we obtain the Noether identities

$$\frac{\delta L}{\delta \Phi^A} \Delta^A_{\ I} - \partial_b (\Delta^{Ab}_{\ \ I} \frac{\delta L}{\delta \Phi^A}) = 0 \ . \tag{8.37}$$

These are r identities relating the Euler-Lagrange derivatives and their first ($s = 1$) derivatives.

Conversely, as mentioned above, identities among the Euler-Lagrange derivatives and their derivatives imply the existence of corresponding local symmetries for which these identities are just the Noether identities. We will establish this claim in section 8.5 below.

Example: Maxwell Theory

For Maxwell theory, the fields are $\phi^A \mapsto A_c$ (so an upper index A is now a lower index c, and this and related substitutions will be indicated by a "maps to" arrow "$\mapsto$" in the following). The local symmetry transformations are the gauge transformations

$$\delta_\theta \Phi^A \mapsto \delta_\theta A_c = \partial_c \theta \ , \tag{8.38}$$

so $r = 1$ (and we can suppress the label I), and the parameters in (8.28) are

$$\Delta^A_{\ I} = 0 \quad , \quad \Delta^{Ab}_{\ \ I} \mapsto \Delta_c^{\ b} = \delta_c^{\ b} \ . \tag{8.39}$$

[If we had also included a minimally coupled complex scalar field, with $\delta_\theta \Phi = i\theta\Phi$, say, then for that field we would have had $\Delta^A_{\ I} \neq 0$.] Moreover, because the Maxwell Lagrangian is strictly invariant under gauge transformations, $F^a_\theta = 0$, and we also have

$$\Pi^a_A \mapsto \Pi^{ac} = -F^{ac} \quad , \quad \frac{\delta L}{\delta \Phi^A} \mapsto \partial_a F^{ac} \tag{8.40}$$

Therefore, the Noether identities (8.37) are

$$\partial_b(\Delta_I^{Ab}\frac{\delta L}{\delta\Phi^A}) = 0 \mapsto \partial_b(\delta_c{}^b\partial_a F^{ac}) = \partial_b(\partial_a F^{ab}) = 0 \quad . \tag{8.41}$$

This is precisely the identity (8.17) we encountered and discussed in the previous section which gives us one ($r = 1$) differential relation among the equations of motion of Maxwell theory.

8.4 Local Symmetries lead to Identically Conserved Noether Currents

We now turn our attention to the Noether currents (8.32)

$$J_\theta^a = \Pi_A^a\delta_\theta\Phi^A - F_\theta^a \quad . \tag{8.42}$$

Since we are only interested in these for solutions to the Euler-Lagrange equations, we now have

$$\frac{d}{dx^a}J_\theta^a = 0 \quad \forall\,\theta^I(x) \quad . \tag{8.43}$$

But actually, we already know much more. Namely, because of the Noether identities (8.37), the right-hand side of (8.36) vanishes identically, and therefore also the left-hand side. But this means that the Noether currents (modulo terms that vanish on solutions) are identically conserved,

$$\text{Noether Identities} \quad \Rightarrow \quad \frac{d}{dx^a}J_\theta^a = 0 \quad \text{identically} \quad \forall\,\theta^I(x) \quad . \tag{8.44}$$

This basically completes the argument, but it is instructive to be a bit more explicit about how this actually comes about, and to find out how one can explicitly show that J_θ has the characteristic total-derivative form (6.13)

$$J_\theta^a(x) = \partial_b U^{ab}(x) \quad \text{with} \quad U^{ab}(x) = -U^{ba}(x) \tag{8.45}$$

of an identically conserved current, and how to obtain the corresponding U^{ab}. The idea[4] will be to expand this equation in the θ^I and their derivatives (upon which it will break up into several equations all of which have to be satisfied individually).

To that end let us first of all take a closer look at F_θ^a. Since L contains at most 1st derivatives of the fields ϕ^A, and $\delta_\theta\phi^A$ at most 1st derivatives of the functions θ^I, $\delta_\theta L$ contains at most 2nd derivatives of the θ^I, and therefore F_θ^a itself contains at most 1st derivatives of the θ^I. We can therefore also expand F_θ^a as

$$F_\theta^a(\Phi) = F_I^a(\Phi)\theta^I + F_I^{ab}(\Phi)\partial_b\theta^I \quad . \tag{8.46}$$

Using (8.28), we can now expand the current J_θ^a itself,

$$J_\theta^a = (\Pi_A^a\Delta_I^A - F_I^a)\theta^I + (\Pi_A^a\Delta_I^{Ab} - F_I^{ab})\partial_b\theta^I \quad . \tag{8.47}$$

[4]See e.g. B. Julia, S. Silva, *Currents and Superpotentials in classical gauge invariant theories I*, `https://arxiv.org/abs/gr-qc/9804029`.

Acting on this with d/dx^a and sorting the terms according to the derivatives of θ^I they contain, we find

$$\begin{aligned} 0 = \frac{d}{dx^a} J^a_\theta &= \left[\frac{d}{dx^a} (\Pi^a_A \Delta^A_I - F^a_I) \right] \theta^I \\ &+ \left[(\Pi^b_A \Delta^A_I - F^b_I) + \frac{d}{dx^a} (\Pi^a_A \Delta^{Ab}_I - F^{ab}_I) \right] \partial_b \theta^I \\ &+ \left[\Pi^a_A \Delta^{Ab}_I - F^{ab}_I \right] \partial_a \partial_b \theta^I \quad . \end{aligned} \tag{8.48}$$

Since this expression has to be zero for arbitrary $\theta^I(x)$, the 3 terms in brackets have to vanish separately. The only thing to pay attention to is that, in the last line, because $\partial_a \partial_b \theta^I$ is symmetric in (a, b), only the symmetrised part of the term in brackets contributes. Thus we have

$$\begin{aligned} (I): &\quad \frac{d}{dx^a} (\Pi^a_A \Delta^A_I - F^a_I) = 0 \\ (II): &\quad (\Pi^b_A \Delta^A_I - F^b_I) + \frac{d}{dx^a} (\Pi^a_A \Delta^{Ab}_I - F^{ab}_I) = 0 \\ (III): &\quad \Pi^a_A \Delta^{Ab}_I - F^{ab}_I = U^{ab}_I \quad , \quad U^{ab}_I = -U^{ba}_I \quad , \end{aligned} \tag{8.49}$$

and we now look at the implications of these conditions in turn.

1. (I) tells us that

$$\frac{d}{dx^a} (\Pi^a_A \Delta^A_I - F^a_I) = 0 \quad , \tag{8.50}$$

   and this is just the statement that the Noether currents J^a_I for constant θ^I,

$$\theta^I \quad \text{constant} \quad \Rightarrow \quad J^a_\theta = (\Pi^a_A \Delta^A_I - F^a_I) \theta^I = J^a_I \theta^I \tag{8.51}$$

   are conserved,

$$\frac{d}{dx^a} J^a_I = 0 \quad . \tag{8.52}$$

2. Using (III) in (II), we now deduce that these Noether currents for constant θ^I have the form

$$J^b_I + \frac{d}{dx^a} U^{ab}_I = 0 \quad \Leftrightarrow \quad J^a_I = \frac{d}{dx^b} U^{ab}_I \quad . \tag{8.53}$$

   Thus the currents have precisely the form (6.13) of identically conserved currents.

3. But this is not the end of the story. With (8.47) we can now write the general Noether current J^a_θ for arbitrary $\theta^I(x)$ as

$$\begin{aligned} J^a_\theta &= (\Pi^a_A \Delta^A_I - F^a_I) \theta^I + (\Pi^a_A \Delta^{Ab}_I - F^{ab}_I) \partial_b \theta^I \\ &= J^a_I \theta^I + U^{ab}_I \partial_b \theta^I \\ &= \left(\frac{d}{dx^b} U^{ab}_I \right) \theta^I + U^{ab}_I \partial_b \theta^I = \frac{d}{dx^b} \left(U^{ab}_I \theta^I \right) \quad . \end{aligned} \tag{8.54}$$

   This shows that also the general Noether current is identically conserved, and the Noether charge is (at best) a surface term at infinity (which, as we have seen, has to be zero in order to be conserved).

Once again, let us look at this in the case of Maxwell theory.

Example: Maxwell Theory (continued)

From the above, we have

$$U_I^{ab} = \Pi_A^a \Delta_I^{Ab} - F_I^{ab} \mapsto -F^{ac}\delta_c{}^b = -F^{ab} \tag{8.55}$$

(which indeed is anti-symmetric, as it should be), and therefore

$$J_\theta^a = \partial_b(-F^{ab}\theta) \ , \tag{8.56}$$

precisely as we found before in (8.5).

8.5 Converse of Noether's 2nd Theorem

We now come to the converse of Noether's 2nd theorem, and we will discuss this at the same level of generality as Noether's 2nd theorem in section 8.3.

Thus assume that there are r identities among the N the Euler-Lagrange derivatives and their first ($s = 1$) derivatives, which we write (similarly to (8.37)) as

$$\frac{\delta L}{\delta \Phi^A}\Gamma_I^A - \Gamma_I^{Ab}\partial_b \frac{\delta L}{\delta \Phi^A} = 0 \ . \tag{8.57}$$

By integration by parts, we can write these identities (now precisely as in (8.37)) as

$$\frac{\delta L}{\delta \Phi^A}\Delta_I^A - \partial_b(\Delta_I^{Ab}\frac{\delta L}{\delta \Phi^A}) = 0 \ . \tag{8.58}$$

where

$$\Delta_I^A = \Gamma_I^A + \partial_b \Gamma_I^{Ab} \quad , \quad \Delta_I^{Ab} = \Gamma_I^{Ab} \ . \tag{8.59}$$

Now multiply these relations by arbitrary functions $\theta^I(x)$,

$$\frac{\delta L}{\delta \Phi^A}\Delta_I^A\theta^I - \theta^I\partial_b(\Delta_I^{Ab}\frac{\delta L}{\delta \Phi^A}) = 0 \ . \tag{8.60}$$

and integrate by parts once more to obtain

$$\frac{\delta L}{\delta \Phi^A}(\Delta_I^A\theta^I + (\partial_b\theta^I)\Delta_I^{Ab}) = \frac{d}{dx^b}(\theta^I\Delta_I^{Ab}\frac{\delta L}{\delta \Phi^A}) \ . \tag{8.61}$$

Thus, defining the local symmetry transformations as in (8.28) by

$$\delta_\theta \Phi^A(x) = \Delta_I^A(\Phi)\theta^I(x) + \Delta_I^{Ab}(\Phi)\partial_b\theta^I(x) \ , \tag{8.62}$$

we have

$$\frac{\delta L}{\delta \Phi^A}\delta_\theta\Phi^A = \frac{d}{dx^b}(\theta^I\Delta_I^{Ab}\frac{\delta L}{\delta \Phi^A}) \ . \tag{8.63}$$

In conjunction with (8.31) this shows that the δ_θ-variation of L is also a total derivative,

$$\delta_\theta L = \frac{\delta L}{\delta \Phi^A}\delta_\theta\Phi^A + \frac{d}{dx^a}\left(\Pi_A^a\delta_\theta\Phi^A\right) = \frac{d}{dx^b}\left(\theta^I\Delta_I^{Ab}\frac{\delta L}{\delta \Phi^A} + \Pi_A^b\delta_\theta\Phi^A\right) \equiv \frac{d}{dx^b}F_\theta^b \ , \tag{8.64}$$

and therefore one has established that δ_θ is a local symmetry of L.

Incidentally note that the corresponding conserved Noether current that we extract from this expression is (modulo the unavoidable ambiguity consisting of the addition of identically conserved currents)

$$J^b_\theta = \Pi^b_A \delta_\theta \Phi^A - F^b_\theta = -\theta^I \Delta_I^{Ab} \frac{\delta L}{\delta \Phi^A} \quad , \tag{8.65}$$

This current is not only manifestly, by (8.63), conserved for a solution to the equations of motion, but it is actually identically zero for a solution to the equations of motion,

$$\frac{\delta L}{\delta \Phi^A} = 0 \quad \Rightarrow \quad J^b_\theta = -\theta^I \Delta_I^{Ab} \frac{\delta L}{\delta \Phi^A} = 0 \ , \tag{8.66}$$

and not just an identically conserved current, as ensured by the general considerations of the previous section and / or Noether's 2nd theorem.

8.6 Epilogue and Outlook

While Noether's 2nd theorem provides considerable insight into the structure of theories with local symmetries, it also shows that the Noether currents associated to such symmetries are essentially devoid of any useful information and are perhaps not the right objects to look at. On the other hand, it is known from examples (e.g. the electric charge in Maxwell theory, or certain definitions of mass in general relativity) that there are physically relevant conserved charges in such theories. Much more recently, therefore, from the mid-90s, the emphasis has shifted from studying such currents (to be integrated over codimension-1 surfaces) to directly studying and defining appropriate charge densities (to be integrated over codimension-2 surfaces) associated to certain local symmetries. Unfortunately to understand this requires a bit more mathematical sophistication than I can develop or explain here.[5]

[5] For an introduction, with references to the original literature, see the lucid account in section 1 of G. Compère, *Advanced Lectures in General Relativity*, `https://arxiv.org/abs/1801.07064`.